



Faculty of Engineering
Department of Electrical and Electronic Engineering

Quantum Annealing-Driven Maximum Likelihood Detection for NOMA in Next-Generation Wireless Networks

A Graduation Project Submitted in Partial Fulfillment of the Requirements for
the Degree of *Bachelor of Science*
in Electronics and Communication Engineering

By:

Mohamed S. Abuain

Supervised by:

Dr. Osama M. Abusaid

FALL 2023

Declaration

Project Title	Quantum Annealing-Driven Maximum Likelihood Detection for NOMA in Next-Generation Wireless Networks
Author	Mohamed S. Abuain
Student ID	2180205390
Supervisor	Dr. Osama M. Abusaid

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements.

Mohamed Abuain

University of Tripoli
Department of Electrical and Electronic Engineering

Date: June, 2024

Acknowledgment

First and foremost, I extend my deepest thanks to Allah for His countless blessings, and for granting me the strength and capability to complete this project.

I express my profound and sincere gratitude to my research supervisor, Dr. Osama Abusaid, for providing me the opportunity to conduct research and for his invaluable guidance throughout this work.

I am deeply indebted to my esteemed professors at the Department of Electrical and Electronic Engineering, University of Tripoli. Their teachings and guidance have been transformative, shaping me into the person I am today.

I am extremely thankful to my parents for their unconditional love, endless prayers, caring and immense sacrifices for educating and preparing me for my future. I would like to say thanks to my friends and relatives for their kind support and care.

Finally, I would like to thank all the people who have supported me to complete the project work directly or indirectly.

Mohamed Abuain

June 2024

Abstract

A central design challenge for future generations of wireless networks is to meet users' ever-increasing demand for capacity, throughput, and connectivity. Recent advances in wireless network design, including the ongoing NextG efforts, are driving the transition from Orthogonal Multiple Access (OMA) to Non-Orthogonal Multiple Access (NOMA). This transition aims to support a high number of users near base stations and maximize spectral efficiency. However, NOMA requires sophisticated signal processing to successfully disentangle mutually interfering streams, a process that becomes less efficient as the number of concurrent streams increases. Theoretically optimal Maximum Likelihood (ML) processing could significantly improve performance over traditional methods but is soon becoming infeasible due to its computational complexity and limitations in processing time. Thus, the computational capacity of base stations is becoming one of the key limiting factors on performance gains in wireless networks.

Quantum computing is a potential tool to address this computational challenge. It exploits unique information processing capabilities based on quantum mechanics to perform fast calculations that are intractable by traditional digital methods.

This work presents the application of Quantum Annealing (QA) to enhance ML processing in NOMA systems within a Centralized Radio Access Network (C-RAN) environment. The proposed QA-aided ML decoder serves as a promising alternative to the conventional Successive Interference Cancellation (SIC) method. Our analysis, which incorporates practical system parameters such as channel randomness and transmit power levels for all involved users, evaluates the Bit Error Rate (BER) performance across various modulation schemes, benchmarking against brute force (BF) solution and SIC methods. The simulation results demonstrate that the quantum annealing-based ML detector achieves near-optimal performance, outperforming SIC and closely approaching the theoretical limits of ML detection. However, the execution time for QA is longer than that for both BF and SIC. Our results set a baseline for future fully integrated quantum-classical systems within centralized RAN architectures. Once integration is optimized, quantum computing should be considered a viable option for designing high-capacity wireless networks.

الملخص

تُمثل إحدى التحديات التصميمية الرئيسية للأجيال المقبلة من شبكات الاتصال اللاسلكية تلبية الطلب المتزايد باستمرار للمستخدمين على السعة والإنتاجية والاتصال. تساهم التطورات الحديثة في تصميم الشبكات اللاسلكية، في دفع التحول من الوصول المتعدد المتعامد (OMA) إلى الوصول المتعدد غير المتعامد (NOMA). يهدف هذا التحول إلى دعم عدد كبير من المستخدمين بالقرب من المحطات الأساسية وتعظيم الكفاءة الطيفية. ومع ذلك، يتطلب NOMA معالجة إشارات معقدة لفك تداخل البث المتبادل بنجاح، وهي عملية تصبح أقل كفاءة مع زيادة عدد البث المتزامنة. من الناحية النظرية، يمكن لعملية المعالجة باستخدام تقدير حسب القيمة العليا لدالة الإمكان (Maximum Likelihood) تحسين الإنتاجية بشكل كبير مقارنة بالطرق التقليدية ولكنها سرعان ما تصبح غير قابلة للتطبيق بسبب تعقيدها الحسابي والقيود المتعلقة بزمان المعالجة. وبالتالي، تصبح القدرة الحسابية للمحطات الأساسية أحد العوامل المحددة الرئيسية للأداء في الشبكات اللاسلكية.

تعد الحوسبة الكمية أداة محتملة لمواجهة هذا التحدي الحسابي. إنها تستغل قدرات معالجة المعلومات الفريدة المستندة إلى ميكانيكا الكم لأداء الحسابات السريعة التي تعجز عنها الطرق الرقمية التقليدية.

يقدم هذا العمل تطبيق التلدين الكمي (QA) لتعزيز معالجة ML في أنظمة NOMA ضمن بيئة شبكة الوصول الراديوي المركزية (C-RAN). يعتبر أن النظام المقترح بديلاً واعداً لطريقة إلغاء التداخل التتابعي (SIC) التقليدية. يقوم تحليلنا، الذي يشمل ظروف النظام الواقعية مثل عشوائية القناة ومستويات قوة الإرسال لجميع المستخدمين المعنيين، بتقييم أداء معدل الخطأ البتي (BER) عبر أنواع التضمين المختلفة، مقارنة بطرق الحل العددي (BF) و SIC. تظهر نتائج المحاكاة أن جهاز الكشف ML المبني على التلدين الكمي يحقق أداءً شبه مثالي، متفوقاً على SIC ويقترب كثيراً من الحدود النظرية لكشف ML. ومع ذلك، فإن زمن التنفيذ لـ QA أطول من زمن التنفيذ لكل من BF و SIC. تُمثل نتائجنا نقطة مرجعية للأنظمة المستقبلية التي تدمج بين التقنيات الكلاسيكية والكمية ضمن بنى شبكات الوصول الراديوي المركزية. بمجرد تحسين هذه التقنية، يجب أن يُنظر إلى الحوسبة الكمية كخيار فعال لتصميم شبكات الاتصال اللاسلكية ذات السعة العالية.

Table of contents

List of figures	ix
List of tables	xii
List of Abbreviations	xiii
1 Introduction	1
1.1 Problem Statement	4
1.2 Project Objectives	5
1.3 Structure of Report	6
2 Towards Non-Orthogonal Multiple Access for Next-Gen Wireless Networks	7
2.1 Enabling Massive Connectivity for Next-Gen Wireless Networks	7
2.2 The Evolution of Multiple Access Techniques in Wireless Networks	9
2.2.1 Considerations for Next Generation Multiple Access Design	11
2.2.2 Possible Candidates for Next Generation Multiple Access	13
2.3 Power Domain Non-Orthogonal Multiple Access (PD-NOMA)	15
2.3.1 Superposition Coding (SC)	15
2.3.2 Successive Interference Cancellation (SIC)	17
2.3.3 Downlink NOMA Transmission	19
2.3.4 Uplink NOMA Transmission	22
2.3.5 Spectral Efficiency and Energy Efficiency	24

2.3.6	Main Advantages of PD-NOMA	26
2.4	Challenges of Multi-User Detection in Uplink	28
2.4.1	Channel Models and Estimation	28
2.4.2	The Shortcomings of SIC and Conventional Detection Methods . .	32
2.4.3	Maximum Likelihood Detection	34
2.5	Centralized Radio Access Networks for Next Generation Wireless Networks	36
3	Introduction to Quantum Annealing	39
3.1	Computational Complexity	39
3.2	Fundamentals of Quantum Computation	41
3.2.1	Quantum Systems and States	42
3.2.2	The Qubit	43
3.2.3	Models of Quantum Computation	45
3.3	Quantum Annealing	47
3.3.1	Theoretical Foundations of Quantum Annealing	48
3.3.2	Quantum Annealing vs. Classical Simulated Annealing	49
3.3.3	Problem Formulation for Quantum Annealing: Ising and QUBO Models	51
3.4	D-Wave Implementation of Quantum Annealing	53
3.4.1	Hardware Architecture of D-Wave Quantum Annealing System . .	53
3.4.2	Topology of D-Wave's Quantum Processing Units	56
3.4.3	The Need for Embedding	58
3.4.4	The Quantum Annealing Process on D-Wave Systems	60
3.4.5	Quantum Annealing Parameters and System Timing	61
3.5	Application of Quantum Annealing in Communication Systems	64
4	Leveraging Quantum Annealing for Maximum-Likelihood NOMA Processing in Centralized Radio Access Networks	66
4.1	System Model	67

Table of contents

4.2	Formulation of the NOMA ML Problem in a QUBO Model	70
4.3	Implementation	76
4.4	Evaluation	78
4.4.1	Understanding Empirical Results	79
4.4.2	Bit-Error Rate Performance for QA Based Maximum-Likelihood Versus. Number of Users Across Various Modulation Schemes . . .	82
4.4.3	Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with BPSK modulation	84
4.4.4	Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with QPSK modulation	86
4.4.5	Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with 16-QAM modulation	88
4.4.6	Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with 64-QAM modulation	90
4.4.7	Comparison of Computation Time	92
5	Conclusion and Future Work	94
5.1	Conclusion	94
5.2	Future Work	96
	References	98
	Appendix A QUBO MODEL COEFFICIENTS	102
A.1	16-QAM	102
A.2	64-QAM	104

List of figures

1	Enter Caption	iv
1.1	A quantum compute-enabled C-RAN system architecture for NextG wireless networks.	3
2.1	Illustrative example of different multiple access schemes [8].	10
2.2	An example of SC encoding (a) signal constellation of user 1 (b) signal constellation of user 2 (c) constellation of superposed signal.	16
2.3	SIC receiver.	17
2.4	An example of SC decoding.	18
2.5	Illustration of two-user downlink NOMA.	21
2.6	Illustration of two-user uplink NOMA.	23
2.7	A trade-off between energy efficiency and spectral efficiency.	26
2.8	Channel gain variation with distance.	29
2.9	The proposed channel model.	31
2.10	Uplink system model [26]	33
2.11	C-RAN architecture for mobile networks. [7]	37
3.1	Sketch representing the relation between the different complexity classes P, NP, NP-complete, and NP-hard.	40
3.2	The comparison of classical and quantum computation.	41
3.3	Qubit represented by Bloch sphere.	44

List of figures

3.4	Illustration of leveraging quantum tunneling phenomena to find the optimal solution of an energy landscape using quantum annealing[38].	50
3.5	The D-Wave quantum computer.	54
3.6	Each qubit is a loop of niobium controlled by Josephson junctions and biases represented by ϕ in this diagram [40].	55
3.7	Graphical representations of the topology graphs of two D-Wave hardware architectures.	58
3.8	Examples of minor embedding.	59
3.9	Visualization of a typical quantum algorithm workflow on a D-Wave quantum annealer[41].	61
3.10	D-Wave software environment with an Overview of execution of a single QMI, starting from a client system.	62
3.11	Breakdown of QPU access time.	63
3.12	The general workflow of QA-based optimization in wireless networks[5].	64
4.1	Proposed system model for evaluation.	68
4.2	Constellations of different modulation schemes.	71
4.3	Equivalent representations of 1×1 QPSK detection: Euclidean distances $\mathcal{D}(v)$ in the I-Q plane (left), and Ising energies $\mathcal{H}(s)$ (right). Shadings highlight likely solutions.	80
4.4	Dwave problem inspector.	81
4.5	BER vs. the number of users per cell under various modulation schemes	82
4.6	The average BER performance vs. the transmit power for a 10 user NOMA scenario employing BPSK, with different decoding techniques.	84
4.7	The BER performance vs. the transmit power per user at different cell positions—near, middle, and far- for BPSK with Quantum Annealing based maximum likelihood decoder.	85
4.8	The average BER performance vs. the transmit power for a 5 user NOMA scenario employing QPSK, with different decoding techniques.	86

List of figures

4.9	The BER performance vs. the transmit power per user at different cell positions—near, middle, and far- for QPSK with Quantum Annealing based maximum likelihood decoder.	87
4.10	The average BER performance vs. the transmit power for a 3 user NOMA scenario employing 16-QAM, with different decoding techniques.	88
4.11	The BER performance vs. the transmit power per user at different cell positions—near, middle, and far- for 16-QAM with Quantum Annealing based maximum likelihood decoder.	89
4.12	The average BER performance vs. the transmit power for a 2 user NOMA scenario employing 64-QAM, with different decoding techniques.	90
4.13	The BER performance vs. the transmit power per user at different cell positions—near, and far- for 64-QAM with Quantum Annealing based maximum likelihood decoder.	91
4.14	Comparison between the decoding techniques in terms of simulation time for different modulations (in <i>ms</i>)	92

List of tables

2.1	Path Loss Exponents for Different Environments[25]	30
3.1	Comparison of problem expressions and their terms in various contexts.	53
3.2	Overview of D-Wave Quantum Annealer Versions[5]	56
4.1	Qubit Requirements of Different NOMA Encoded Configurations	76
4.2	D-Wave System Parameters	77
4.3	Simulation Parameter Settings	83

List of Abbreviations

Acronyms / Abbreviations

1G	First Generation
2G	Second Generation
3G	Third Generation
4G	Fourth Generation
5G	Fifth Generation
6G	Sixth Generation
APs	Access Points
AQC	Adiabatic Quantum Computation
AR	Augmented Reality
BBU	Baseband Unit
BER	Bit Error Rate
BF	Brute Force
BS	Base Station
CDMA	Code Division Multiple Access

List of Abbreviations

CD-NOMA Code-Domain Non-Orthogonal Multiple Access

C-RAN Centralized Radio Access Networks

CSI Channel State Information

DoFs Degrees of Freedom

EE Energy Efficiency

eMBB Enhanced Mobile Broadband

FDMA Frequency Division Multiple Access

IoE Internet of Everything

IoT Internet of Things

KPIs Key Performance Indicators

LDPC Low-Density Parity Check

MIMO Multiple Input Multiple Output

MLD Maximum Likelihood Detection

mMTC Massive Machine-Type Communications

MnAC Many-Access Channel

MUD Multi-User Detection

MUST Multiuser Superposition Transmission

NGMA Next Generation Multiple Access

NISQ Noisy-Intermediate-Scale Quantum

NOMA Non-Orthogonal Multiple Access

List of Abbreviations

NP	Non-deterministic Polynomial
OFDM	Orthogonal Frequency Division Multiplexing
OMA	Orthogonal Multiple Access
PA	Power Allocation
PDMA	Pattern Division Multiple Access
PD-NOMA	Power-Domain Non-Orthogonal Multiple Access
P	Polynomial
QAM	Quadrature Amplitude Modulation
QAOA	Quantum Approximate Optimization Algorithm
QA	Quantum Annealing
QMI	Quantum Machine Instruction
QPSK	Quadrature Phase Shift Keying
QPU	Quantum Processing Units
QUBO	Quadratic Unconstrained Binary Optimization
RB	Resource Block
RF	Radio Frequency
RRH	Remote Radio Heads
RSMA	Rate Splitting Multiple Access
SAPI	Solver Application Programming Interface
SCMA	Sparse Code Multiple Access

List of Abbreviations

SC Superposition Coding

SDMA Space Division Multiple Access

SE Spectral Efficiency

SIC Successive Interference Cancellation

SINR Signal-to-Interference-Plus-Noise Ratio

SNR Signal-to-Noise Ratio

SQUID Superconducting Quantum Interference Devices

TDMA Time Division Multiple Access

THz Terahertz

UE User Equipment

UL-NOMA Uplink Non-Orthogonal Multiple Access

URLLC Ultra-Reliable Low-Latency Communications

VR Virtual Reality

Chapter 1

Introduction

As wireless networks evolve through 5G, scaling up spectral density through the use of millimeter-wave frequency bands, Massive MIMO, Non-Orthogonal Multiple Access (NOMA) and dense cells, network designers are looking ahead to the 6G roadmap, anticipating an even more data-driven society where wireless brain-computer interfaces, extended reality, and connected robotics drive 6G networks to handle data rates $10\text{--}1000\times$ greater than 5G [1]. To scale up spectral efficiency, designers will consider transitioning from traditional orthogonal multiplexing to advanced NOMA schemes, employing innovative air interface multiplexing techniques, more robust forward error correction coding, and even higher-density network deployments in wider bandwidths at higher carrier frequencies. As spectral efficiency is scaled up, 6G system designers will strive to improve *key performance indicators* (KPIs) such as latency, reliability, and energy efficiency of terminals and base-stations while also trying to not compromise one KPI to achieve another. The implementation of algorithms for 6G to optimize data throughput, spectral efficiency, user density, reliability, and latency, operating in wider bandwidths will lead to exponentially more computation than current 5G systems.

In the base station (BS) and cellular infrastructure, 5G RF modem signal processing is based on classical compute concepts which are typically implemented in ASIC, FPGA and GPU/CPU fabrics. However, improvements in classical compute performance are not

advancing at an exponential rate as in past years, but are plateauing due to transistors reaching atomic limits [2]. Since the design of efficient and fast computational structures is now competing with wireless communication as the most significant challenge for many high-capacity wireless communication systems, it is doubtful that conventional silicon will be able to implement the high spectral performance, low latency, and high-reliability optimization algorithms needed to achieve 6G's KPIs.

Quantum computing is a potentially valuable compute-acceleration tool to address future tradeoffs between performance, latency, and reliability in wireless networks as the 6G roadmap evolves. If quantum computing enables optimal algorithms for computationally heavy optimization problems currently bottleneck achievable network throughput, spectral efficiency could then benefit. The numerous hardware platforms capable of quantum information processing could then combine with other scalable technologies such as millimeter wave and small cells, further increasing spectral efficiency. Due to the linearity of quantum mechanics, quantum computing is fundamentally constrained to rely on reversible operations, which dissipate no heat, except in the initialization and readout phase of the computation. While noisy quantum computation has elements of irreversibility, in the long term quantum computation can in principle reach arbitrarily low power consumption for computations that would be power-hungry if performed classically. Over the last few years, due to advances in nanotechnology and engineering, real-world quantum computers have become commercially available. For wireless networks, recent studies first made use of a quantum annealer, a certain type of analog quantum computing processor, and showed promising results for a quantum-based MIMO detector in centralized radio access networks (C-RAN) [3] and for a quantum-based Low-Density Parity Check (LDPC) error control decoding [4], providing guidance of how to make use of the machine and baseline performance metrics. In wireless networks, there are representative optimization problems, including but not limited to the previously studied applications, which suffer from well-known conventional trade-offs between throughput and compute complexity, where optimal solvers are known but very difficult to practically implement considering available hardware and processing time limitations.

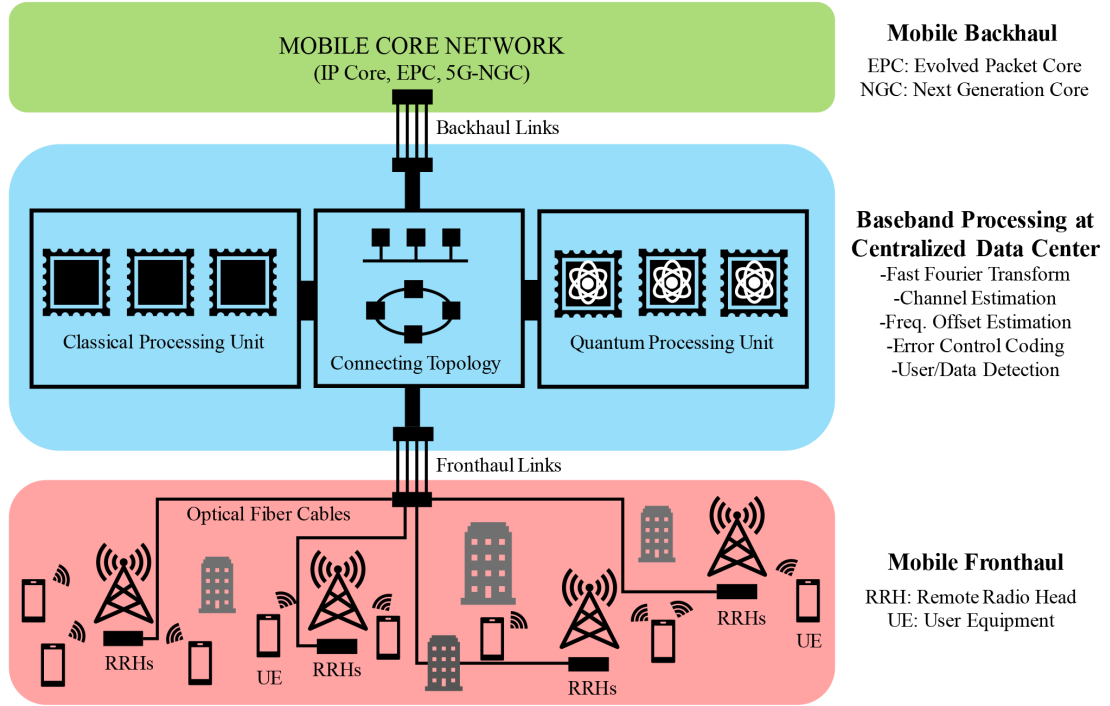


Fig. 1.1 A quantum compute-enabled C-RAN system architecture for NextG wireless networks.

We expect that overcoming practical issues of these optimization problems existing on current algorithms and/or computer architectures will lead to drastic improvements in wireless communications and networks, achieving optimal performance within a short processing time, making investigation of quantum computing as a potential accelerator a compelling research objective. In our envisioned scenario shown in Figure 1.1, quantum processors will be co-located with C-RAN computational resources in a data center, partitioning the compute roles for hundreds of base stations or Remote Radio Heads (RRH) connected via low-latency optical fiber, where quantum computing will take care of computationally-demanding optimization processing that bottlenecks achievable throughput and latency, while classical processors will take care of otherwise tractable processing [5, 6]. We believe that the C-RAN embedded quantum processors will have optimized interfaces and information processing architecture to maximize wireless system performance.

1.1 Problem Statement

The demand for faster and more capable video, audio, and teleconferencing applications over the past decade has led to sharp and unprecedented increases in wireless traffic loads at base stations. To address this, *Non-orthogonal multiple access* (NOMA) has emerged as a promising technique to maximize spectral efficiency in modern wireless systems like 5G New Radio cellular networks and beyond. In NOMA, the use of overlapping or the same time and frequency slots for all users has allowed for a significant improvement in the utilization of network resources compared to traditional orthogonal multiple access (OMA) techniques.

From a design standpoint, one of the most efficient architectures to implement advanced 5G technologies is the C-RAN architecture[7]. C-RAN offloads most of the physical-layer processing that currently takes place at the BS to a centralized data center, where it is aggregated with other BSs' processing on the same hardware, supporting hundreds or thousands of access points. This centralized computation helps in managing the increased complexity and computational load that NOMA introduces, especially in handling multi-user interference, which is crucial for maintaining high data rates and system reliability in NOMA networks.

In practice, however, to fully realize NOMA's potential for throughput and spectral efficiency gains, the system must effectively and efficiently demultiplex mutually-interfering information streams as they arrive. NOMA requires sophisticated signal processing techniques to disentangle these overlapping signals, which often involve the use of successive interference cancellation (SIC)[8]. the practical implementation of NOMA, especially in uplink scenarios, faces significant challenges. SIC suffer from high implementation complexity, error propagation in iterative decoding, while the optimal maximum likelihood (ML) suffer from stringent latency requirements that can hinder the performance of NOMA systems.

To address these challenges, this project proposes the use of Quantum Annealing (QA) for uplink signal detection in NOMA networks. QA, which runs on specialized quantum computing hardware, offers a novel approach to solving the computationally intensive task of maximum likelihood signal detection, traditionally a bottleneck in high-density NOMA

scenarios. By reformulating the signal detection problem as an Ising spin energy minimization problem, QA can be leveraged to perform this task more efficiently and with potentially lower latency than classical algorithms[9, 10].

1.2 Project Objectives

The project objectives are summarized as follows:

- Study and understand the existing multiplexing schemes, focusing particularly on Non-Orthogonal Multiple Access (NOMA), and compare their efficiencies and applications.
- Investigate the current techniques proposed for implementing power-domain NOMA (PD-NOMA) and identify their limitations and drawbacks in practical scenarios.
- Explore the principles of Quantum Annealing (QA) and its applicability in solving complex computational problems, especially those related to signal processing in wireless communications.
- Reformulate the Maximum Likelihood (ML) detection problem for NOMA systems by describing it through the Quadratic Unconstrained Binary Optimization (QUBO) model to enable integration with Quantum Processing Units (QPUs).
- Study the Bit Error Rate (BER) performance of QA-aided ML Multi-User Detection (MUD) across various modulation schemes and assess how it scales with an increasing number of users.
- Compare the QA method with other conventional signal detection techniques, including Successive Interference Cancellation (SIC) and brute force solutions for maximum likelihood detection, highlighting the advantages and potential improvements offered by QA.

1.3 Structure of Report

This project is systematically organized into the following chapters to provide a comprehensive exploration of the integration of quantum computing with wireless communications:

- **Chapter 1: Introduction** — Provides an overview of wireless communications, delineates the problem statement, outlines project objectives, and describes the project's organizational structure.
- **Chapter 2: Towards Non-Orthogonal Multiple Access for Next-Gen Wireless Networks** — Discusses the technology, performance, and challenges of Power-Domain NOMA in uplink multi-user detection, critiques the limitations of Successive Interference Cancellation (SIC), and promote for Maximum Likelihood detection as a superior alternative.
- **Chapter 3: Introduction to Quantum Annealing** — Introduces key concepts of quantum computation, offers a tutorial on using Noisy-Intermediate-Scale Quantum (NISQ) technology, and details quantum annealing (QA), particularly the application of D-Wave systems for practical purposes in wireless communications.
- **Chapter 4: Leveraging Quantum Annealing for Large NOMA Processing in Centralized Radio Access Networks** — Starts with a detailed system model for NOMA, explains the conversion of the NOMA problem into a Quadratic Unconstrained Binary Optimization (QUBO) format, describes its implementation on quantum annealing hardware, and evaluates the system's performance in optimizing NOMA decoding.
- **Chapter 5: Conclusions and Future Work** — Summarizes the findings of the project and outlines the potential future directions for extending this research.

Chapter 2

Towards Non-Orthogonal Multiple Access for Next-Gen Wireless Networks

Non-Orthogonal Multiple Access (NOMA) represents a shift in multiple access technologies, enhancing spectrum efficiency and user capacity over traditional methods. This chapter explores Power-Domain NOMA, addressing its technology, performance, and challenges in uplink multi-user detection. It critiques the limitations of Successive Interference Cancellation (SIC) and proposes Maximum Likelihood detection as a superior alternative.

2.1 Enabling Massive Connectivity for Next-Gen Wireless Networks

The current landscape of wireless communications is dominated by fourth-generation (4G) and fifth-generation (5G) networks. 4G has been the backbone of modern mobile internet, providing substantial improvements in speed and connectivity over its predecessors. However, as technology progresses, 5G has begun to take the spotlight, offering even faster speeds, reduced latency, and greater capacity. 5G networks support a variety of new applications, including enhanced mobile broadband, massive machine-type communications (mMTC), and ultra-reliable low-latency communications (URLLC).

Despite these advancements, the rapid increase in the number of mobile users and Internet-enabled devices poses a significant challenge. By the end of 2024, Cisco predicts that the number of mobile users will reach 13.1 billion, and the number of Internet-enabled devices will soar from 18.4 billion in 2018 to 29.3 billion[11]. This explosive growth will gradually overwhelm the connectivity capabilities of both 4G and 5G networks. Additionally, this increasing trend shows no sign of slowing down in the next decade since this demand is caused only by around 70% of the global population. Therefore, providing massive access is one of the most important targets for next generation wireless networks, namely the sixth-generation (6G). Currently, the research on 6G is being carried out on the global stage, e.g., the “6G Flagship” in Finland[12], “6G Hubs” in Germany, “Broadband Communications and New Networks” in China, Terahertz (THz) communication studies in the USA, to name a few. To support future Internet of Everything (IoE) and Mobile Internet, 6G has several critical performance targets[13]: a) The peak data rate needs to be at least one terabit per second; b) the air interface latency shall range from 0.01 to 0.1 millisecond for users with high mobility; c) the connectivity density shall be 10 times larger than in current 5G systems; d) the spectral efficiency (SE) and energy efficiency (EE) have to be 5-10 and 10-100 times higher than for 5G, respectively; and e) the reliability has to be higher than 99.99999%.

In order to achieve these stringent targets, one of the most fundamental advancements needed is the design of sophisticated multiple access techniques, referred to as Next Generation Multiple Access (NGMA). The transition from Orthogonal Multiple Access (OMA) to Non-Orthogonal Multiple Access (NOMA) is already expected to occur in the eMBB use cases of 5G to increase system throughput. This evolution will be even more crucial for 6G to handle the vast number of devices and the high data rate requirements. NGMA will play a pivotal role in enabling the scalability, efficiency, and performance required for the next wave of global connectivity demands. The following section will explore into the currently employed multiple access techniques alongside the emerging Next Generation Multiple Access (NGMA)[14].

2.2 The Evolution of Multiple Access Techniques in Wireless Networks

Wireless communication fundamentally relies on the efficient use of radio resources—time and frequency—to transmit data from one device to another. The fundamental physical radio resources are time and frequency, which are usually interpreted as physical degrees of freedom to transmit data. The problem of multiple access arises when multiple users need to be served with limited (or scarce) degrees of freedom in the radio resource. It is intuitive to consider dividing the available degrees of freedom in an orthogonal way so that each user's transmission will not interfere with another.

Looking back at the development history of wireless communications, multiple access techniques have been key enablers. Depending on whether each user is allocated orthogonal resources or not, the current multiple access techniques can be loosely classified into two categories, namely orthogonal transmission strategies and non-orthogonal transmission strategies. In particular, given their advantages of low complexity and interference-avoidance, orthogonal transmission strategies have been extensively employed in practical wireless communication systems. These include Frequency Division Multiple Access (FDMA) in the first generation (1G), where the available spectrum is partitioned into non-overlapped frequency sub-bands, each accommodating one digital data stream. Time Division Multiple Access (TDMA) in the second generation (2G) partitions time into slots, each serving a digital data stream in a round-robin fashion[15]. Code Division Multiple Access (CDMA) with orthogonal codes in the third generation (3G) and Orthogonal Frequency Division Multiple Access (OFDMA) in 4G allocate orthogonal frequency/time/code resource blocks[16]. OFDMA is a multi-carrier multiple access scheme based on the Orthogonal Frequency Division Multiplexing (OFDM) waveform, which enables tight and orthogonal frequency-domain packing of the subcarriers with a subcarrier spacing inverse to the symbol duration. With OFDMA, the time and frequency plane is divided into a two-dimensional raster, each transmitting a modulated symbol that belongs to one data stream.

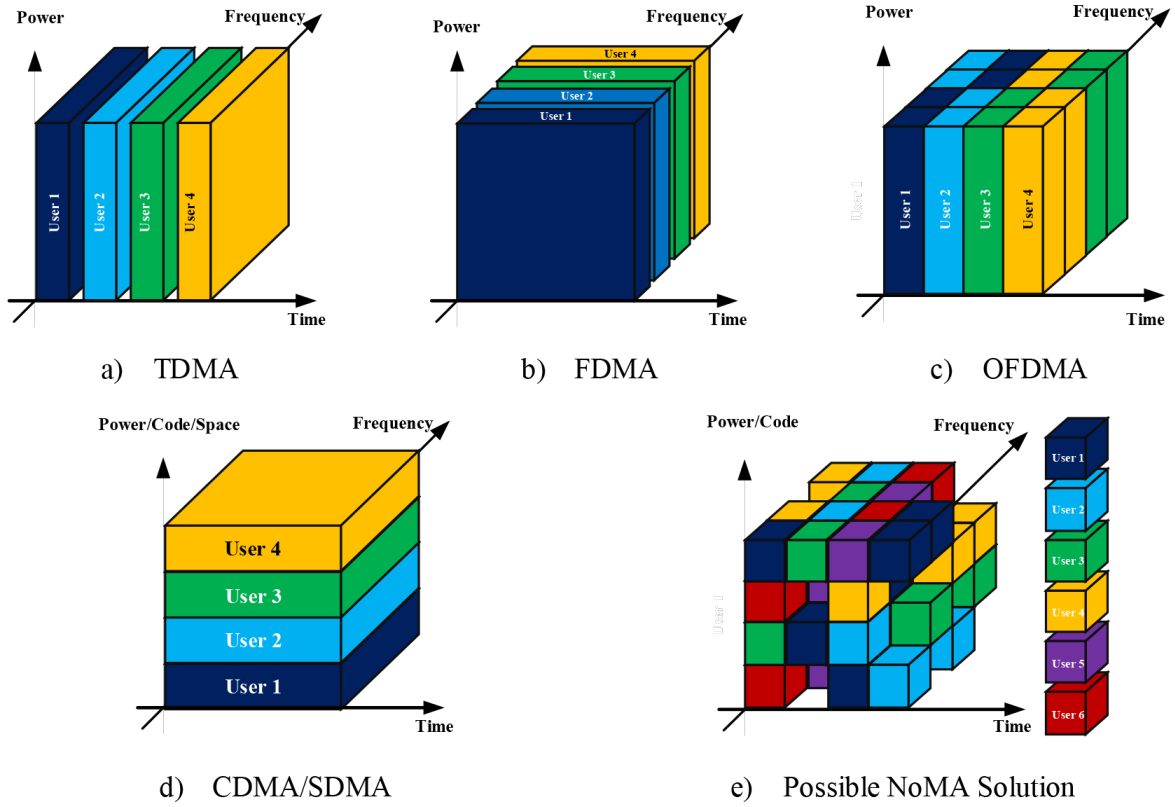


Fig. 2.1 Illustrative example of different multiple access schemes [8].

The introduction of code and spatial domains has further expanded the degrees of freedom in resource allocation. Code Division Multiple Access (CDMA) utilizes unique code signatures to spread each symbol across a wider bandwidth, effectively using the code domain for multiple access. CDMA can also be taken as one type of OMA schemes where the number of users that can be simultaneously supported is less than or equal to N . However, it is also possible to trade off orthogonality for higher system throughput to accommodate more users simultaneously. In this sense, CDMA can also be considered a type of non-orthogonal transmission strategy[17]. Similarly, Spatial Division Multiple Access (SDMA) uses the spatial domain to separate users based on their physical location, which can be managed either orthogonally or non-orthogonally through various precoding techniques[?]. Despite the clear benefits of OMA schemes, such as simplified transceiver design and reduced co-channel interference, they have notable limitations. Primarily, the number of users that can be served simultaneously is capped by the available radio resources, and maintaining orthogonality

requires complex user scheduling and feedback mechanisms, leading to increased signaling overhead[8]. However, given the ever-increasing number of wireless devices and the limited amount of available spectrum, conventional orthogonal transmission strategies have become inefficient due to their low spectral efficiency (SE) and the limited number of users/devices that can be supported due to inflexible resource allocation. As a result, to handle the challenging emerging heterogeneous services and applications (e.g., virtual reality (VR), augmented reality (AR), and Industry 4.0), non-orthogonal transmission strategies are widely considered to be a promising solution and have drawn significant research interests in recent years[18].

The goal of NAGMA is to enable a tremendous number of users/devices to be efficiently and flexibly connected with the network over the given wireless radio resources. Different from orthogonal transmission strategies, the key idea of non-orthogonal transmission strategies is to allow different users to share the same resource blocks. To deal with the resulting interference caused by the non-orthogonal resource allocation, additional techniques have to be employed at the transmitters and receivers, including but not limited to superposition coding (SC), rate splitting (RS), successive interference cancellation (SIC), and message passing (MP). Although the non-orthogonal transmission strategies increase the transmitter and receiver complexity, significant benefits can be achieved, such as supporting massive connectivity, achieving high SE and EE, and guaranteeing user fairness. As a result, compared with orthogonal transmission strategies, the aforementioned benefits make non-orthogonal transmission strategies promising candidates for NGMA. Nevertheless, the investigation of NGMA is still in its infancy stage. Besides achieving high bandwidth efficiency and supporting high connectivity, NGMA has to facilitate diverse application scenarios and be compatible with other promising physical layer techniques[14].

2.2.1 Considerations for Next Generation Multiple Access Design

In this subsection, we identify several new considerations that need to be taken into account for *NGMA design*, addressing the specific needs and challenges that emerge with the evolution of cellular networks and IoT integration.

1. **Massive Access:** With the increasing demand for expanding cellular connectivity, as well as the burgeoning Internet of Things (IoT) market, next-generation cellular networks will be characterized by an exceptionally large number of users—a scenario also referred to as *massive connectivity* or *massive access*. Traditional information-theoretic results on multiple access schemes mostly focus on finding capacity limits for an infinite coding blocklength and a fixed number of users. However, for massive access channels, where the number of users grows with the blocklength, these fundamental limits need to be revisited. The authors of a key study proposed the notion of a *many-access channel* (MnAC), modeling a channel with a single receiver and multiple transmitters, the number of which grows unboundedly with the blocklength[18]. New notions of achievable message length and symmetric capacity were introduced, where separate identification and decoding were shown to be capacity-achieving. These insights provide valuable performance upper bounds for NGMA in the many-user asymptotic regime, setting a new benchmark for what is achievable as user numbers continue to climb.

2. **Short-Packet Transmission:** Another critical feature to consider for NGMA is *short-packet transmission*, particularly relevant in IoT applications where the coding blocklength is finite and short due to the low latency requirements. Research in this area has derived the maximum channel coding rate for a given blocklength and error probability, highlighting the rate gap compared to scenarios with infinite coding blocklength[14]. Extending these results to a regime with a large yet finite number of users and a finite coding blocklength remains an open problem. This gap indicates a significant area for future research, as approaching these fundamental limits with practical transmission and multiple access schemes will be crucial for the successful deployment of NGMA technologies in real-world scenarios.

Both of these considerations—*massive access* and *short-packet transmission*—point towards a future where NGMA must be adaptable, robust, and capable of handling increasingly complex communication environments.

2.2.2 Possible Candidates for Next Generation Multiple Access

As discussed in the previous subsections, the trend of *NGMA* is to transition from orthogonality to non-orthogonality, serving multiple users/devices by allowing them to share the same resources instead of allocating them dedicated orthogonal resources. This shift is driven by the fact that orthogonal transmission schemes are strictly suboptimal compared to non-orthogonal transmission schemes, especially given the explosively growing number of users/devices in next-generation wireless networks. This Rapidly growing user base can only be supported to a limited extent by traditional orthogonal resources, calling for the development of non-orthogonal transmission schemes for *NGMA*. Below, we introduce some potential candidates for *NGMA* based on the concept of non-orthogonality, along with their technical foundations:

1. *Power-Domain NOMA (PD-NOMA)*: The key idea of PD-NOMA is to serve multiple users within the same time/frequency/code resources and distinguish them in the power domain. Superposition coding (SC) and successive interference cancellation (SIC) are two key technologies in PD-NOMA, which have been proven to be capacity-achieving in the single-antenna broadcast channel (BC) and multiple access channel (MAC)[15]. PD-NOMA can be straightforwardly integrated with OFDM, as demonstrated by the multiuser superposition transmission (MUST) technique, which has been incorporated into LTE-A for supporting two users on the same OFDMA subcarrier. Another application of PD-NOMA is layered division multiplexing (LDM), included in the digital TV standard (ATSC 3.0) for delivering multiple superpositioned data streams for TV broadcasting[17].
2. *Code-Domain NOMA (CD-NOMA)*: Inspired by CDMA, CD-NOMA serves multiple users via the same time/frequency resources but employs user-specific spreading sequences which may be sparse or have low cross-correlation. At the receiver, multiuser detection (MUD) is usually carried out in an iterative manner using message passing (MP) based algorithms. The family of CD-NOMA schemes includes Low-Density

Signature (LDS)-CDMA, LDS-OFDM, Sparse Code Multiple Access (SCMA), and Pattern Division Multiple Access (PDMA)[8].

3. *Space Division Multiple Access (SDMA)*: With the advancement of multi-antenna techniques, MIMO communication has become a cornerstone in current 5G standards and future wireless networks. By utilizing multiple antennas at transmitters and/or receivers, additional spatial degrees of freedom (DoFs) can be exploited, distinguishing multiple users/devices in the same time/frequency/code domain in the spatial domain. The most commonly employed method for SDMA is linear precoding (LP), which effectively mitigates inter-user interference by exploiting spatial DoFs to design suitable transmit and/or receive beamformers[16].
4. *Rate Splitting Multiple Access (RSMA)*: RSMA is a newer non-orthogonal multi-antenna transmission scheme that relies on the rate splitting technique. The transmitter splits each user's message into a common message intended for all served users and private messages. The constructed common message and the remaining private messages are transmitted via beamformers, similar to SDMA. At the receivers, the common message is decoded first with private messages treated as interference, followed by the subtraction of the common message from the received signal (using SIC), and then decoding the intended private message. RSMA provides more flexibility in mitigating interference compared to SDMA, offering significant advantages in overloaded network scenarios[18].

Given these potential candidates, this project focuses on exploring PD-NOMA to develop NGMA, as NOMA offers a higher degree of compatibility and flexibility. This enables the seamless integration of NOMA with other components of next-generation networks, such as multi-antenna techniques, challenging application scenarios, and other physical layer techniques, details of which will be elaborated in the following section.

2.3 Power Domain Non-Orthogonal Multiple Access (PD-NOMA)

In the evolving landscape of wireless communications, efficiently managing the scarce spectral resources to accommodate the exponential growth in the number of devices and data traffic is paramount. PD-NOMA emerges as a transformative solution in this context, fundamentally changing how spectrum resources are utilized by allowing multiple users to share the same spectral resources distinguished by different power levels. This technique diverges from traditional orthogonal access methods, where resources such as time, frequency, and code are allocated exclusively to individual users, often leading to under-utilization of these precious assets. By assigning different power levels to user signals that overlap in both time and frequency, PD-NOMA can significantly increase the spectral efficiency of a network. Additionally, it also improves user fairness by ensuring that users with poor channel conditions are not left behind. This capability is especially critical in high-demand environments such as urban centers and for emerging technologies that require massive connectivity, such as the Internet of Things (IoT).

In the following sections, we will explore the key components and operational aspects of PD-NOMA, including superposition coding, successive interference cancellation, and its application in uplink and downlink scenarios.

2.3.1 Superposition Coding (SC)

Superposition Coding (SC) is a fundamental technique in PD-NOMA, which enables a single transmitter to simultaneously communicate information to several receivers by varying the power levels assigned to each user's signal. This method is essential in broadcasting scenarios where multiple users receive the same channel's information, such as in digital television or certain types of data streaming.

To implement SC, the transmitter employs two point-to-point encoders that map the input bits of each user to output bit sequences. These sequences are then transmitted over the same

channel but are modulated at different power levels. The encoding functions for the users are defined as:

- User 1: $f_1 : \{0, 1\}^{R_1} \rightarrow \mathcal{C}^T$
- User 2: $f_2 : \{0, 1\}^{R_2} \rightarrow \mathcal{C}^T$

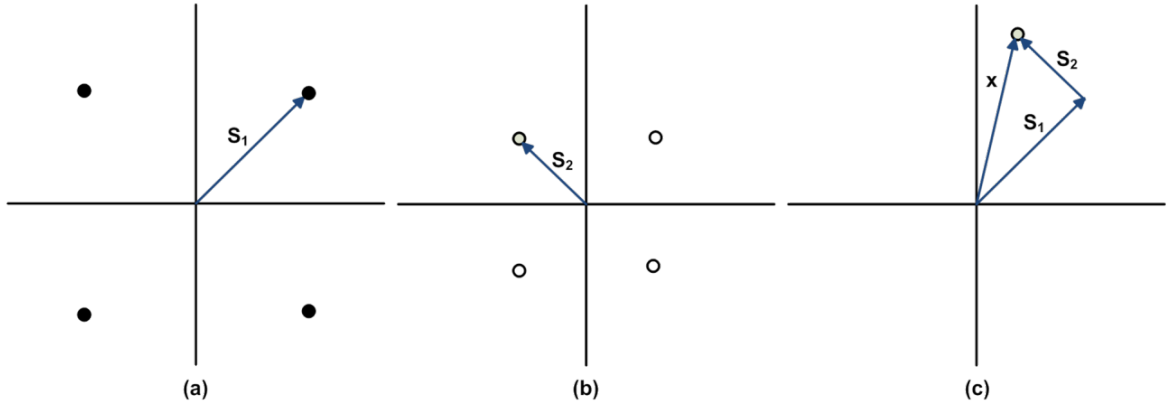


Fig. 2.2 An example of SC encoding (a) signal constellation of user 1 (b) signal constellation of user 2 (c) constellation of superposed signal.

where R_1 and R_2 denote the transmission rates for User 1 and User 2, respectively, T represents the block length, and $\mathcal{C}$ is the coding library used. The encoded sequences $S_1(n)$ and $S_2(n)$ are combined using the superposition principle:

$$X(n) = \sqrt{P\alpha_1}S_1(n) + \sqrt{P\alpha_2}S_2(n) \quad (2.1)$$

Here, α_1 and α_2 are the power fractions allocated to User 1 and User 2, respectively, with the constraint $\alpha_1 + \alpha_2 = 1$ [17].

Superposition coding is noted for its capacity-achieving properties on scalar Gaussian broadcast channels and its ability to maximize network throughput by efficiently using the available bandwidth and power. It adjusts power levels according to the varying channel conditions of different users, optimizing the overall communication efficiency. SC has been integrated into various practical applications, including LTE-A through Multiuser Superposition Transmission (MUST), supporting multiple users on the same OFDMA subcarrier,

and in digital TV standards like ATSC 3.0, enabling the delivery of multiple data streams simultaneously [19].

2.3.2 Successive Interference Cancellation (SIC)

Successive Interference Cancellation (SIC) is a decoding technique critical in environments where signals from multiple users are superposed, such as in PD-NOMA. SIC exploits the differences in signal strengths to sequentially decode overlapping signals transmitted over the same channel. The process involves initially ordering the signals by strength, decoding the strongest signal first while treating the others as noise, and then subtracting the decoded signal from the combined signal to facilitate the decoding of the next strongest signal, this process is done iteratively until all users are decoded[20]. this is outlined in Figure 2.3.

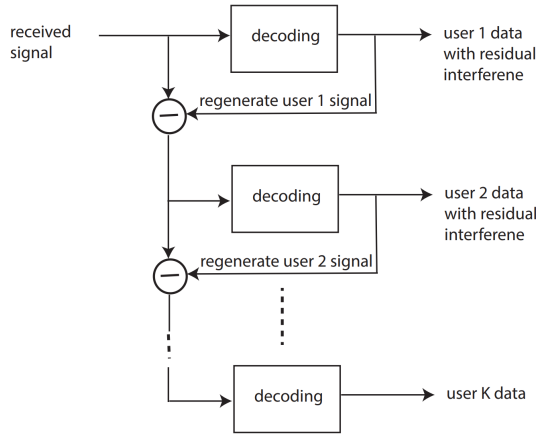


Fig. 2.3 SIC receiver.

The effectiveness of SIC in enhancing decoding accuracy and spectral efficiency makes it essential for managing interference among signals in dense user environments. The procedural implementation for a two-user scenario involves the following mathematical steps:

1. At user 1, the decoder $g_1 : \mathcal{C}^T \rightarrow \{0, 1\}^{R_1}$ decodes the message $S_1(n)$ while considering $S_2(n)$ as noise.

2. User 2's decoder successively recovers its message by:

- Decoding user 1's message $S_1(n)$ using g_1 .
- Subtracting the decoded signal, adjusted for the channel gain and power $\sqrt{P\alpha_1 h_2} S_1(n)$, from the received signal $Y_2(n)$ to get:

$$Y'_2(n) = Y_2(n) - \sqrt{P\alpha_1 h_2} S_1(n), \quad (2.2)$$

where h_2 represents the complex channel gain at user 2.

- Decoding user 2's message $S_2(n)$ using another decoder $g_2 : \mathcal{C}^T \rightarrow \{0, 1\}^{R_2}$ applied to $Y'_2(n)$.

Figure 2.4 presents the technique for decoding the superposed signal at the receiving side. Here, the constellation point of user 1 is decoded first from the received signal. Then, the decoding of the constellation point of user 2 is performed with respect to decoded constellation point of user 1.

The success of SIC depends on the perfect cancellation of the signals in the iteration steps. The transmitter should accurately split the power between the user information waveforms and superimpose them. The methodology for power split differs for uplink and downlink channels.

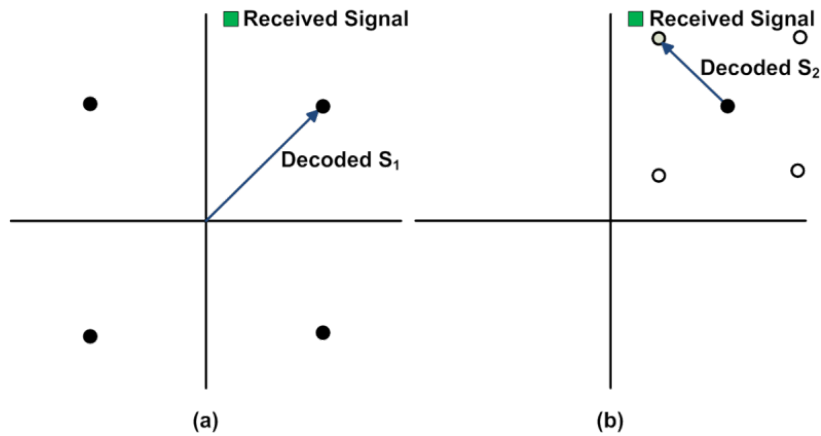


Fig. 2.4 An example of SC decoding (a) decoding the signal of user 2 (b) decoding the signal of user 1.

2.3.3 Downlink NOMA Transmission

We consider a simple NOMA system consisting of a single base station (BS) and two users, each equipped with a single antenna. Suppose that x_1 and x_2 are the signals to be transmitted from the BS to users 1 and 2, respectively. The BS transmits the superposition coded signal as:

$$s = \sqrt{P_1}x_1 + \sqrt{P_2}x_2, \quad (2.3)$$

where P_i , for $i = 1, 2$, is the transmit power for user i , and the message signal x_i , for $i = 1, 2$, is of unit power, i.e., $E\{|x_i|^2\} = 1$, with $E\{\cdot\}$ as the expectation operator. The total transmit power for users 1 and 2 can then be written as $P = P_1 + P_2$. In practice, for a particular system setting, P is predefined and is thus divided into P_1 and P_2 according to the adopted power allocation (PA) scheme. The received signal at user i can be represented as:

$$y_i = h_i s + n_i, \quad (2.4)$$

where h_i is the channel gain between the BS and user i , and n_i represents the Gaussian noise plus interferences with power spectral density $N_{f,i}$. For a multi-cell scenario, the inter-cell interference is also included in n_i . Successive Interference Cancellation (SIC) is employed at the receivers to separate the signals of different users. The optimal decoding order for SIC is determined by the decreasing order of the strength of the users' channels, represented as $\frac{|h_i|^2}{N_{f,i}}$. This order allows each user to effectively eliminate interferences from signals of users whose decoding orders follow theirs.

In this scheme, user 1, having the highest channel strength $\frac{|h_1|^2}{N_{f,1}}$, is referred to as the strong user, and user 2, with the least channel strength $\frac{|h_2|^2}{N_{f,2}}$, as the weak user. The base station (BS) periodically updates the SIC order based on channel state information (CSI) feedback from users. Users are informed of the latest SIC order by the BS. Generally, a user with weaker channel strength (weak user) receives a higher power allocation than a user with stronger channel strength (strong user) to enhance its signal-to-interference-plus-noise ratio (SINR).

In a typical 2-user NOMA scenario with $\frac{|h_1|^2}{N_{f,1}} > \frac{|h_2|^2}{N_{f,2}}$ (and hence $P_1 < P_2$), only user 1 performs SIC. It decodes the signal of user 2 (x_2) first and subtracts it from its received signal y_1 before decoding its own signal x_1 . User 2 directly decodes its own signal y_2 , treating x_1 as noise and thus not using SIC.

If SIC is perfectly executed, the achievable data rate for the NOMA user i , $R_{i,\text{NOMA}}$, for a transmission bandwidth of 1 Hz can be written as:

$$R_{1,\text{NOMA}} = \log_2 \left(1 + \frac{P_1 |h_1|^2}{N_{f,1}} \right), \quad (2.5a)$$

$$R_{2,\text{NOMA}} = \log_2 \left(1 + \frac{P_2 |h_2|^2}{P_1 |h_2|^2 + N_{f,2}} \right). \quad (2.5b)$$

Equations (3) and (4) suggest that the BS can adjust the data rate of each user by tuning the power allocation coefficients $\alpha_1 = \frac{P_1}{P}$ and $\alpha_2 = \frac{P_2}{P}$.

Comparative Analysis with OMA: In the 2-user FDMA scenario, the transmission bandwidth of 1 Hz is divided such that user 1 uses W Hz and user 2 uses the remaining $1 - W$ Hz. The power allocation ratio remains the same as in the NOMA scheme. The achievable data rates for OMA users are expressed as:

$$R_{1,\text{OMA}} = W \log_2 \left(1 + \frac{P_1 |h_1|^2}{W N_{f,1}} \right), \quad (2.6a)$$

$$R_{2,\text{OMA}} = (1 - W) \log_2 \left(1 + \frac{P_2 |h_2|^2}{(1 - W) N_{f,2}} \right). \quad (2.6b)$$

Figure 2.5 illustrates broadcast (BC) capacity/rate region comparison. The sum capacity for both OMA and NOMA can thus be written as:

$$R_T = \sum_{k=1}^K R_k \quad (2.7)$$

This demonstrates that unlike NOMA, no OMA user suffers from the interference from the signal of the other user. Note that we considered asymmetric channel condition (different channel gains for users 1 and 2) to perform the above comparisons on the data rate regions.

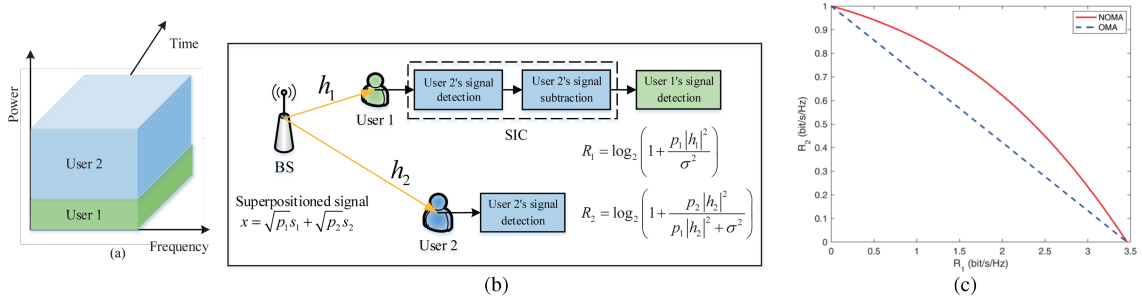


Fig. 2.5 Illustration of two-user downlink NOMA. (a) The signals of two users are multiplexed in the power domain using the same time/frequency resource. (b) decoding NOMA transmission. (c) BC capacity/rate region comparison, $0 \leq p_1 + p_2 \leq 1W$.

We can find that OMA and NOMA provide identical data rate regions in a symmetric channel environment—multiplexing in power domain does not become advantageous as the signals of both users undergo the same interference and noise [7]. Therefore, compared to OMA, NOMA in cellular downlink provides higher sum capacity and performs a better tradeoff between system efficiency and user fairness when the channel conditions are different among users. It is demonstrated that other forms of OMA techniques including TDMA and orthogonal frequency-division multiple access (OFDMA) are also significantly outperformed by NOMA in terms of throughput under different settings [8].

Generalization to Multi-User Scenario: For a K -user downlink NOMA with $\frac{|h_1|^2}{N_{f,1}} > \frac{|h_2|^2}{N_{f,2}} > \dots > \frac{|h_K|^2}{N_{f,K}}$ (and hence $P_1 < P_2 < \dots < P_K$), the data rate for user 1 remains as in equation (3), while the data rate for any other user k is:

$$R_{k,\text{NOMA}} = \log_2 \left(1 + \frac{P_k |h_k|^2}{\sum_{j=1}^{k-1} P_j |h_k|^2 + N_{f,k}} \right). \quad (2.8)$$

This model illustrates that while NOMA theoretically supports numerous users, practical implementations usually limit it to a few users in a cluster to maintain manageable bit error rates and computational demands, particularly for resource-constrained devices[21].

2.3.4 Uplink NOMA Transmission

In uplink NOMA, multiple users transmit their signals to the base station (BS) within the same resource block. The BS employs Successive Interference Cancellation (SIC) to detect all the messages of the users. Notable differences between uplink NOMA and downlink NOMA include[22]:

- **Transmit Power:** Unlike downlink NOMA, the transmit power of users in uplink NOMA may not differ significantly. It is primarily dictated by the channel conditions of each user. If the users' channel conditions vary greatly, their received Signal-to-Interference-plus-Noise Ratio (SINR) at the BS can be markedly different, regardless of their transmit power levels.
- **SIC Operations:** The mechanisms and interference experienced in uplink and downlink NOMA are distinct. In downlink NOMA, the SIC process involves canceling the interference from weaker users by first detecting the stronger user's signal, remodulating it, and subtracting it from the combined signal. Conversely, in uplink NOMA, the BS detects the stronger user's signal first by treating the other user as interference, then it remodulates and subtracts the detected signal to decode the weaker user's signal.
- **Performance Gain:** The performance advantage of NOMA over Orthogonal Multiple Access (OMA) varies between uplink and downlink scenarios. While downlink NOMA generally offers superior throughput performance, uplink NOMA provides improvements mainly in terms of fairness, particularly when channel conditions between users differ significantly.

Considering these aspects, the received signal at the BS is given by:

$$y = \sum_{i=1}^2 \sqrt{P_i} h_i x_i + n, \quad (2.9)$$

where P_i is the transmit power for user i , with $E\{|x_i|^2\} = 1$, and n represents the Gaussian noise plus interferences with power spectral density N_f . Typically, the BS broadcasts a

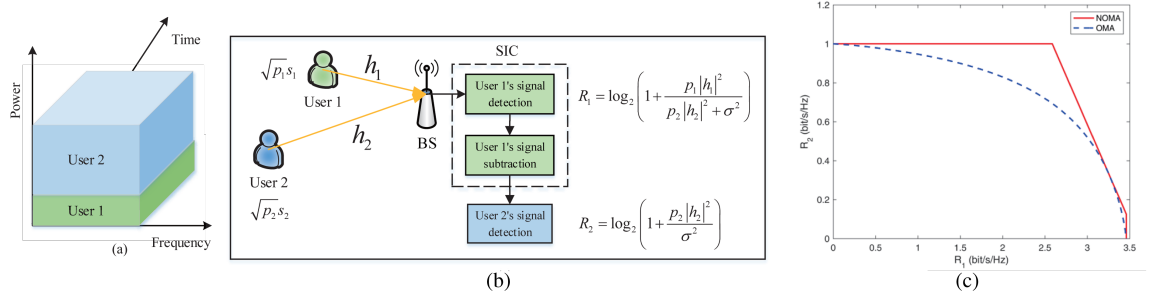


Fig. 2.6 Illustration of two-user uplink NOMA. (a) The signals of two users are multiplexed in the power domain using the same time/frequency resource (b) Uplink NOMA transmission. (c) MAC capacity/rate region comparison.

downlink reference signal, based on which each user performs channel estimation and regulates their transmit power to either P_1 or P_2 , depending upon their channel gains.

In this setting, user 1 is marked as the strong user experiencing a higher channel gain compared to user 2, the weak user. Upon receiving the superposed signal, the BS performs Successive Interference Cancellation (SIC) to separate the users' signals. The BS first decodes the signal of user 1 by treating the signal of user 2 as noise, then subtracts the decoded signal of user 1 ($\hat{x}_1$) from the received signal (y). From the remaining signal, the signal of user 2 ($\hat{x}_2$) is then decoded, Figure 2.6 illustrates this scenario.

Therefore, in uplink NOMA, user 1 experiences interference from user 2, whereas user 2 receives no interference from user 1 since user 1's signal is removed prior to decoding user 2's signal. This is in contrast to downlink NOMA, where user 2 experiences interference from user 1, while user 1 does not suffer interferences from user 1 since user 2's signal is removed prior to decoding user 1's signal. If SIC is performed perfectly, the achievable data rate of the NOMA user i , $R_{i,\text{NOMA}}$, for a transmission bandwidth of 1 Hz is expressed as:

$$R_{1,\text{NOMA}} = \log_2 \left(1 + \frac{P_1 |h_1|^2}{P_2 |h_2|^2 + N_f} \right), \quad (2.10a)$$

$$R_{2,\text{NOMA}} = \log_2 \left(1 + \frac{P_2 |h_2|^2}{N_f} \right). \quad (2.10b)$$

The achievable sum capacity is:

$$R_{\text{NOMA}} = R_{1,\text{NOMA}} + R_{2,\text{NOMA}}.$$

As a generalization of the 2-user case, for a K -user uplink NOMA with $|h_1|^2/N_f > |h_2|^2/N_f > \dots > |h_K|^2/N_f$ (and hence $P_1 < P_2 < \dots < P_K$) and 1 Hz BW, the data rates of user k , for $k = 1, 2, \dots, K-1$, and K can be written as:

$$R_k = \log_2 \left(1 + \frac{P_k |h_k|^2}{\sum_{j=k+1}^K P_j |h_j|^2 + N_f} \right), \quad (2.11a)$$

$$R_K = \log_2 \left(1 + \frac{P_K |h_K|^2}{N_f} \right). \quad (2.11b)$$

Unlike downlink NOMA, uplink NOMA can accommodate a relatively larger number of users because the BS can usually be equipped with the required computing power and energy. Moreover, the BS can apply more computing-intensive decoding schemes to reduce the impact of interference, making uplink NOMA preferable for massive machine-type communications[16].

This project primarily focuses on the study and analysis of uplink NOMA due to its potential for accommodating a larger number of users with varied channel conditions, and the possibility of employing more sophisticated decoding techniques at the BS to enhance overall system performance

2.3.5 Spectral Efficiency and Energy Efficiency

In the pursuit of optimizing *Non-Orthogonal Multiple Access* (NOMA) for 5G networks, both *spectral efficiency* (SE) and *energy efficiency* (EE) are critical. The previous discussion generally revolved around throughput performance, but a comprehensive analysis must also consider the power dynamics involved. This section investigate the energy aspects of NOMA, acknowledging the static power consumption from power amplifiers and the dynamic consumption from the information waveform. The total power at the transmitter is thus the

sum of the power for the information signal and the power used by circuitry components, particularly power amplifiers. Considering the downlink, the total power consumed by the BS can then be written as

$$P_{\text{total}} = P_T + P_{\text{static}} \quad (2.12)$$

Where P_T represents the signal power and P_{static} accounts for circuitry power consumption. Energy efficiency is quantitatively defined as the ratio of the sum rate to the total power consumed by the base station:

$$\text{EE} = \frac{R_{\text{total}}}{P_{\text{total}}} = \text{SE} \times \frac{W}{P_{\text{total}}} \quad (2.13)$$

In the context of increasing demand for green communications, reducing energy consumption has become a priority. The information capacity theorem by Shannon indicates that minimizing energy consumption and maximizing SE cannot be achieved simultaneously, necessitating a trade-off. This trade-off is evident in NOMA systems, where circuit power considerations reveal that there always exists an optimal point on the EE-SE curve[17]. Shannon theory suggests a monotonic relationship between EE and SE, where increasing SE typically reduces EE. However, this does not account for circuit power, which when considered, reveals a more complex relationship. Initially, as SE increases, so does EE, up until a peak—often referred to as the "green point"—beyond which EE declines as SE continues to rise. This peak represents the optimal trade-off point, achieving maximum energy efficiency[20].

NOMA inherently introduces controlled interference through its non-orthogonal resource allocation, allowing it to support overloading with only a slight increase in receiver complexity. This capability leads to higher SE for 5G applications. However, while SE measures how efficiently a limited spectral resource is utilized, it does not provide insights into energy utilization.

NOMA's approach to power management, including dynamic power allocation between users, enables it to achieve any desired point on the EE-SE curve for maximum EE. This is

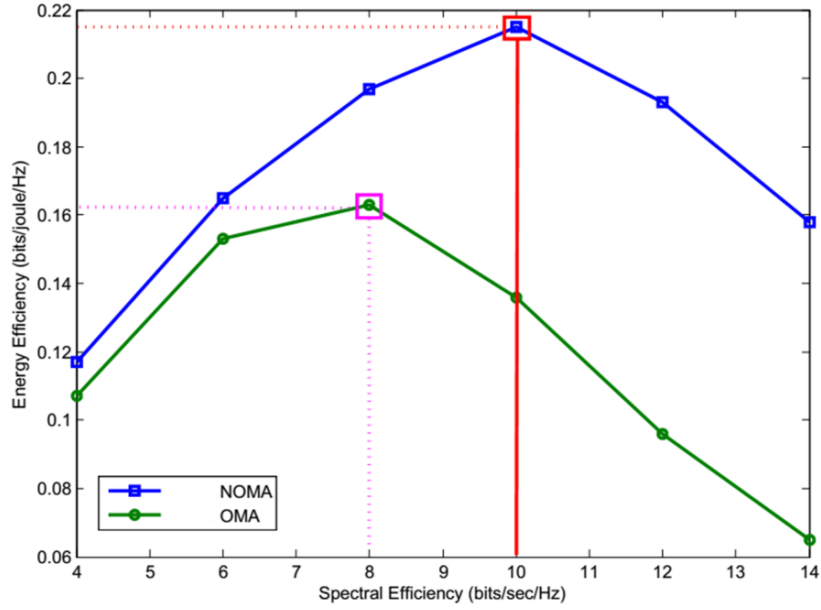


Fig. 2.7 A trade-off between energy efficiency and spectral efficiency, The peak of the curve (or the corresponding derivative of the EE-SE relationship) is where the system has the maximum energy efficiency.

facilitated by varying the total power through power-control schemes, allowing the system to adjust efficiency levels based on specific requirements. Statistical channel state information (CSI) models and MIMO configurations further enhance NOMA's ability to optimize EE under various constraints, including total transmit power and minimum required rates for users[16].

Comparative analyses with OFDMA have demonstrated that NOMA outperforms in terms of EE-SE trade-offs, especially in high Signal to Noise Ratio (SNR) conditions. depicted in Figure 2.7.

2.3.6 Main Advantages of PD-NOMA

As NOMA continues to evolve in both theoretical and practical aspects, it is increasingly being proposed for next-generation networks. Unlike traditional OMA schemes where the strongest user typically benefits from the highest power, NOMA structures its transmissions differently to provide several distinct advantages summarized as:

- **High Bandwidth Efficiency:** NOMA exhibits high bandwidth efficiency, enhancing the system's overall throughput. This improvement is due to NOMA's ability to allow each Resource Block (RB), such as time or frequency slots, to be shared by multiple users[17].
- **Fairness:** A fundamental aspect of NOMA is its power allocation policy, which tends to favor weaker users, thereby ensuring a more equitable distribution of throughput among all users. This is supported by sophisticated power allocation (PA) policies and the cooperative NOMA schemes that will be further discussed in this paper[16].
- **Ultra-high Connectivity:** With the anticipated rise of the *Internet of Things* (IoT) and the expected connection of billions of smart devices, NOMA offers a promising solution for efficiently managing such extensive connectivity. Unlike conventional OMA, which requires as many frequency/time RBs as there are devices, NOMA can serve a much larger number of devices using fewer RBs by exploiting its non-orthogonal characteristics[8].
- **Compatibility:** From a theoretical standpoint, NOMA can be integrated as an “add-on” technique alongside existing OMA techniques such as TDMA, FDMA, CDMA, and OFDMA. This compatibility is feasible because NOMA operates in a new dimension—the power-domain—complementing the well-established SC and SIC techniques[23].
- **Flexibility:** PD-NOMA is conceptually simpler and generally incurs a lower design complexity compared to other NOMA variants like *Multi-User Shared Access* (MUSA), *Pattern Division Multiple Access* (PDMA), and *Sparse Code Multiple Access* (SCMA). Although these schemes share the fundamental principle of assigning multiple users to a single RB, NOMA's approach, particularly when compared to SCMA, offers straightforward integration of coding, modulation, and subcarrier allocation, making it notably flexible[14].

These advantages position PD-NOMA as a highly effective strategy for enhancing the capabilities of Next-Gen networks, addressing both capacity and performance challenges in

densely populated user environments and device-heavy areas such as those anticipated with the expansion of IoT.

2.4 Challenges of Multi-User Detection in Uplink

Multi-user detection in uplink scenarios presents numerous challenges that are critical to the performance and reliability of communication systems, especially in environments with high user density and diverse service requirements. This section explores these challenges, firstly by emphasizing the pivotal role of channel state information (CSI) and modeling. Then we discuss the traditional and optimal detection methods

2.4.1 Channel Models and Estimation

Accurate channel modeling is crucial for studying multi-user detection in uplink scenarios. This modeling involves understanding various channel impairments such as noise, fading, and interference, which affect the quality and reliability of transmitted signals.

Rayleigh Fading

Rayleigh fading is a statistical model used for describing the impact of multipath reception in wireless communications. It assumes that the magnitude of a signal that has undergone multipath propagation will vary randomly according to a Rayleigh distribution. This model is particularly applicable in environments without a direct line of sight between transmitter and receiver.

$$p_R(r) = \frac{2r}{\Omega} e^{-\frac{r^2}{\Omega}} \quad \text{for } r \geq 0 \quad (2.14)$$

Where Ω is the expected value of R^2 , representing the total power of the signal received through multiple paths. Rayleigh fading is assumed when there is no dominant propagation

path, and the small-scale variations in the channel are modeled as Gaussian processes [24]. Figure 2.8 illustrates the impact of these small-scale variations on the channel gain.

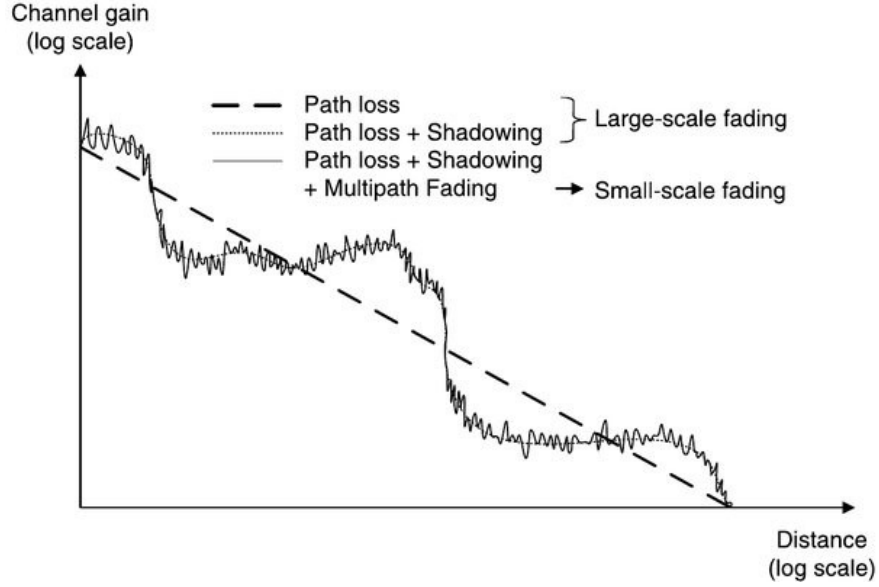


Fig. 2.8 Channel gain variation with distance.

Path Loss and Propagation Models

Path loss is a critical factor in determining the quality and reliability of the signal received in a mobile communication system. It is essential to use accurate path loss models to predict how the signal degrades over distance, which in turn helps in estimating the Signal-to-Noise Ratio (SNR) for various transmission scenarios.

One of the common path loss models is the Log-distance Path Loss Model. This model is based on empirical measurements and predicts that the average received signal power decreases logarithmically with distance. The path loss can be expressed as:

$$\overline{PL}(dB) = \overline{PL}(d_0) + 10n \log \left(\frac{d}{d_0} \right) \quad (2.15)$$

The bars in this equation denote the ensemble average of all possible path loss values for a given value of d . When plotted on a log-log scale, the modeled path loss is a straight line with a slope equal to $10n$ dB per decade, this can be observed in Figure 2.8 where the

dashed line represents the channel gain large-scale variation caused by the path-loss with distance. This modeling approach allows for a simplified yet effective representation of path loss across various environments[25]. Here, $\overline{PL}(dB)$ is the path loss in decibels, $\overline{PL}(d_0)$ is the path loss at a reference distance d_0 , n is the path loss exponent that characterizes how the path loss increases with distance, and d is the transmitter-receiver (T-R) separation distance. The value of n varies based on specific environmental conditions.

Below is a table that lists typical path loss exponents obtained in various mobile radio environments, formatted according to your specified design:

Table 2.1 Path Loss Exponents for Different Environments[25]

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

Additive White Gaussian Noise (AWGN)

Additive White Gaussian Noise (AWGN) is a fundamental noise model used in information theory to represent the effect of many random processes that occur in nature. The term "additive" indicates that this noise is added to any intrinsic noise within the information system. "White" suggests that it has uniform power spectral density across the frequency band, similar to white light which contains all visible frequencies. "Gaussian" refers to the normal distribution of the noise's amplitude over time, centered around zero, meaning it is a Gaussian process.

AWGN arises from various natural sources like thermal vibrations of atoms in conductors (thermal noise), shot noise, and radiation from celestial bodies. It is predominantly used as a channel model where the primary impairment to communication is the linear addition of noise with constant spectral density and a Gaussian distribution of amplitude. Although

simple, AWGN models are foundational because they provide insight into system behavior under idealized conditions without considering complexities like fading or interference.

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(-\frac{x^2}{2\sigma_x^2}\right) \quad (2.16)$$

Here, σ_x^2 represents the variance of the noise. The noise in AWGN is characterized as:

$$y_n = x_n + w_n \quad (2.17)$$

Where x_n is the transmitted signal, typically following a specific modulation scheme, and w_n represents the AWGN. The effectiveness of the AWGN model is often quantified by the Signal-to-Noise Ratio (SNR), which compares the level of the desired signal to the level of background noise.

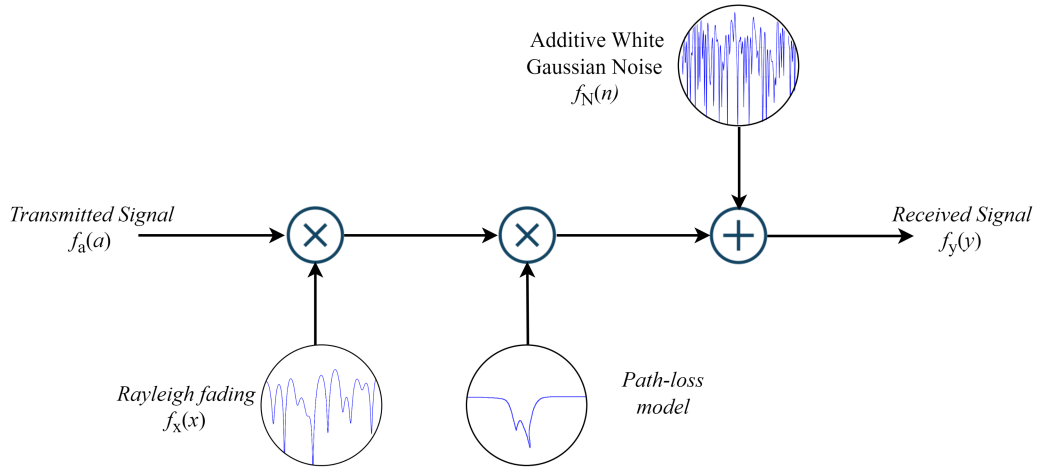


Fig. 2.9 The proposed channel model.

These models combined -as illustrated in Figure 2.9- collectively contribute to our understanding of the uplink channel in multi-user environments.

Importance of Channel Estimation in Uplink Communications

Channel estimation plays a more significant role in NOMA than in OMA systems, especially for Uplink. At the base station, precise channel estimation is required to predict and mitigate the effects of the channel as signals arrive. any inaccuracy in estimating the channels between the antennas of all users and the base station can significantly degrade the performance of *Multi-User Detection* (MUD) systems, since the channel estimation errors will result in ambiguous user ordering as well as inaccurate power control, which will in turn affect the SIC decoding accuracy. Naturally, perfect CSI cannot be obtained in practice. High performance near-optimal conventional channel estimation algorithms impose an unacceptable system overhead and computational complexity on NOMA systems, especially for MIMO-NOMA scenarios.

In our study, particularly in the simulations discussed in Chapter 4, we use the aforementioned channel model for constructing the signal and we assume perfect channel estimation at the BS to focus on evaluating the performance of NOMA MUD under ideal conditions. This assumption allows us to isolate the impact of channel estimation accuracy on system performance and explore potential optimizations in a controlled environment.

2.4.2 The Shortcomings of SIC and Conventional Detection Methods

Multi-user detection (MUD) in wireless communication systems is a significant challenge, particularly pronounced in the context of NOMA and Orthogonal OMA systems. In OMA systems like Code Division Multiple Access (CDMA), Orthogonal Frequency Division Multiple Access (OFDMA), Single-Carrier Frequency Division Multiple Access (SC-FDMA), or Time Division Multiple Access (TDMA), users are allocated resources either in a round-robin fashion or allowed to share orthogonal resources simultaneously. For instance, TDMA allocates the entire bandwidth to a single user for specific time slots, while in CDMA, all users simultaneously use the entire bandwidth with unique orthogonal spreading codes.

In contrast, NOMA allows users to share the same frequency and time resources to enhance cell throughput and support more simultaneous users. This necessitates advanced MUD techniques at the base station to accurately extract each user's signal from the received superposition of signals¹. Each user transmits its own symbol based on its constellation, and these signals are combined at the receiver, where they are also influenced by Additive White Gaussian Noise (AWGN).

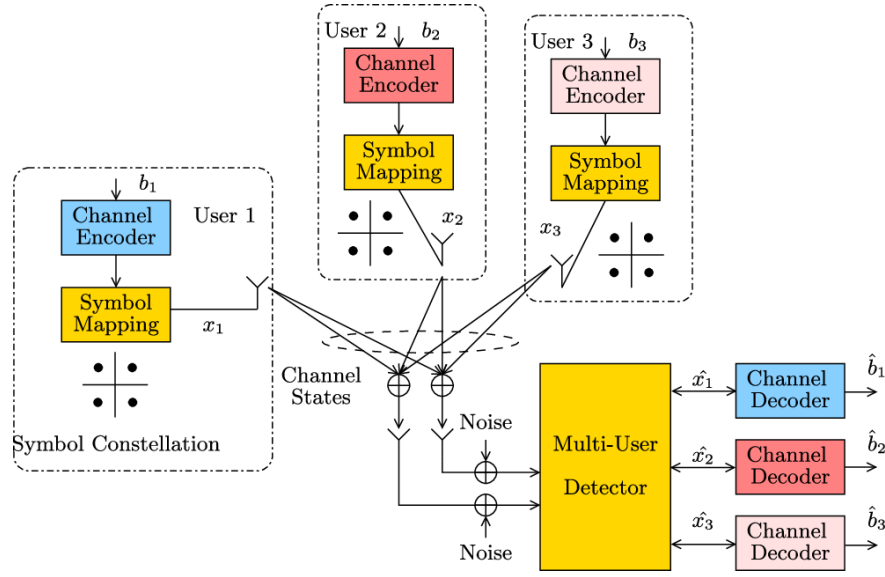


Fig. 2.10 Uplink system model [26]

Challenges with Successive Interference Cancellation (SIC): As previously discussed, SIC has been recognized as a potential decoding solution for NOMA. However, it presents several well-known drawbacks, which become exacerbated as the number of users increases:

- The error made in each iteration of SIC can propagate through subsequent iterations, compounding inaccuracies.
- Complete channel information must be known at the receiver to perform SIC effectively.

¹Please note that the mentioned superposition of signals is their addition in the classical domain, since in a synchronous system the signals arrive simultaneously. It should not be confused with the superposition of quantum states explained in detail in chapter 3.

- The decoding time increases with the number of users, potentially violating strict latency requirements.
- A significant difference in the power levels of each signal is required for successful detection[10].

Despite its potential, the implementation of SIC in current wireless systems remains impractical due to high computational costs, latency, and error propagation issues. These challenges have been noted by industry experts and scholars, and even in the latest 5G 3GPP Release 15, NOMA was thoroughly considered but not continued as a work item. The major reasons cited include high implementation complexity required by SIC receivers, significant error propagation in iterative decoding, and high latency were noted as the major reasons for their decision[23].

Efforts to mitigate the load on SIC receivers include user grouping to create parallel channels, which can then be decoded separately using techniques such as beamforming, multicarrier transmission, or Multiple Input Multiple Outputs (MIMO) support[18].

Given these significant challenges with SIC, the next logical step is to consider Maximum Likelihood (ML) decoding. ML decoding supports high-quality decoding by effectively managing the complexities associated with multi-user environments without the drawbacks of SIC. This approach is considered next as it offers a robust alternative capable of addressing the stringent demands of modern wireless communication systems.

2.4.3 Maximum Likelihood Detection

The *Maximum Likelihood detection* method represents an optimal approach for MUD in communication systems, particularly effective in distinguishing among the signals transmitted by multiple users in both Orthogonal Multiple Access (OMA) and Non-Orthogonal Multiple Access (NOMA) environments. It is optimal in the sense that it minimizes the error probability of the detection.

Operational Principle of ML Detection

ML detection seeks to identify the most likely symbol vector transmitted by users, based on the received signal alongside channel estimates. It achieves this by evaluating all conceivable multi-user symbol combinations that could have resulted in the observed reception, thereby selecting the combination that maximizes the likelihood function for a symbol vector $\mathbf{a}_n$ given the received signal $\mathbf{y}_n$ $p(y_n|a_n)$ and the channel gains $\mathbf{h}_n$:

$$\hat{\mathbf{a}}_n = \arg \max_{\mathbf{a}_n \in \mathcal{X}^U} \left\{ -\|\mathbf{y}_n - \mathbf{H}_n \mathbf{a}_n\|^2 \right\} \quad (2.18)$$

Here, $\mathcal{X}$ denotes the constellation set, U is the number of symbol vectors considered (number of users), $\mathbf{a}_n$ includes the symbols on the n -th subcarrier, and $\mathbf{H}_n$ represents the corresponding channel gain matrix. The optimal symbol vector is then forwarded to the channel decoder for subsequent processing[27].

The search set of $\mathbf{a}_n$ (i.e., the search space $\mathbb{S} \subseteq \mathcal{X}^U$) is the set of possible solutions that the optimizer can take into account. Each element $\mathbf{a}_n$ in the search space is a candidate solution with which the values of the *ML objective function* $\mathcal{D}(\mathbf{a}_n) = \|\mathbf{y}_n - \mathbf{H}_n \mathbf{a}_n\|^2$ (or $\text{Obj}_{\text{ML}}(\mathbf{a}_n)$) in Eq. 2.18 (i.e., Euclidean distances) are measured and compared with each other. The best candidate, with the minimum value, becomes the ML solution $\hat{\mathbf{a}}_n$. Further, $\mathbb{S}$ is an indicator of complexity. In principle, the ML search involves all possible candidates (i.e., $\mathbb{S} = \mathcal{X}^U$), which makes the brute-force approach intractable for large MIMO sizes with high-order modulations.

Complexity and Performance

The computational complexity of ML MUD is notably high, for a system supporting U users, each transmitting symbols from a set $\mathcal{X}$, scales exponentially with the number of users and symbol choices, making it highly complex. For example, if each of the $U = 20$ users transmits symbols using 4-ary Quadrature Phase Shift Keying (QPSK), then the ML detector must evaluate 4^{20} (more than one trillion) potential symbol vectors to determine

the most likely one. Generally, the complexity can be described as $O(|\mathcal{X}|^U)$, where $|\mathcal{X}|$ is the cardinality of the symbol set and U is the length of the symbol vector. In contrast, in OMA systems where each received signal typically conveys information from a single user, the complexity of ML MUD is considerably lower, approximately $O(\mathcal{X})$, which is manageable given that $\mathcal{X}$ corresponds to the size of the symbol set used by individual users. Comparatively, the complexity of Successive Interference Cancellation (SIC) scales linearly with the number of users and symbol length as $O(|\mathcal{X}| \cdot U)$. Although SIC has lower computational demands, it performs poorly under conditions where channels have similar characteristics or are rank-deficient, and it is prone to error propagation across decoding stages[28].

ML detection substantially outperforms SIC regarding Bit Error Rate (BER) performance due to its exhaustive approach in evaluating all symbol combinations and its superior handling of channel and noise effects. However, the trade-off involves significantly higher computational demands, particularly challenging in NOMA systems with many users or extensive symbol constellations. Therefore, while ML detection offers enhanced accuracy and superior performance, its practical deployment is often limited by computational capabilities, especially in large-scale user environments, which is the issue we seek to solve in this project[29].

2.5 Centralized Radio Access Networks for Next Generation Wireless Networks

Centralized Radio Access Networks (C-RAN) represent a significant evolution in cellular network architecture, designed to meet the increasing demands of modern wireless communications. Unlike traditional Radio Access Networks (RAN) where each base station houses its own baseband processing and radio functionalities, C-RAN centralizes these baseband resources in high-capacity data centers. This centralization allows the resources to

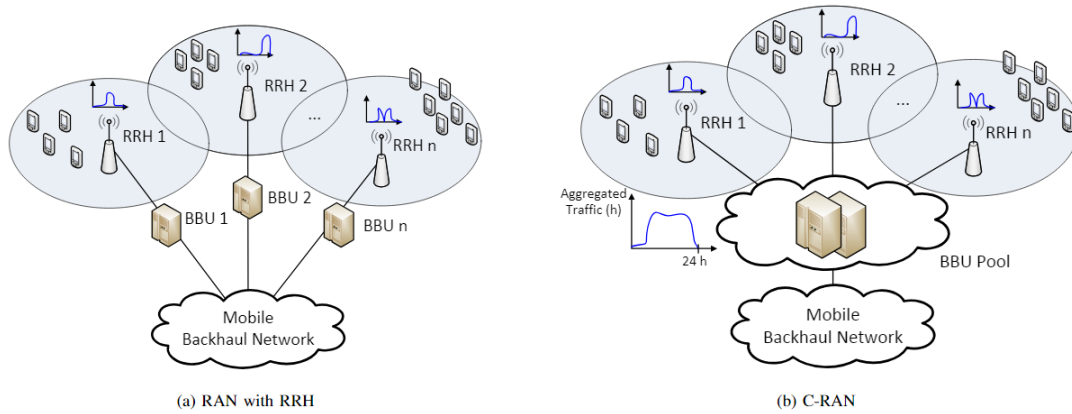


Fig. 2.11 C-RAN architecture for mobile networks. [7]

be dynamically shared and allocated among various base stations. Figure 2.11 provides an overview of the overall C-RAN architecture along with typical RAN .

In traditional cellular networks, each base station is responsible for serving all the mobile users within its cell coverage, integrating both baseband processing and radio functionalities. These functionalities include tasks such as coding, modulation, and executing Fast Fourier Transforms (FFT). On the other hand, C-RAN pools these baseband processing tasks into a few, strategically located data centers that are equipped with advanced computational resources. This pooling not only simplifies the physical infrastructure at individual cell sites but also enhances the network's ability to handle large volumes of data and complex processing tasks required for next-generation wireless technologies[7].

The benefits of C-RAN are multifaceted. Firstly, the high concentration of computational power in centralized data centers allows for the efficient handling of advanced communication techniques such as mMIMO, NOMA and beamforming, which are integral to modern networks but computationally intensive. Secondly, this architecture offers remarkable scalability and flexibility. As the network's user density and service requirements change, resources can be reallocated efficiently across the network without the need for physical adjustments at the base station level. Moreover, By centralizing the most energy-demanding components, the overall energy usage of the network is lowered, which, in turn, reduces the carbon footprint and operational expenses associated with energy. The centralized nature of C-RAN also

facilitates easier upgrades and maintenance, as these tasks can be performed at centralized locations rather than at numerous individual sites[30].

In the context of enhancing C-RAN's capabilities, our envisioned scenario, as shown in Figure 1.1, introduces quantum processors co-located with classical computational resources in data centers. These quantum processors are tasked with managing computationally intensive optimization processes that currently bottleneck throughput and latency, while classical processors handle more tractable processing tasks. The next chapter explores the fundamentals of quantum computation paving the way to integrate NOMA uplink user detection to be solvable using QPUs.

Chapter 3

Introduction to Quantum Annealing

Quantum computing represents a revolutionary approach to computational technology, utilizing principles of quantum mechanics like superposition and entanglement to perform operations far beyond the capabilities of classical computers. This chapter introduces quantum computing, emphasizing its potential to solve complex, intractable problems for traditional systems.

The chapter begins by introducing foundational quantum computation concepts, laying the groundwork for understanding this cutting-edge technology. It also provides a brief tutorial on using currently available Noisy-Intermediate-Scale Quantum (NISQ) technology, enabling practical experience in this emerging field.

Furthermore, the chapter explores quantum annealing (QA), a promising quantum computing approach. It explains QA problem formulation and annealing algorithms, particularly highlighting the use of D-Wave systems for practical quantum annealing applications.

3.1 Computational Complexity

Computational complexity theory, a branch of theoretical computer science, classifies problems into different classes based on the computational effort required to solve them, primarily focusing on time and space. Key complexity classes for combinatorial optimization problems

As shown in Figure 3.1 include P, NP, NP-complete, and NP-hard. The P (Polynomial) class encompasses decision problems solvable efficiently with deterministic polynomial-time algorithms. Conversely, the NP (Non-deterministic Polynomial) class includes decision problems that, although challenging to solve in polynomial time, have solutions that are verifiable within polynomial time.

NP-complete problems represent the most challenging problems within the NP class, being both verifiable in polynomial time and reducible from any other NP problem in polynomial time. NP-hard problems, while as challenging as NP-complete problems, may not be decision problems and might not have verifiable solutions in polynomial time[31].

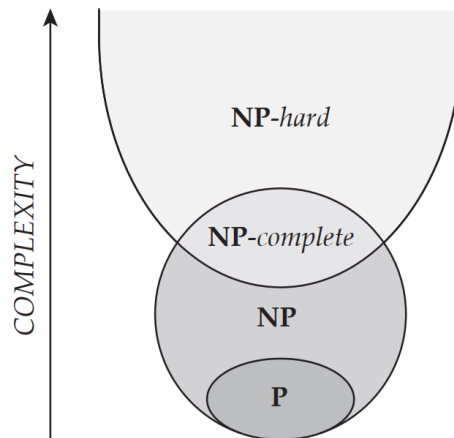


Fig. 3.1 Sketch representing the relation between the different complexity classes P, NP, NP-complete, and NP-hard.

Combinatorial optimization problems, which require finding an optimal arrangement of a finite set of objects to minimize or maximize a cost function, are typically NP-hard. Such problems pervade fields like artificial intelligence and machine learning.

Quantum computing holds potential for addressing some NP-hard problems more effectively than classical computing by utilizing quantum phenomena such as superposition and entanglement. These capabilities allow quantum algorithms to explore solution spaces more comprehensively, potentially accelerating the convergence to optimal or near-optimal solutions[32].

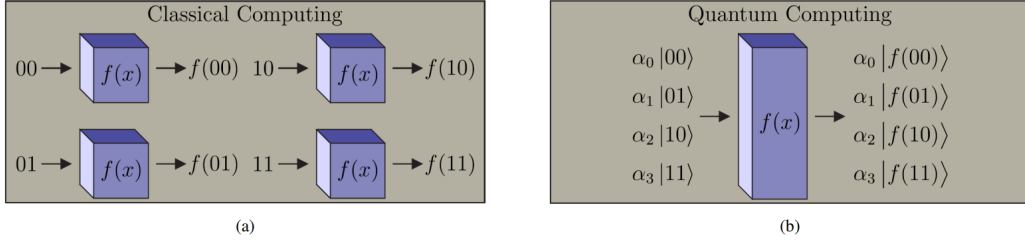


Fig. 3.2 The comparison of classical and quantum computation[33].

This thesis utilizes quantum annealing to tackle the Maximum Likelihood Detection (MLD) problem in communication systems—an NP-hard problem. By representing the MLD as a combinatorial optimization problem and solving it using a quantum annealer, this work aims to demonstrate how quantum computing can significantly enhance communication system performance and efficiency. As quantum technology progresses, it is anticipated to complement classical computing in resolving complex optimization challenges across various domains, including communication systems.

3.2 Fundamentals of Quantum Computation

Quantum computation offers a significant advance over traditional computing by employing quantum mechanics' distinctive properties—superposition, entanglement, and quantum tunneling. Additionally, The inherent parallelism of quantum information processing intimated in Figure 3.2 equips quantum computers with immense computational power. Unlike conventional computers that operate with bits represented solely by 0s or 1s, quantum computers utilize qubits. These qubits can represent multiple states simultaneously due to superposition, enhanced by the interconnectedness of states via entanglement and the non-classical movement of quantum tunneling. This capacity allows quantum computers to process complex data configurations and solve problems that are currently infeasible for classical systems.

3.2.1 Quantum Systems and States

In quantum mechanics, the behavior of quantum systems is governed by Schrödinger's equation, the quantum analogue of Newton's second law in classical mechanics. This fundamental equation describes how the wavefunction of a physical system evolves over time, dictated by the Hamiltonian, a Hermitian operator that determines the system's discrete energy levels. Schrödinger's equation is given by:

$$\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle \quad (3.1)$$

where $\hbar$ is the reduced Planck's constant, i is the imaginary unit, $|\psi(t)\rangle$ represents the state vector of the system at time t , and $\hat{H}$ is the Hamiltonian operator[34].

The energy levels of a quantum system are determined by the eigenvalues of the Hamiltonian. Since the Hamiltonian is *Hermitian*, all eigenvalues are real numbers, making these energy levels physically observable. The smallest eigenvalue corresponds to the ground state energy, the second smallest to the first excited state, and so forth. Quantum systems exist only in these discrete energy states—either the ground state or an excited state—but not in between, in other words are "quantized." This concept underlies the term *quantum*, derived from *quanta*.

Quantum states are represented as vectors in a complex vector space known as Hilbert space. In Dirac notation, a quantum state is denoted as $|\alpha\rangle$, where α includes complex numbers that are probability amplitudes.

Superposition is a distinguishing feature of quantum systems is their ability to exist in states of superposition. Unlike classical systems, which are confined to a single state at any given time, quantum systems can simultaneously inhabit multiple states. For instance, a state $|\phi\rangle$ might be a superposition of two basis states $|a\rangle$ and $|b\rangle$, expressed as:

$$|\phi\rangle = c_a |a\rangle + c_b |b\rangle \quad (3.2)$$

where c_a and c_b are complex coefficients, and the probabilities of finding the system in either state upon measurement are $|c_a|^2$ and $|c_b|^2$ respectively.

Entanglement is another pivotal phenomenon is entanglement, which occurs when pairs or groups of particles interact such that the state of one particle cannot be described independently of the others, regardless of the distance between them. A classic example is a two-particle system in a Bell state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|aa\rangle + |bb\rangle) \quad (3.3)$$

In this state, if one particle is measured and found in state $|a\rangle$ or $|b\rangle$, the other particle will instantaneously be found in the same state, demonstrating quantum non-locality. This effect, where entangled particles affect each other instantaneously over any distance without direct physical connection, has no equivalent in classical physics and illustrates the unique, interconnected nature of quantum systems[34].

3.2.2 The Qubit

The qubit is the fundamental unit of quantum information, analogous to the classical bit in conventional computing. However, unlike classical bits, which can only exist in one of two states (0 or 1), a qubit can exist in a superposition of these states. The quantum state of a qubit, represented in an orthogonal basis commonly referred to as the computational basis $\{|0\rangle, |1\rangle\}$, is expressed as:

$$|q\rangle = \alpha|0\rangle + \beta|1\rangle \quad (3.4)$$

where α and β are complex numbers ($\alpha, \beta \in \mathbb{C}$) representing the amplitudes of the qubit state on the computational basis, and the condition $\|\alpha\|^2 + \|\beta\|^2 = 1$ ensures the normalization of the state.

In the special cases where $\alpha = 0$ and $\beta = 1$, the qubit state simplifies to $|q\rangle = |1\rangle$, corresponding to the classical bit value 1. Conversely, when $\alpha = 1$ and $\beta = 0$, the state becomes $|q\rangle = |0\rangle$, analogous to the classical bit value 0. For the case of equal superposition,

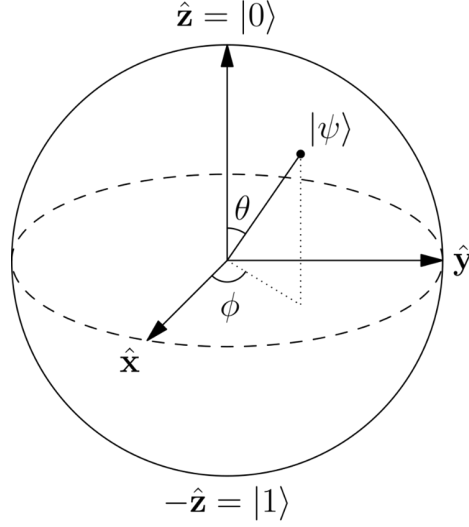


Fig. 3.3 The Bloch sphere is a geometrical representation of the pure state space of a two-level quantum mechanical system (qubit), named after the physicist Felix Bloch[35].

where $\alpha = \beta = 1/\sqrt{2}$, the qubit is in the state:

$$|q\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (3.5)$$

This state does not favor either of the orthogonal states $|0\rangle$ or $|1\rangle$, and is widely utilized in quantum algorithms due to its symmetric properties[31].

Geometrically, the state of a qubit can be depicted using a 2-D or 3-D representation. For real-valued amplitudes, the qubit state simplifies to:

$$|q\rangle = \cos(\theta)|0\rangle + \sin(\theta)|1\rangle \quad (3.6)$$

and can be visualized on a 2-D plane. For complex-valued amplitudes, a more comprehensive representation on the 3-D Bloch sphere is used shown in Figure 3.3, where the qubit state is expressed as:

$$|q\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle \quad (3.7)$$

Measuring the state of a qubit collapses the superposition to one of the orthogonal states $|0\rangle$

or $|1\rangle$, with probabilities $\|\alpha\|^2$ and $\|\beta\|^2$ respectively. This process, akin to a quantum-to-classical conversion, provides critical insights into the quantum system's state[32].

Post-measurement, the quantum state of the system collapses irreversibly to the observed state. For example, if a measurement of a qubit in an equal superposition state yields $|1\rangle$, the post-measurement state of the qubit becomes $|q'\rangle = |1\rangle$. This phenomenon of wave function collapse highlights the non-deterministic nature of quantum measurements and is central to quantum mechanics.

Moreover, qubits can exhibit quantum entanglement, a property allowing multiple qubits to demonstrate correlations exceeding those possible in classical systems. Consider two qubits in the entangled Bell state:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (3.8)$$

Here, if one qubit is measured and found to be $|0\rangle$ or $|1\rangle$, the state of the other qubit is instantaneously known, regardless of the distance separating them. This perfect correlation, demonstrating quantum non-locality, is one of the many counterintuitive aspects of quantum physics[26].

To physically implement qubits, various platforms like superconducting quantum interference devices (SQUIDs), ion traps, and nuclear magnetic resonance (NMR) systems are utilized. A simple physical example of a qubit could be a spin -1/2 particle, where the eigenstates of spin up and spin down form the computational basis of the system, represented in a two-dimensional complex Hilbert space. These states are manipulated using spin operators, typically represented by 2x2 Pauli matrices, which are both Hermitian and unitary[36].

3.2.3 Models of Quantum Computation

Quantum computing encompasses several distinct models and hardware implementations, notably **gate-based quantum computing** and **adiabatic quantum computation (AQC)**. In the *gate model*, often considered akin to digital computing, computation is achieved

by applying a series of *unitary gates* to quantum bits (*qubits*), which are measured at the conclusion of the operations. This model mirrors the programmable logic gates used in classical digital computing, but employs operations that leverage quantum mechanical properties [36].

Adiabatic quantum computation, or quantum annealing, can be thought of as more analogous to analog computing. It begins with the quantum state prepared as the ground state of a simple Hamiltonian and involves a continuous evolution of the system towards a final Hamiltonian. The process seeks the lowest-energy state that represents the solution to an optimization problem, employing a gradual adjustment of parameters in a manner similar to how analog systems handle continuous variations [37].

Alongside these traditional models, **Noisy Intermediate-Scale Quantum (NISQ)** technologies have emerged, characterized by both gate model processors and quantum annealers. NISQ technologies are crafted to operate within the limitations of current technology and are focused on achieving quantum advantages for specific problem domains, even though they are not fully scalable. For example, gate model processors developed by companies like Google and IBM feature a range of programmable gate-sets designed to address various algorithms. However, these devices are still hindered by noise, which limits the number of qubits capable of effective gate operations, and the short coherence times currently achievable [26].

The **Quantum Approximate Optimization Algorithm (QAOA)** is another approach being explored on gate-model quantum computers, aiming to optimize computational solutions. Although it is a promising development, QAOA is currently limited by the relatively small number of qubits available compared to the capacities of quantum annealers.

Quantum annealers apply the adiabatic principle to find low-energy configurations within a high-dimensional energy landscape. Computational problems are formatted using either the *Quadratic Unconstrained Binary Optimization (QUBO)* method or the *Ising model*. These formats translate problems into a real-valued matrix where diagonal and off-diagonal elements denote variable biases and coupling strengths, respectively. Annealers continuously

adjust the transverse field and problem energies, aiming to achieve a state that represents the optimal solution [35].

In practical terms, both gate-based NISQ devices and quantum annealers are operated in environments filled with physical noise sources such as thermal baths and electromagnetic cross-talk, which significantly impact their performance. Progress in coherent materials, fabrication processing, and circuit design is essential for enhancing the robustness and scalability of these quantum computing technologies. Future designs are anticipated to allow significant enhancements, potentially enabling thousands of gate operations on hundreds to thousands of qubits [38].

3.3 Quantum Annealing

Quantum annealing (QA) is a heuristic optimization algorithm that leverages quantum mechanics to solve complex combinatorial optimization problems. Analogous to the metallurgical process of *annealing*, which involves heating and then slowly cooling to remove defects, quantum annealing utilizes quantum-mechanical fluctuations, notably **quantum tunneling**, to navigate the solution space. Quantum tunneling allows a system to transition through potential energy barriers even when it does not possess sufficient classical energy to overcome these barriers. This property is critical in quantum annealing, as it enables the exploration and escape from local minima in pursuit of the global minimum of an objective function across a set of candidate solutions.

Quantum annealing has become particularly relevant with the advent of small- and intermediate-scale quantum processors, such as those developed by *D-Wave Systems*, which allow for programmable quantum annealing in various research and industrial applications. These processors are used to tackle problems characterized by many local minima, such as finding the ground state of a spin glass or optimizing routes in the traveling salesman problem. The concept of quantum annealing was initially proposed in 1988 and has since

been developed into a robust computational strategy that manipulates quantum fluctuations to resolve optimization challenges that are impractical for classical computers.

3.3.1 Theoretical Foundations of Quantum Annealing

Quantum Annealing (QA) is fundamentally grounded in the adiabatic theorem of quantum mechanics, which posits that a quantum system will remain in its ground state throughout a process if the Hamiltonian governing it changes sufficiently slowly. This condition hinges critically on the minimum energy gap between the ground state and the first excited state, often referred to as the *minimum gap*.

In quantum mechanics, the Hamiltonian is an operator that defines the total energy of a system at any given state, analogous to its counterpart in classical mechanics. It maps specific states, known as eigenstates, to their corresponding energy levels, termed eigenenergies. The system's state is fully described by its wavefunction, with the Hamiltonian dictating the evolution of this wavefunction[39].

In the context of QA, the process begins with the system in the ground state of an initial Hamiltonian H_i , typically simple and well-understood:

$$H_i = \sum_{i \in V} \sigma_x^i \quad (3.9)$$

where σ_x^i denotes the Pauli-X matrix acting on the i -th qubit. This configuration prepares all qubits in a superposition of computational basis states, facilitating quantum tunneling necessary for exploring various state configurations.

The system is then adiabatically evolved towards a final Hamiltonian H_f , which is more complex and whose ground state encodes the solution to the optimization problem:

$$H_f = \sum_{i \in V} h_i \sigma_z^i + \sum_{(i,j) \in E} J_{ij} \sigma_z^i \sigma_z^j \quad (3.10)$$

Here, h_i represents the magnetic fields at each qubit site, and J_{ij} the interaction strengths between qubits. The adiabatic transition is described mathematically by:

$$H(t) = A(t)H_i + B(t)H_f \quad (3.11)$$

where $A(t)$ and $B(t)$ are functions that manage the annealing schedule, starting with $A(0) = 1$ and $B(0) = 0$, transitioning to $A(T_a) = 0$ and $B(T_a) = 1$ by the end of the annealing process at time T_a [33].

The objective is for the Hamiltonian to evolve such that the quantum system continuously resides in its lowest energy state throughout the annealing, transitioning from the simple problem encoded in H_i to the target problem in H_f . This approach ensures that, at the process's conclusion, the system's configuration aligns with the optimal solution of the problem.

The Hamiltonian's pivotal role in quantum annealing is not merely to dictate the energy landscape but also to define the trajectory that the system's wavefunction follows during the annealing. This trajectory must be carefully managed to maintain the system in its lowest energy state, despite potential disruptions from environmental noise and other perturbations.

3.3.2 Quantum Annealing vs. Classical Simulated Annealing

Quantum Annealing (QA) and Classical Simulated Annealing (SA) are optimization methods inspired by the physical process of annealing, where materials are heated and then slowly cooled to achieve a low-energy crystalline state. While both approaches aim to find global minima within complex energy landscapes, their operational principles diverge significantly due to the underlying physics.

Classical Simulated Annealing mimics this thermal process by utilizing a *temperature* parameter that gradually decreases during the optimization. This temperature governs the probability of accepting higher-energy states during the search, thus enabling the algorithm to escape local minima and eventually settle in the global minimum. The effectiveness of

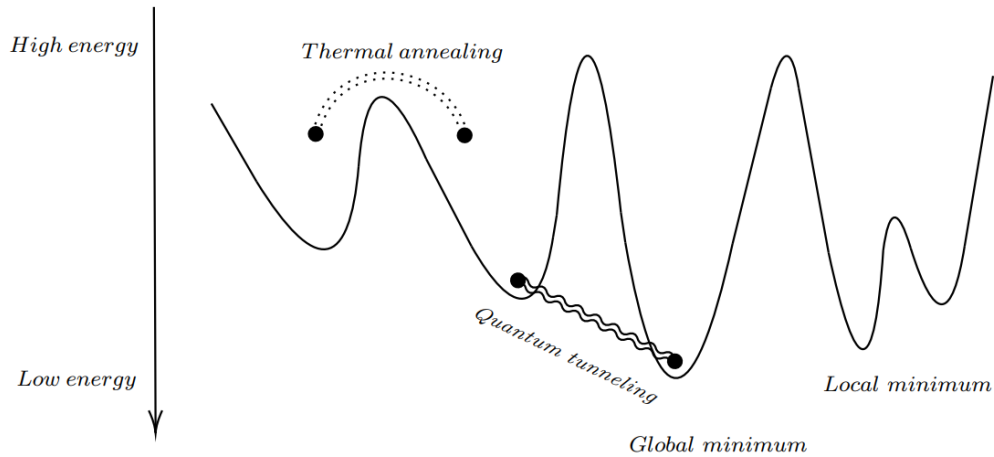


Fig. 3.4 Illustration of leveraging quantum tunneling phenomena to find the optimal solution of an energy landscape using quantum annealing[38].

SA is heavily dependent on the *cooling schedule* and the randomness inherent in thermal fluctuations.

In contrast, Quantum Annealing leverages quantum mechanical phenomena such as *tunneling* and *superposition* to enhance the search process. Unlike SA, which must climb over energy barriers -illustrated in Figure 3.4, quantum tunneling allows QA to pass directly through these barriers. This capability enables QA to potentially reach the global minimum more efficiently by exploring the solution space more thoroughly and rapidly. Furthermore, quantum superposition permits the simultaneous evaluation of multiple states, introducing a level of parallelism to the search process that is absent in classical algorithms[38]. The advantages of Quantum Annealing over Classical Simulated Annealing include faster convergence and better scalability, particularly for problems characterized by a rugged energy landscape with many local minima. Additionally, quantum effects such as *entanglement* and *superposition* may offer computational benefits not accessible in classical settings. However, the real-world performance of QA also hinges on maintaining *coherence* and managing *quantum decoherence*, which pose significant technical challenges.

Overall, while QA presents theoretical and practical advantages over SA under certain conditions, the selection between these methods should be informed by the specific characteristics of the problem and the computational resources available.

3.3.3 Problem Formulation for Quantum Annealing: Ising and QUBO Models

Quantum Annealing (QA) requires the problem of interest to be defined as an objective function, typically a quadratic polynomial of binary variables. This objective function is then expressed in one of two equivalent forms: the Ising model or the Quadratic Unconstrained Binary Optimization (QUBO) model, which are standard in QA literature.

Ising Model Formulation In the Ising model, solution variables are spins $s_i \in \{+1, -1\}$ arranged in a lattice:

$$s_1, \dots, s_{N_v} = \arg \min_{s_1, \dots, s_{N_v}} \left(\sum_{i < j}^{N_v} g_{ij} s_i s_j + \sum_{i=1}^{N_v} f_i s_i \right) \quad (3.12)$$

where N_v is the number of spin variables, and g_{ij} and f_i are the *Ising model parameters* that characterize the problem. The f_i specifies the external magnetic fields affecting each spin, favoring either +1 or -1 states.: positive indicates a preference for -1 while negative indicates a preference for $+1$, with the magnitude corresponding to the magnitude of the preference for either state. The g_{ij} capture preferred correlations between spins: positive causes the QA to prefer $s_i \neq s_j$, while negative causes the QA to prefer $s_i = s_j$ in its optimization outcome. Analogously to f_i , the magnitude of g_{ij} corresponds to the magnitude of its preference. Accordingly, the Ising energy function E of the model (i.e., cost function):

$$E(s_1, \dots, s_{N_v}) = \sum_{i=1}^{N_v} M_{i,i} s_i + \sum_{i < j}^{N_v} M_{i,j} s_i s_j \quad (3.13)$$

where $M \in \mathbb{R}^{N_v \times N_v}$ is an upper triangular matrix, each element representing an Ising coefficient that is decided by the problem of interest:

$$M = \begin{bmatrix} f_1 & g_{12} & \cdots & g_{1N_v} \\ 0 & f_2 & \cdots & g_{2N_v} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & f_{N_v} \end{bmatrix} \quad (3.13b)$$

QUBO Formulation The QUBO model uses classical binary bits $q_i \in \{0, 1\}$ to represent the problem:

$$q_1, \dots, q_{N_v} = \arg \min_{q_1, \dots, q_{N_v}} \left(\sum_{i \leq j}^{N_v} Q_{i,j} q_i q_j \right) \quad (3.14)$$

where Q_{ij} includes both *linear terms* (on the diagonal) and *quadratic terms* (off-diagonal), where $Q \in \mathbb{R}^{N_v \times N_v}$ is upper triangular. The off-diagonal matrix elements Q_{ij} ($i \neq j$) correspond to g_{ij} in Equation 3.12, and the diagonal elements correspond to f_i .

The two forms are equivalent, their solutions related by:

$$q_i \leftrightarrow \frac{1}{2}(s_i + 1), \quad (3.15)$$

leading to

$$g_{ij} \leftrightarrow \frac{1}{4}Q_{ij} \quad \text{and} \quad f_i \leftrightarrow \frac{1}{2}Q_{ii} + \frac{1}{4} \sum_{k=1}^{i-1} Q_{ki} + \frac{1}{4} \sum_{k=i+1}^{N_v} Q_{ik}. \quad (3.16)$$

This equivalence allows quantum annealers to seamlessly switch between problem representations, optimizing the same objective function formatted differently depending on the computational or contextual needs[3]. This flexibility is particularly useful in practical applications where the choice of model may depend on the specific characteristics of the optimization problem. Table 3.1 lists terms associated with the context.

Table 3.1 Comparison of problem expressions and their terms in various contexts.

	Terms			
	Linear coefficient	Quadratic coefficient	Variable	States
QUBO (scalar)	a_i	b_{ij}	q_i	$\{0, 1\}$
QUBO (matrix)	Q_{ii}	Q_{ij}	x_i	$\{0, 1\}$
Ising	h_i	I_{ij}	s_i	$\{-1, 1\}$
Graph	Node Weight	Edge Strength	Node	-
QPU	Qubit Bias	Coupling Strength	Qubit State	{Spin Up, Spin Down}

3.4 D-Wave Implementation of Quantum Annealing

D-Wave Systems, based in Canada, leads the field in developing and commercializing quantum computers through the method of quantum annealing. This approach leverages quantum mechanical properties, such as superposition and entanglement, to address optimization problems more effectively than traditional computers in specific applications.

The company's core technology, the Quantum Processing Unit (QPU), employs qubits designed to identify low-energy states that correspond to optimal solutions, thus efficiently solving complex optimization challenges. Since introducing the first commercial quantum annealer, the D-Wave One, in 2011, equipped with a 128-qubit processor, D-Wave has consistently advanced the capabilities of its technology[40].

3.4.1 Hardware Architecture of D-Wave Quantum Annealing System

The D-Wave quantum computer is a commercially available computing hardware device specifically designed to implement *quantum annealing*. It consists of several components: a **quantum processor**, a **magnetically shielded box**, and **cooling equipment**, all housed inside a ten square meter shielded room (see Figure 3.5). The core component, the quantum processor, comprises superconducting flux qubits. These qubits are based on niobium superconducting quantum interference devices (SQUIDs), protected from environmental interference by the shielded box and maintained at near absolute zero temperature by the cooling equipment to preserve their quantum states.

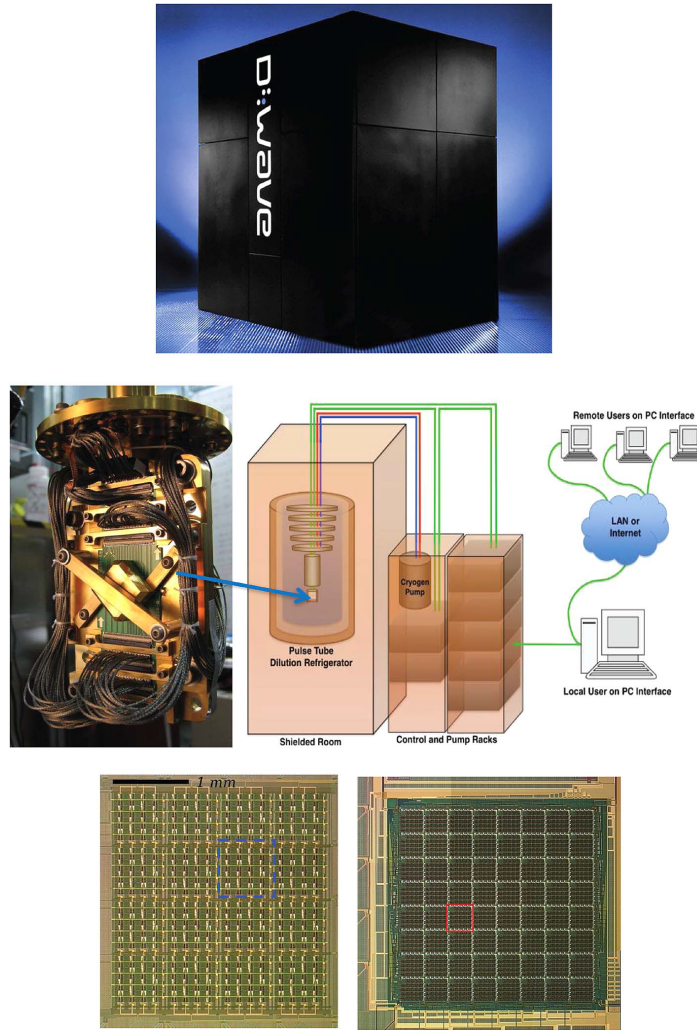


Fig. 3.5 The D-Wave quantum computer in the top panel, its inside structural illustration in the middle panel, and 128 and 512 qubit quantum chips of D-Wave One and Two devices, respectively, in the bottom panel.

The superconducting flux qubits in D-Wave's quantum annealers are niobium-based *radio-frequency SQUIDs* (**rf-SQUIDs**) that utilize superconducting current loops interrupted by two *Josephson junctions*. These junctions are controlled by variable magnetic fluxes as illustrated in Figure 3.6. Each qubit is a double-ring shaped niobium structure with two Josephson junctions that regulate the flow of current within the ring. The direction of the current, which can be clockwise, counterclockwise, or a superposition of both when the qubit is in a quantum superposition state, determines the magnetic forces represented as spin-up, spin-down, or a combination of both [40].

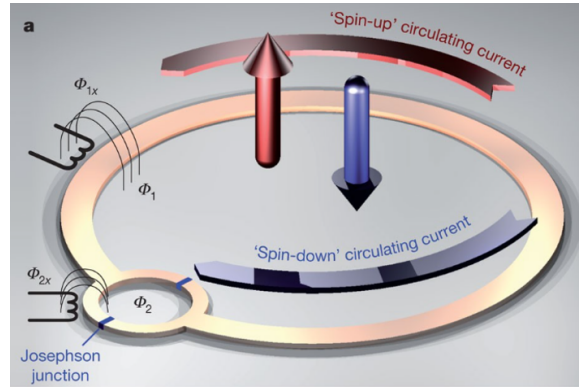


Fig. 3.6 Each qubit is a loop of niobium controlled by Josephson junctions and biases represented by ϕ in this diagram [40].

Couplers, which connect pairs of qubits, consist of simpler niobium loops with less control circuitry compared to the qubits. Both qubits and couplers are etched onto *silicon wafer circuits* with four niobium layers separated by insulation layers. To achieve the necessary conditions for quantum superposition, the niobium must be cooled to temperatures below 20 mK. Although this technology is robust against various types of noise, it remains sensitive to electromagnetic interference, necessitating that the quantum processing unit (QPU) operates in a shielded chamber with a temperature below 15 mK, a magnetic field 50,000 times weaker than Earth's, and an atmospheric pressure 10 billion times lower than the ambient environment.

The *cryogenic subsystem* of the D-Wave quantum annealer features an enclosure with five layers of magnetic isolation, consuming about 16 kW of the total 25 kW energy consumption, with the remaining 9 kW utilized by the classical control system outside the cryostat[41].

Table 3.2 presents the recent versions of D-Wave quantum annealers and their features. The **D-Wave Advantage** quantum annealer, released in 2020, offers around 5000 sparsely connected qubits, while digital annealers can operate with up to 8192 fully connected variables at 16-bit precision and 4096 variables at 64-bit precision. The table also shows the evolution of the qubit connectivity topology, from *Chimera* in the D-Wave 2000Q to *Pegasus* in the D-Wave Advantage, and the upcoming *Zephyr* topology in the D-Wave Advantage 2 prototype and full-size versions, expected to be released in 2024 and 2025, respectively.

Table 3.2 Overview of D-Wave Quantum Annealer Versions[5]

	DW2Q	DWAdv	DWAdv2 (prototype)	DWAdv2 (full-size)
Release Year	2017	2020	2022	(exp.) 2025
Topology	Chimera	Pegasus	Zephyr	Zephyr
Qubits	2000+	5000+	500+	7000
Connectivity	6	15	20	20

3.4.2 Topology of D-Wave's Quantum Processing Units

D-Wave's Quantum Processing Units (QPUs) are distinguished by their unique hardware graph topologies, defining the physical connections between qubits and their couplers. The evolution from the **Chimera** to the **Pegasus topology** represents a significant advancement in QPU design, enhancing the capability and efficiency of quantum annealing processes[42]. Below is description for each type:

Chimera Topology Initially, D-Wave utilized the Chimera topology, characterized by a two-dimensional (2D) lattice of unit cells, each forming a small complete bipartite graph, typically denoted as $K_{N,N}$ [43]. Key features of the Chimera topology include:

- Qubits are oriented vertically or horizontally.
- It includes two types of couplers: internal couplers connecting pairs of orthogonal qubits and external couplers connecting colinear pairs of qubits.
- Each qubit is connected to four orthogonal qubits through internal couplers and has a degree of connectivity of six.
- Composed of a 2D lattice of tiles or unit cells, denoted as C_X , where X is the length of one side of the square lattice. Each tile is a $K_{N,N}$ bipartite graph with full connectivity between its partitions.
- The D-Wave 2000Q model features a lattice size of C_{16} and tiles of $K_{4,4}$.

The structural details of a typical Chimera unit cell are illustrated in Figure 3.7(a). the highlighted green square is a $K_{4,4}$ unit cell. Blue and red nodes are vertically and horizontally connected qubits to the highlighted unit cell.

Pegasus Topology The newer Pegasus topology, introduced with the D-Wave Advantage series, provides significantly enhanced connectivity[43]. Distinctive attributes of the Pegasus topology include:

- Qubits oriented vertically or horizontally.
- Incorporates three types of couplers: internal couplers, external couplers, and odd couplers (connecting parallel qubit pairs in adjacent rows or columns).
- Qubits have a nominal length of 12, connected to 12 orthogonal qubits through internal couplers, with a total degree of connectivity of 15.
- Configurations containing $N \times N$ unit cells are referred to as P_N and consist of $24N(N - 1)$ qubits.
- Supports a significantly more complex structure, enabling denser graph structures and the facilitation of more challenging optimization problems.

An example of the Pegasus graph, highlighting its complex unit cell structure composed of multiple subgraphs, is displayed in Figure 3.7(b). the highlighted region represents the equivalent unit cell, composed of K_4 and $K_{6,6}$ subgraphs. Blue and red nodes here represent vertical and horizontal pairs of qubits connected to the highlighted cell.¹

Despite the sophisticated designs, not all qubits and couplers may be operational due to manufacturing imperfections or calibration issues, with the hardware yield typically around 97%. The working graph, representing the functional part of the full hardware graph, directly impacts the size and complexity of problems that the QPU can effectively address[44].

¹Each qubit has at most degree 15. junctions of horizontal and vertical loops (qubits) indicate internal couplers. External couplers connect adjacent qubits that are oriented the same (e.g., the green vertical qubit is coupled to the two adjacent vertical blue qubits). Additional odd couplers connect similarly aligned pairs of qubits (e.g. the green qubit is coupled to the red qubit by an odd coupler).

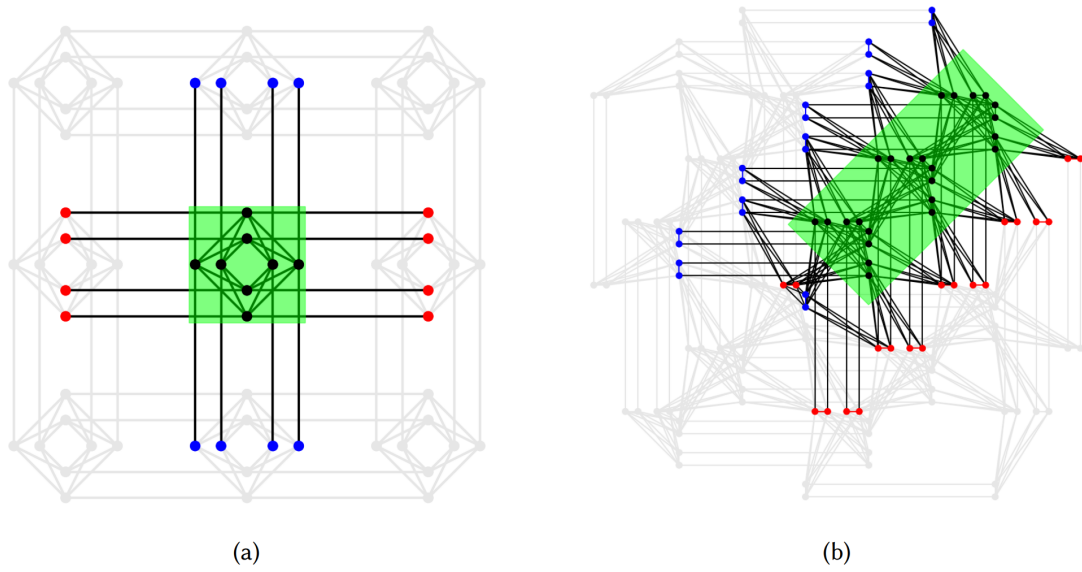


Fig. 3.7 Graphical representations of the topology graphs of two D-Wave hardware architectures. (a) Chimera graph, (b) Pegasus graph. Each qubit is represented by a node in the graph, and every coupler is an edge between nodes.

Implications of Topology on Quantum Annealing The specific topology of a QPU significantly influences the quantum annealing process. The Chimera topology, while foundational, has proven less conducive to quantum speedups for natively formulated problems due to its sparse connectivity. Conversely, the Pegasus topology, with its higher connectivity and non-bipartite structure, is better suited for handling more complex and larger-scale optimization problems. This enhanced connectivity represents a critical shift in the design philosophy of quantum annealers, reflecting ongoing improvements in QPU architecture aimed at broadening the range of solvable problems and improving the efficiency of quantum annealing operations.

3.4.3 The Need for Embedding

In many cases, QUBO models have high connectivity, which makes them challenging to map directly onto QPU hardware topologies. To facilitate this mapping, a process known as *minor-embedding* is employed. This involves linking chains of physical qubits using large

negative couplings to represent each logical qubit, a method essential due to the NP-hard nature of minor-embedding [44].

The *MinorMiner* (**MM**) algorithm, implemented in D-Wave’s *Ocean SDK*, is one heuristic approach used for this purpose. MM is designed for finding graph minors suitable for embedding into the quantum hardware. For more uniform embeddings, particularly of dense graphs, the *Native Clique Embedding* (**CLIQUE**) algorithm is preferable [45]. CLIQUE is optimized for the Chimera (C_{16}) and Pegasus (P_{16}) topologies, with maximum clique sizes of 64 and 180, respectively [46].

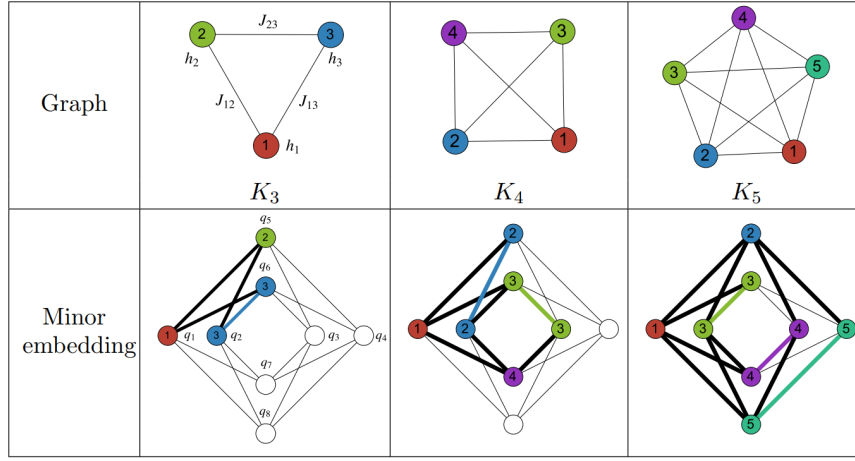


Fig. 3.8 Examples of minor embedding into the (1, 1, 4)-Chimera of the complete graphs K_3 , K_4 , and K_5 . The numbers and colors of the vertices in the graph are the same as in the minor embedding. **Bold black lines** correspond to the mapped edges and **bold color lines** correspond to chain of qubits.

Manufacturing imperfections and the intrinsic sparse connectivity of C_{16} and P_{16} topologies sometimes restrict the effective use of available qubits, impacting the embedding process. In practice, automated embedding tools in the Ocean SDK mitigate these challenges by optimizing the placement of logical qubits within the physical qubit architecture. The effectiveness of these embeddings directly influences the quantum annealer’s performance, making the quality of embedding—as measured by chain length—a crucial factor in achieving solutions that closely approximate the original problem’s requirements.

3.4.4 The Quantum Annealing Process on D-Wave Systems

Solving industry-relevant problems using quantum annealing (QA) as implemented by D-Wave Systems involves several distinct steps, Figure 3.9 visualizes this process. Each step is critical in transforming a real-world application problem into a format suitable for quantum computation and ultimately obtaining a solution. The process is outlined as follows:

1. **Definition of a QUBO Formulation and Graph Representation:** The real-world problem is first converted into a QUBO (Quadratic Unconstrained Binary Optimization) formulation, which has become the standard input format for quantum annealers. This QUBO is then transformed into a logical graph, where each node represents a variable and each edge represents the interaction between variables.
2. **Minor-Embedding:** The logical interaction graph is translated onto the physical hardware graph of the Quantum Processing Unit. This involves selecting sets of physical qubits to represent a single logical node and identifying the couplings between physical qubits to achieve the correct interactions between logical variables.
3. **Programming:** Programming the quantum annealer involves setting the parameters that define the final Hamiltonian of the problem. This includes specifying the weights for each qubit bias (to control the magnetic field acting on the qubit) and coupler strength (to control the interaction between qubits).
4. **Initialization:** Once programmed, the spin configuration of the QPU is initialized to the lowest energy configuration of an easily implementable initial Hamiltonian. In this state, the qubits are placed in an equal superposition of all possible states.
5. **Annealing Process:** The system transitions from the initial Hamiltonian to the final Hamiltonian according to predefined annealing functions, aiming to minimize energy. This phase may include a hybrid approach where part of the calculation is performed by a quantum processor, or it may involve reverse annealing, starting from a known (classical) solution and searching the state space around this local optimum.

6. **Readout of the Solution:** At the end of the annealing phase, the qubits are either in an eigenstate or a superposition of eigenstates in the computational basis. Each eigenstate represents a potential minimum of the final Hamiltonian. The individual spin values of the qubits are read out and recorded, representing a candidate solution to the original problem.
7. **Resampling:** Since quantum annealing is heuristic, it often requires multiple iterations to increase the probability of finding the ground state of the system. Therefore, the anneal-readout cycle is repeated numerous times to collect a variety of candidate solutions.

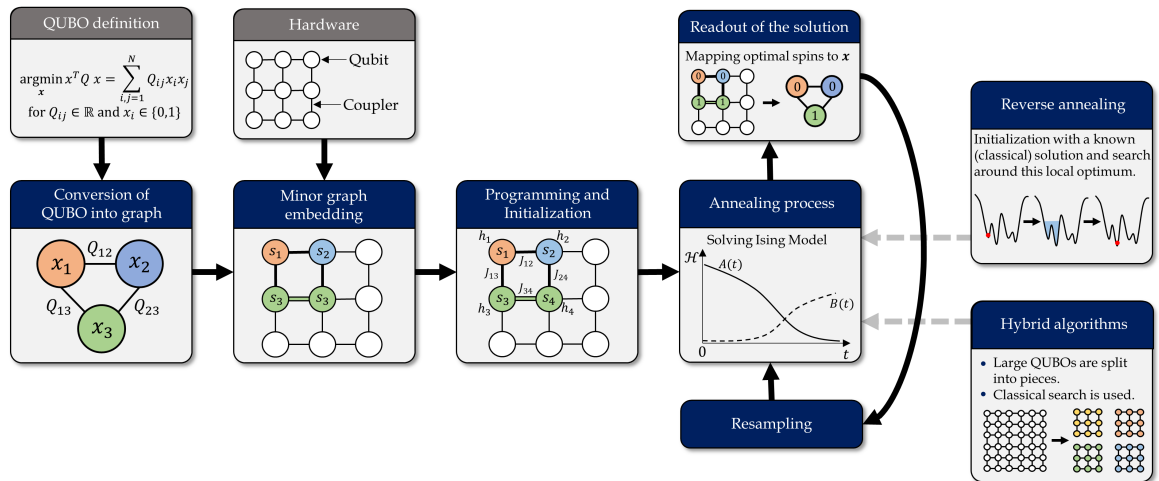


Fig. 3.9 Visualization of a typical quantum algorithm workflow on a D-Wave quantum annealer[41].

3.4.5 Quantum Annealing Parameters and System Timing

Quantum annealing relies on the principles of quantum mechanics to solve optimization problems, with the performance and quality of solutions dependent on several critical parameters[47]:

1. **Annealing Time (T_a):** This parameter sets the duration of each annealing cycle in microseconds, specified using the `annealing_time` property in the solver configuration.

T_a ranges vary based on the annealing protocol used, standard or fast (activated with the *fast_anneal* property). For standard anneals, times are rounded to two decimal places ($0.01 \mu\text{s}$), whereas for fast-anneal, times can be set to a maximum resolution of 5%.

2. **Number of Requested Samples (N_a):** N_a determines the number of sampling cycles executed by the Quantum Processing Unit (QPU). This is set using the *num_reads* property, indicating the number of output solutions to read from the solver. N_a must be a positive integer within the range specified by the *num_reads_range* property.
3. **Chain Strength:** Governs the cohesion of physical qubit chains. Setting the chain strength too high may lead to lost precision in QUBO coefficients, while too low a setting can cause broken chains. Experiments with various chain strengths, like 0.3 and 0.8, help determine the optimal value for specific problems.

Empirical exploration of the parameter space is often necessary to identify the most effective combinations of T_a , N_a , and chain strength for particular problems[45].

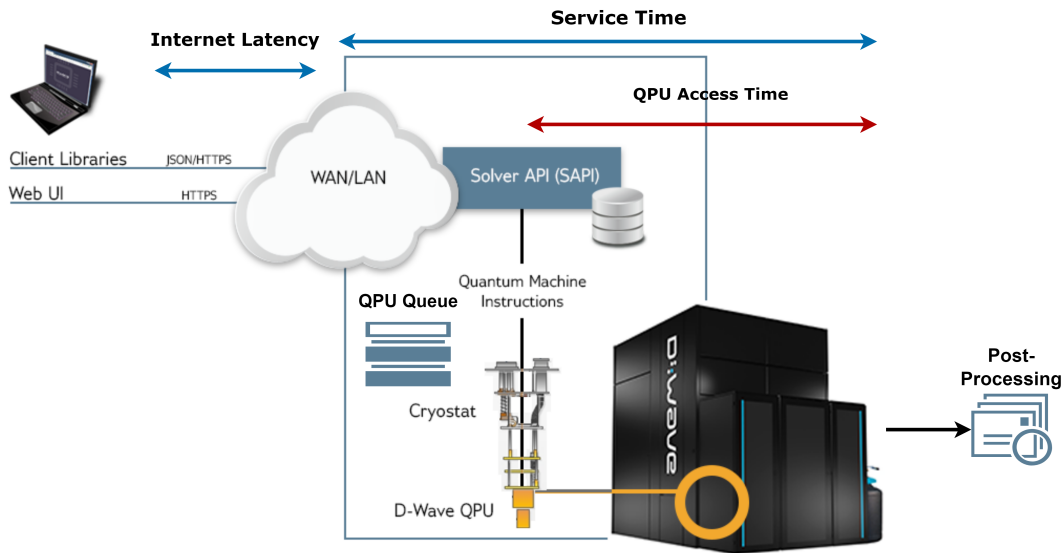


Fig. 3.10 D-Wave software environment with an Overview of execution of a single QMI, starting from a client system.

Problem Submission Process Submitting problems to a D-Wave system involves a secure connection over the internet through the Leap platform, whether running demos via the Leap website or sending problems using the Ocean cloud-client. All interactions occur through the Solver API (SAPI), which schedules and manages the problems submitted to the quantum processing units (QPUs).

Each quantum machine instruction (QMI) comprises a single input with parameters and is transmitted to the SAPI server, joining a queue. Each QMI in the queue is handled by potentially multiple parallel workers that prepare the QMI for the QPU, execute it, and post-process the results. The total service time for a QMI includes the QPU access time and additional factors such as queue wait times and post-processing. The client-observed execution time also encompasses internet latency[48].

QPU Access Time Breakdown The QPU access time is divided into:

- T_p (programming time)
- Δ (initialization time not included in the `qpu_access_time` reported by SAPI, typically 10-20 ms for Advantage systems)
- T_s (sampling time, broken down further into anneal time T_a , readout time T_r , and delay time T_d)

QPUs, as shared resources, process problems within milliseconds, typically delivering solutions back to users within seconds depending on system and network usage.

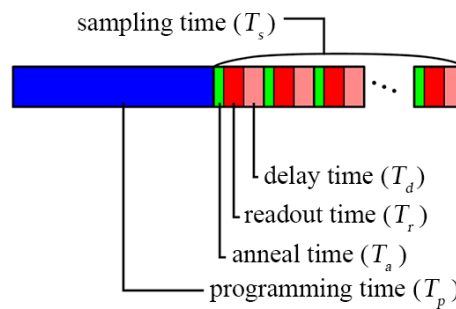


Fig. 3.11 Breakdown of QPU access time.

3.5 Application of Quantum Annealing in Communication Systems

As the field of communications continues to confront challenges associated with *heavy optimization processing in wireless networks*, quantum annealing (QA) presents a transformative approach for addressing these complexities. The technique's ability to efficiently solve optimization problems makes it particularly suitable for the demanding computations required in various aspects of wireless communications, ranging from Multiuser detection to resource allocation.

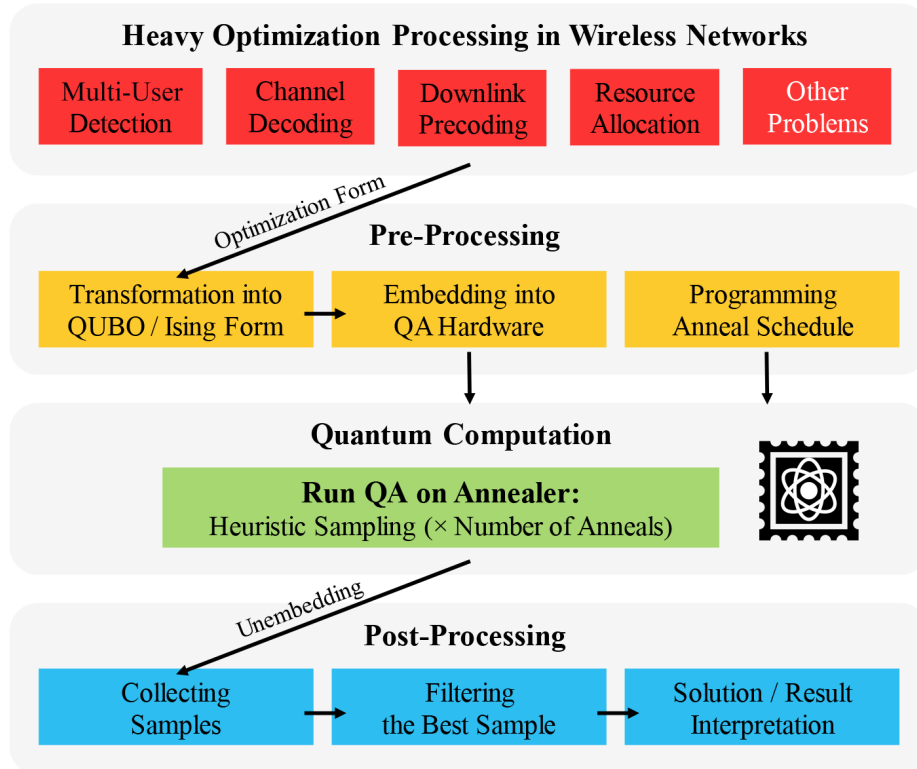


Fig. 3.12 The general workflow of QA-based optimization in wireless networks[5].

In the upcoming chapter, we will focus specifically on the application of QA in a critical area of wireless communications: multiuser detection for Non-Orthogonal Multiple Access (NOMA). NOMA plays a crucial role in enhancing spectral efficiency and supporting a

higher density of users in network environments. However, the complexity of decoding overlapping signals in NOMA presents a significant computational challenge. Traditional algorithms often struggle with efficiency, particularly as network scales increase.

By leveraging QA, we aim to harness its computational power to explore potential solutions more deeply and efficiently than is possible with classical approaches. The next chapter will explore how quantum annealing is applied to optimize multiuser detection in NOMA, examining both the theoretical and practical benefits of using this advanced computational technique in real-world scenarios.

Chapter 4

Leveraging Quantum Annealing for Maximum-Likelihood NOMA Processing in Centralized Radio Access Networks

This chapter explores the application of **Quantum Annealing (QA)** to expedite the computations required for **Non-Orthogonal Multiple Access (NOMA)** within the framework of *Centralized Radio Access Networks (C-RAN)*. We consider a scenario in which a quantum computer, integrated with C-RAN computational resources in a data center, is interconnected with Access Points (APs) via high-speed, low-latency fiber or millimeter-wave links.

Optimization has been identified as one of the key short-term applications where quantum computing may offer significant benefits, even before quantum processors evolve into scalable, error-corrected, and universally capable devices. Although the effectiveness of *Near-Intermediate-Scale Quantum (NISQ)* devices for optimization tasks remains to be conclusively proven, there is potential for these devices to achieve polynomial or exponential speedups over the best known classical algorithms. Insights from current quantum computing prototypes are crucial for designing practical systems, as real-world performance is difficult to predict or simulate, especially in the presence of device-specific noise.

This chapter outlines the application of QA to the complex task of NOMA decoding within a centralized RAN architecture. Initially, the problem of ML NOMA decoding is reduced to a format processable by a QA solver. The chapter progresses by discussing the practical implementation and performance evaluation of this approach.

Chapter Roadmap: The discussion begins with Section 4.1, which details the *system model for practical NOMA*, setting the stage for subsequent technical developments. Section 4.2 describes the *formulation of the NOMA machine learning problem into a Quadratic Unconstrained Binary Optimization (QUBO) format*, suitable for QA. Section 4.3 focuses on the *implementation* of this formulation on quantum annealing hardware, and Section 4.4 presents an *evaluation* of the system's performance, analyzing how effectively quantum annealing can optimize NOMA decoding.

By the end of this chapter, the discussions will have demonstrated the potential and practical considerations of applying quantum annealing to improve the efficiency and capability of NOMA systems in real-world telecommunications environments.

4.1 System Model

We consider a system that exemplifies an uplink non-orthogonal multiple access (NOMA) configuration involving a base station (BS) and N end-users, denoted by U_k , where $k \in \mathcal{A} = \{1, 2, \dots, N\}$. Each node, including the BS and end-users, is equipped with a single antenna. In this NOMA setup, all users simultaneously transmit their data to the BS. The received signal at the BS is a composite of all transmitted signals, modulated by their respective channel gains and added noise, expressed as:

$$y = \sum_{k=1}^N s_k h_k + n, \quad (4.1)$$

where s_k is the transmitted symbol of user k , with the transmit power defined by an average power constraint $\mathbb{E}\{s_k s_k^*\} = P_k$, assumed equal for simplicity, thus $P = P_1 = P_2 = \dots = P_N$. The wireless channel between each user U_k and the BS is denoted by h_k . These channel

gains are independent and identically distributed random variables that follow Rayleigh fading, and subject to log-distance path-loss model capturing the stochastic nature of wireless communications:

$$h_k = \frac{\tilde{h}_k}{\sqrt{d_k^\tau}}, \quad (4.2)$$

- $\tilde{h}_k$ represents the Rayleigh fading component, modeled as independent and identically distributed (i.i.d.) random variables (RVs). This component introduces the necessary randomness to account for the multipath effects and other variations in the channel.
- d_k is the distance of the k -th user from the BS, which influences the signal attenuation due to spreading and absorption over the path.
- τ is the path-loss exponent, a parameter that depends on the specific environment (e.g., urban, suburban) and dictates how quickly the signal power decreases with distance.

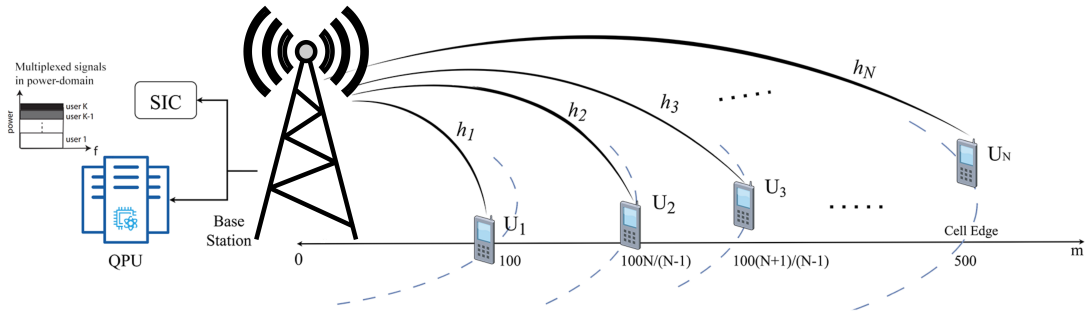


Fig. 4.1 Proposed system model for evaluation.

The term n denotes the additive white Gaussian noise (AWGN) at the BS with zero mean and variance σ_n^2 .

The power domain NOMA requires that each user's signal arrives at the BS with distinct power levels, typically ensured through varying transmission powers or path losses in the downlink. In our model of uplink NOMA, the distinction in power levels is inherently

achieved through the differences in the channel gains h_k , assuming users are geographically spaced such that their signals inherently arrive at different power levels at the BS. We consider users geographically distributed such that the closest and farthest users are located 100 m and 500 m from the BS, respectively, with others evenly spaced in between. This setup ensures that signals from different users arrive at the BS with distinct power levels.

The detection at the BS is performed using Maximum Likelihood (ML) detection to decode the received superimposed signal. The goal of ML detection is to estimate the transmitted symbols from all users by minimizing the squared difference between the received signal and the hypothesized signals based on possible transmit symbol vectors. The ML estimation for each user symbol vector $\hat{s}_1, \hat{s}_2, \dots, \hat{s}_N$ that minimizes the residual sum of squares is given by:

$$\hat{\mathbf{s}}_{ML} = \arg \min_{\hat{s}_1, \hat{s}_2, \dots, \hat{s}_N} \|\mathbf{y} - \mathbf{H}\mathbf{s}\|^2, \quad (4.3)$$

where $\mathbf{s} = [s_1, s_2, \dots, s_N]$ is the vector of transmitted symbols, and $\mathbf{H} = [h_1, h_2, \dots, h_N]$ is the vector of channel gains. We assume perfect channel estimation at the BS, i.e., the channel gains of each user are known.

Maximum Likelihood (ML) detection is particularly effective in environments with high noise and interference, as it robustly seeks the most probable transmit vector that could have resulted in the observed received signal and aims to minimize the bit error rate (BER), thus increasing throughput.

This system model provides a comprehensive framework for analyzing the performance of uplink NOMA networks under realistic channel conditions and user distributions. It forms the basis for subsequent simulations and performance evaluations discussed in the following sections, focusing on the application of quantum annealing to enhance NOMA decoding processes.

4.2 Formulation of the NOMA ML Problem in a QUBO

Model

This section outlines the transformation of the NOMA Maximum Likelihood (ML) detection problem, as previously described in the system model, into a QUBO framework suitable for Quantum Annealing. Utilizing the system model enables the mapping of the complex NOMA scenario into a computationally feasible quantum framework. This transformation is essential for harnessing the quantum computational advantages in solving the NOMA detection problem, which involves optimizing a highly nonlinear and complex objective function under real-world communication constraints.

To solve an optimization problem using quantum annealing, the objective function must be reformulated into the form of a QUBO model. Both QUBO and the equivalent *Ising model* share the same mathematical foundation, allowing for straightforward conversion between the two. The essence of this transformation lies in encoding the problem such that the solution with the lowest energy state corresponds to the optimal solution of the original problem.

The QUBO formulation is typically represented by an upper-diagonal matrix Q , where each element q_i within the matrix is a binary variable, and the relationship between these variables is governed by:

$$\hat{q}_1, \hat{q}_2, \dots, \hat{q}_M = \arg \min_{\hat{q}_1, \hat{q}_2, \dots, \hat{q}_M} \sum_{i \leq j}^M Q_{ij} q_i q_j \quad (4.4)$$

Here, M indicates the number of qubits, and the quadratic and linear coefficients are encoded within the matrix Q , with diagonal elements Q_{ii} and off-diagonal elements Q_{ij} respectively. Utilizing the property that $q_i^2 = q_i$.

The ML detection problem, specifically for NOMA, involves finding the symbol vector that minimizes the Euclidean distance between the predicted and actual received signals. To effectively handle the complexities introduced by the superposition of signals in a NOMA

system, our approach focuses on the received signal model at the base station (BS). Both the received signal y and the channel coefficients h_k are expressed in terms of their real and imaginary components due to broadcast transmission, allowing for a more tractable representation in the quantum domain:

$$y = y_R + jy_I \quad (a)$$

$$h_k = h_{k,R} + jh_{k,I} \quad (b)$$

Each symbol s_k , transmitted by user k , is then transformed into a binary qubit form. The exact representation of these symbols in binary form, and the subsequent construction of the QUBO model, depends on the type of modulation used -described by constellation points as illustrated in Figure 4.2.

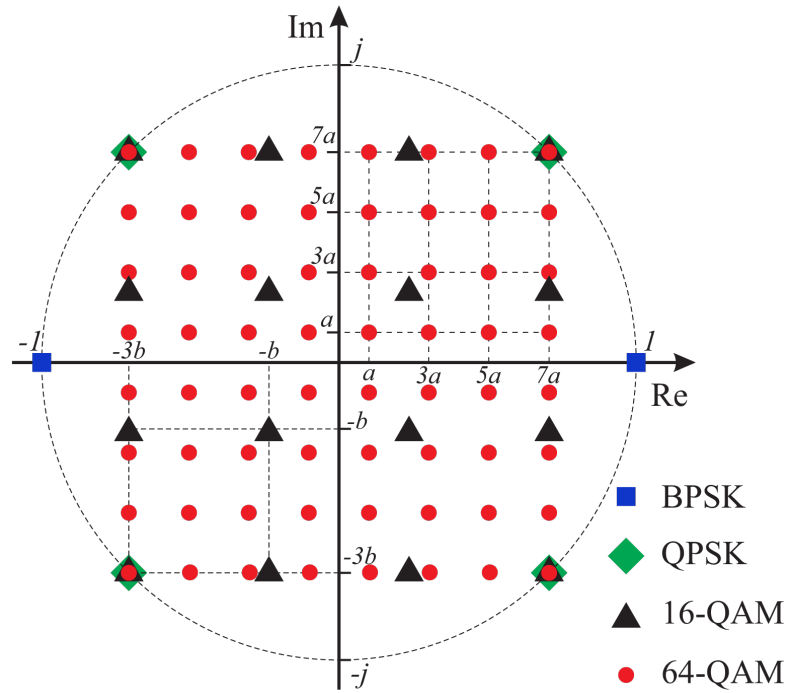


Fig. 4.2 Constellations of different modulation schemes.

We represent each of the k senders' candidate symbols $s_k \in O$ (where $1 \leq i \leq k$), with $\log_2(|O|)$ QUBO solution variables, naturally requiring $M = k \cdot \log_2(|O|)$ QUBO variables

for k transmitters, and form these QUBO variables into a vector q_i for each sender k : $q_i = [q_{(i-1) \cdot \log_2(|O|)+1}, \dots, q_{i \cdot \log_2(|O|)}]$, This directly affects the complexity and structure of the QUBO matrix[3].

The subsequent sections will detail specific transformations for different modulation schemes such as BPSK, QPSK, 16-QAM, and 64-QAM, and explore how each scheme's characteristics influence the QUBO formulation and strategies for efficient quantum annealing.

BPSK Modulation

In the BPSK modulation scheme, each symbol s_k can take values from the set $\{+1, -1\}$. These symbols can be reformulated into the QUBO form by using the binary transformation $s_k = 2q_k - 1$, where $q_k \in \{0, 1\}$ represents the binary state of user k 's symbol. Here, M in the QUBO equation corresponds to N , the number of users.

To derive the QUBO coefficients, we start by considering the objective function that arises from the maximum likelihood detection of the received signals[10], expanding the norm:

$$\begin{aligned}
 ||y - \mathbf{sH}||^2 &= \left\| y - \sum_{k=1}^N s_k h_k \right\|^2 \\
 &= \left\| y - \sum_{k=1}^N (2q_k - 1) h_k \right\|^2 \\
 &= \left\| \left(y + \sum_{k=1}^N h_k \right) - 2 \sum_{k=1}^N h_k q_k \right\|^2 \\
 &= \left(y + \sum_{k=1}^N h_k \right)^2 + 4 \sum_{k=1}^N h_k^2 q_k^2 - 4 \left(y + \sum_{k=1}^N h_k \right) \sum_{k=1}^N h_k q_k + 8 \sum_{i=1}^N \sum_{j=i+1}^N h_i h_j q_i q_j.
 \end{aligned} \tag{4.5}$$

Simplifying and rearranging this expression for a QUBO model involves breaking down the received signal y and the channel coefficients h_k into their real and imaginary parts, where $y = y_R + jy_I$ and $h_k = h_{k,R} + jh_{k,I}$:

$$\begin{aligned}
 ||y - \mathbf{sH}||^2 &= \left(y_R + \sum_{k=1}^N h_{k,R} \right)^2 + \left(y_I + \sum_{k=1}^N h_{k,I} \right)^2 \\
 &\quad + 4 \sum_{k=1}^N (h_{k,R}^2 + h_{k,I}^2) q_k \\
 &\quad - 4 \left(y_R + \sum_{k=1}^N h_{k,R} \right) \sum_{k=1}^N h_{k,R} q_k - 4 \left(y_I + \sum_{k=1}^N h_{k,I} \right) \sum_{k=1}^N h_{k,I} q_k \\
 &\quad + 8 \sum_{i=1}^N \sum_{j=i+1}^N (h_{i,R} h_{j,R} + h_{i,I} h_{j,I}) q_i q_j.
 \end{aligned} \tag{4.6}$$

¹ This leads us to the QUBO coefficients[9]:

$$\begin{aligned}
 Q_{ii}^B &= P \left[-4h_{i,R} \left(\sum_{l=1}^N h_{l,R} - h_{i,R} \right) - 4h_{i,I} \left(\sum_{l=1}^N h_{l,I} - h_{i,I} \right) \right] \\
 &\quad - \sqrt{P} (4y_R h_{i,R} + 4y_I h_{i,I}),
 \end{aligned} \tag{4.7}$$

$$Q_{ij}^B = P (8h_{i,R} h_{j,R} + 8h_{i,I} h_{j,I}). \tag{4.8}$$

QPSK Modulation

Quadrature Phase Shift Keying (QPSK) modulation uses four points on the constellation diagram, equally spaced around a circle. Each symbol in QPSK, s_k , is represented by a complex number of the form $\{\frac{\pm 1 \pm j}{\sqrt{2}}\}$, as illustrated in Figure 4.2 residing on a unit circle with 90° phase differences between consecutive symbols. This phase difference allows each symbol to be represented by two qubits. Therefore, for the formulation in the QUBO model, M becomes $2N$, where N is the number of users, effectively doubling the number of binary variables required.

¹Note that the equations involving the positive integer variables i and j must satisfy the condition $i < j$.

The representation of QPSK symbols in QUBO is given by:

$$s_k = [(2q_{2k-1} - 1) + j(2q_{2k} - 1)] / \sqrt{2} \quad (4.9)$$

The corresponding QUBO matrix coefficients for the QPSK modulation are derived from the real and imaginary parts of the received signals and the channel coefficients, following the method previously discussed for BPSK[9]:

$$\begin{aligned} Q_{(2i-1),(2i-1)}^Q = & \frac{P}{2} \left[-4h_{i,R} \left(\left(\sum_{l=1}^N h_{l,R} - h_{i,R} \right) - \left(\sum_{l=1}^N h_{l,I} - h_{i,I} \right) \right) \right. \\ & \left. -4h_{i,I} \left(\left(\sum_{l=1}^N h_{l,R} - h_{i,R} \right) + \left(\sum_{l=1}^N h_{l,I} - h_{i,I} \right) \right) \right] \\ & - \sqrt{2P} (2y_R h_{i,R} + 2y_I h_{i,I}), \end{aligned} \quad (4.10)$$

$$\begin{aligned} Q_{(2i),(2i)}^Q = & \frac{P}{2} \left[-4h_{i,R} \left(\left(\sum_{l=1}^N h_{l,R} - h_{i,R} \right) + \left(\sum_{l=1}^N h_{l,I} - h_{i,I} \right) \right) \right. \\ & \left. -4h_{i,I} \left(- \left(\sum_{l=1}^N h_{l,R} - h_{i,R} \right) + \left(\sum_{l=1}^N h_{l,I} - h_{i,I} \right) \right) \right] \\ & - \sqrt{2P} (-2y_R h_{i,I} + 2y_I h_{i,R}). \end{aligned} \quad (4.11)$$

$$Q_{(2i-1),(2i)}^Q = 0, \quad \forall i \geq 1, \quad (4.12)$$

$$Q_{(2i-1),(2j-1)}^Q = Q_{(2i),(2j)}^Q = \frac{P}{2} (8h_{i,R}h_{j,R} + 8h_{i,I}h_{j,I}), \quad (4.13)$$

$$Q_{(2i-1),(2j)}^Q = \frac{P}{2} (8h_{i,I}h_{j,R} - 8h_{i,R}h_{j,I}), \quad (4.14)$$

$$Q_{(2i),(2j-1)}^Q = \frac{P}{2} (-8h_{i,I}h_{j,R} + 8h_{i,R}h_{j,I}). \quad (4.15)$$

16-QAM Modulation

The 16-QAM modulation scheme encodes four information bits into one complex symbol, resulting in a constellation with 16 distinct points. These symbols are represented as $s_k \in \{\pm b \pm jb, \pm b \pm j3b, \pm 3b \pm jb, \pm 3b \pm j3b\}$, where $b = 1/(3\sqrt{2})$. To incorporate these symbols into the QUBO model, each symbol s_k is encoded using four qubits:

$$s_k = [(4q_{4k-3} + 2q_{4k-2} - 3) + j(4q_{4k-1} + 2q_{4k} - 3)]/3\sqrt{2}. \quad (4.16)$$

Here, M in (4.4) is extended to four times the number of users, i.e., $4N$. The coefficients for the QUBO model will be detailed in Appendix A1, following the method previously discussed.

64-QAM Modulation

64-QAM, an even more densely packed modulation scheme, encodes six bits per symbol, creating a constellation with 64 points. The symbols are defined as:

$$s_k \in \left\{ \begin{array}{cccc} \pm a \pm ja & \pm a \pm j3a & \pm a \pm j5a & \pm a \pm j7a \\ \pm 3a \pm ja & \pm 3a \pm j3a & \pm 3a \pm j5a & \pm 3a \pm j7a \\ \pm 5a \pm ja & \pm 5a \pm j3a & \pm 5a \pm j5a & \pm 5a \pm j7a \\ \pm 7a \pm ja & \pm 7a \pm j3a & \pm 7a \pm j5a & \pm 7a \pm j7a \end{array} \right\},$$

where $a = 1/(7\sqrt{2})$. Each symbol requires six qubits for representation in the QUBO model:

$$s_k = a(8q_{6k-5} + 4q_{6k-4} + 2q_{6k-3} - 7) + ja(8q_{6k-2} + 4q_{6k-1} + 2q_{6k} - 7). \quad (4.17)$$

The variable M in (4.4) increases to six times the number of users, i.e., $6N$. The detailed QUBO coefficients are included in Appendix A2.

Table 4.1 Qubit Requirements of Different NOMA Encoded Configurations

Number of Users	Modulation Type			
	BPSK	QPSK	16-QAM	64-QAM
10	10	20	40	60
20	20	40	80	120
30	30	60	120	180
40	40	80	160	240
80	80	160	320	480
100	100	200	400	600
120	120	240	480	720
140	140	280	560	840
160	160	320	640	960
180	180	360	720	1080

Given a number M of binary variables (i.e., logical qubits) in QUBO form q_i , the embedding process represents each with a chain of $\lceil M/4 \rceil + 1$ qubits, and requires a total of $M \cdot (\lceil M/4 \rceil + 1)$ qubits. Recall that $M = k \cdot \log_2(|O|)$.

Table 4.1 summarizes the size of the embedding in logical qubits, as a function of the problem’s parameters—number of users with single antenna, and modulation type. Color coding indicates feasibility, whether or not the given parameters fit into the number of qubits available on current D-Wave machines.

4.3 Implementation

The implementation of the quantum annealing (QA) method for signal decoding in a power-domain uplink Non-Orthogonal Multiple Access (UL-NOMA) system was executed using D-Wave’s Advantage QPU. The experiments took advantage of the QPU’s parallelization capabilities to optimize efficiency and reduce simulation time significantly. The annealer was programmed to execute multiple annealing cycles with consistent parameters across each cycle to collect robust statistical data. The configuration with the lowest energy from these cycles was selected as the most likely solution.

Table 4.2 D-Wave System Parameters

Parameter	Value
Annealing Time (T_{anneal})	20 μs
Total Number of Samples (N_{total})	1000
Version	D-Wave Advantage 5.2
Embedding Type	Native Clique Embedding
Chain Strength	0.3-0.8

To manage computation efficiently, the QA setup included running multiple instances simultaneously on the QPU. This parallelization substantially reduced computation time by a factor approximately given by $P_f \approx \frac{N_{\text{tot}}}{N(\lfloor N/4 \rfloor + 1)}$, where N_{tot} is the total number of instances and N is the number of qubits involved[3]. For example, a 16-qubit problem, which represents a 16-user BPSK or an 8-user QPSK, could potentially be executed over 20 times in parallel on the Advantage system.

The waveform generation and signal processing were implemented in Python, except for the parts directly executed by the QA. Conventional decoding techniques such as Successive Interference Cancellation (SIC) and a brute-force solution of Maximum Likelihood (ML) decoding were performed using MATLAB on a machine equipped with an AMDTMRyzen 7 processor, 16GB RAM, and an RTX 3060 GPU. These traditional methods provided a basis for comparing the effectiveness and efficiency of quantum annealing against established algorithms.

The annealing time, T_{anneal} , was set to 20 μ , and the total number of samples, N_{total} , was fixed at 1000 for each run. Most runs utilized Native Clique Embedding due to its ability to produce uniform chains, leading to more stable outcomes compared to heuristic embedding methods.

The decoding time was also recorded to determine the real-time computing demands for NOMA using QA and to compare these demands with other decoding methods. The results and comparisons are discussed in detail in the following evaluation section.

4.4 Evaluation

The evaluation of our quantum annealing (QA) implementation for Maximum Likelihood (ML) detection in Non-Orthogonal Multiple Access (NOMA) primarily utilized the Bit Error Rate (BER) as the main performance metric. BER is defined as the ratio of incorrectly decoded bits and is given by:

$$\text{BER}(\hat{b}) = \frac{\|b - \hat{b}\|}{n}, \quad (4.18)$$

where n is the length of the transmitted bit string b , and $\|b - \hat{b}\|$ represents the number of bits that differ between the transmitted bit string b and the decoded bit string $\hat{b}$. This metric was pivotal in assessing the QA's effectiveness in managing the complexities inherent in multi-user detection for PD-NOMA systems. An increase in problem complexity, indicated by higher numbers of users or larger constellations, was expected to yield higher BERs.

A substantial number of simulation runs, in the order of thousands, were executed to ensure statistical significance of the results. The experiments spanned a variety of symmetric user setups typical in commercial telecommunications to confirm the practical relevance of the findings. The quantum annealer was programmed to execute multiple annealing cycles with uniform parameters to compile a robust dataset, where the configuration with the lowest energy among these cycles was considered the optimal solution, the process is outlined in Algorithm 1.

BER evaluations were conducted against two main variables: the number of users and the transmit power for different user distances. This methodical approach allowed thorough examination of the QA's scalability and performance under varying configurations. Additionally, the performance of the QA-based ML detection was benchmarked against conventional decoding techniques, such as Successive Interference Cancellation (SIC) and brute-force ML decoding performed on MATLAB.

Furthermore, decoding times were meticulously recorded to evaluate the real-time computational demands of NOMA using quantum annealing, highlighting its practical applicability in live telecommunications environments. Results from these comprehensive tests demon-

strated that the advanced QPU architectures not only accommodate larger problem sizes but also enhance performance for smaller configurations, validating our hypothesis and underscoring the potential of quantum computing to elevate NOMA systems.

Algorithm 1 Simulation of NOMA on Quantum Annealer

```

1: Input: Number of users  $N$ , Modulation type, Transmit power  $P$ 
2: Output: Bit Error Rate (BER)
3: procedure SIMULATENOMA( $N$ , Modulation,  $P$ )
4:   Generate bits according to  $N$  and Modulation type
5:   Generate channel gains  $h$  with i.i.d. Rayleigh fading
6:   Calculate path loss based on users' distances
7:   Generate AWGN  $n$  with specified variance
8:   Compute received signal  $y$  at BS using  $h$ , path loss,  $P$ , and  $n$ 
9:   Calculate QUBO coefficients using  $h$ ,  $y$ , and  $P$ 
10:  if Number of users and modulation type are unchanged then
11:    Use existing embedding
12:  else
13:    Generate embedding using MinorMiner
14:  end if
15:  Send coefficients and embedding to D-Wave for annealing
16:  Obtain SampleSet from D-Wave QPU
17:  Select first sample from SampleSet (lowest energy)
18:  Transform QUBO variables back to bits or symbols
19:  Compare the decoded bits/symbols against the original
20:  Calculate error and accumulate over many iterations
21:  Compute BER
22:  return BER
23: end procedure

```

4.4.1 Understanding Empirical Results

When running a problem on a quantum annealer, the machine returns a sample set containing solutions ranked by their energy levels, with the lowest energy solution being the most optimal. The number of reads performed during the quantum annealing process determines the size of the sample set. Each solution in the sample set may occur multiple times, as the same solution can be found in different reads.

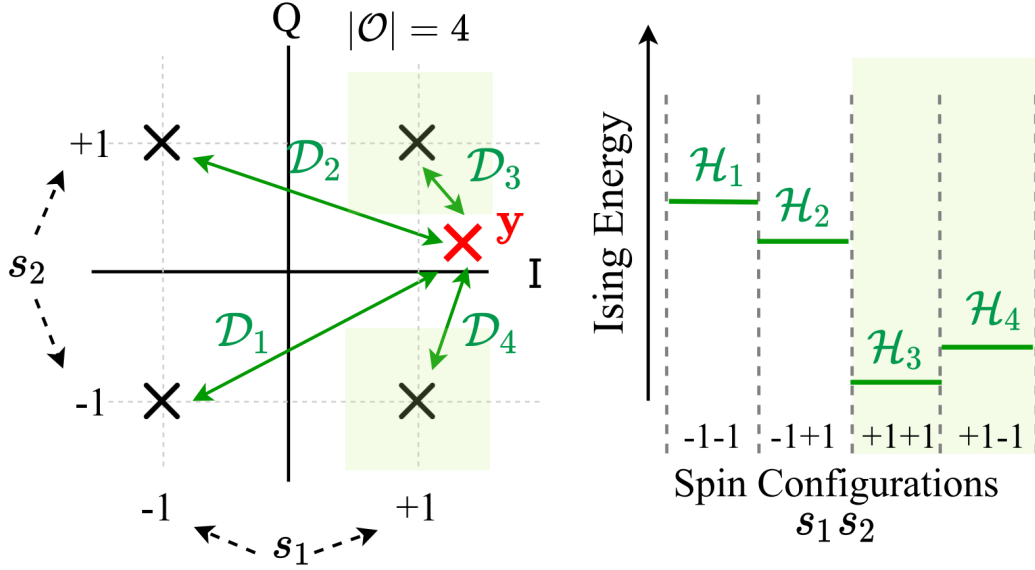


Fig. 4.3 Equivalent representations of 1×1 QPSK detection: Euclidean distances $\mathcal{D}(v)$ in the I-Q plane (left), and Ising energies $\mathcal{H}(s)$ (right). Shadings highlight likely solutions.

Among these, multiple occurrences of a certain spin configuration (with agreeing spin variables) are very likely to be observed, which could be used to identify spins easy (or hard) to detect (i.e., variables that are very likely to be assigned a certain value in the unknown optimal ML solution). Figure 4.3 shows an illustrative example of detecting a received 1×1 QPSK signal y in two equivalent representations, one in the I-Q plane with Euclidean distance $\mathcal{D}$ (left), and the other with Ising energies $\mathcal{H}(s)$ (right). In this example, the first spin variable s_1 (corresponding to the symbol's real part) is likely to be detected as $+1$ for most PMIS runs, since the difference in Ising energy from all configurations that have $s_1 = +1$ (resulting in $\mathcal{H}_3$ or $\mathcal{H}_4$ in Figure 4.5, right) and $s_1 = -1$ (resulting in $\mathcal{H}_1$ or $\mathcal{H}_2$ in Figure 4.3, right) is significant. The spin s_1 is easy to detect compared to spin s_2 (corresponding to the symbol's imaginary part). Multiple occurrences of QA runs agreeing on $s_1 = +1$ indicate this, while QA runs will disagree on the value of s_2 , because the two most frequent spin configurations' energies ($\mathcal{H}_3$ and $\mathcal{H}_4$) themselves disagree on the value of s_2 . This phenomenon is even more pronounced in higher-order modulations like 16-QAM, where each symbol is represented by multiple spins. In such cases, the mapping complexity increases as each spin or group of

spins corresponds to different segments of the encoded symbol. For instance, odd and even-numbered spins might determine different quadrants and positions within these quadrants in the symbol constellation. This differentiation helps in identifying which spins are more susceptible to noise and which are more robust, thereby aiding in more accurate symbol reconstruction.

The D-Wave inspection tool provides valuable insights into the quantum annealing results. It allows users to view the histogram of the sample set, which displays the distribution of solutions and their corresponding energy levels. This histogram helps in understanding the frequency of occurrence for each solution and the relative energy differences between them. Furthermore, the inspection tool visualizes how the problem is embedded in the quantum processing unit (QPU) as visualized in Figure 4.4. As explained earlier, the logical qubits of the problem are mapped onto the physical qubits of the QPU, taking into account the connectivity constraints of the hardware. The inspection tool provides a graphical representation of this embedding, enabling users to see how the problem is physically represented on the QPU.

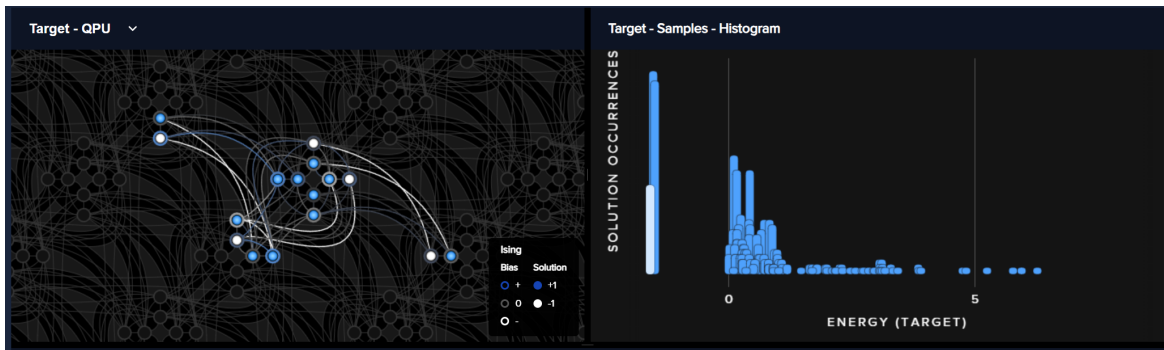


Fig. 4.4 Dwave problem inspector displaying the embedded problem(left), and the returned energies histogram(right), for the case of 10-user BPSK.

In addition to the histogram and embedding visualization, the inspection tool also provides execution timing information. This includes details such as the total execution time, the time spent on problem initialization, the actual annealing time, and the time taken for readout. These timing metrics can be useful for analyzing the performance of the quantum annealing process and identifying potential bottlenecks.

4.4.2 Bit-Error Rate Performance for QA Based Maximum-Likelihood Versus. Number of Users Across Various Modulation Schemes

In Figure 4.5, The impact of user density per cell on the Bit Error Rate (BER) when using Quantum Annealing (QA) for maximum likelihood decoding across various modulation schemes is evaluated within a Non-Orthogonal Multiple Access (NOMA) system. The test conditions were standardized with the transmit power set at 10 dBm and the Additive White Gaussian Noise (AWGN) power fixed at -40 dBm. As expected, the average BER

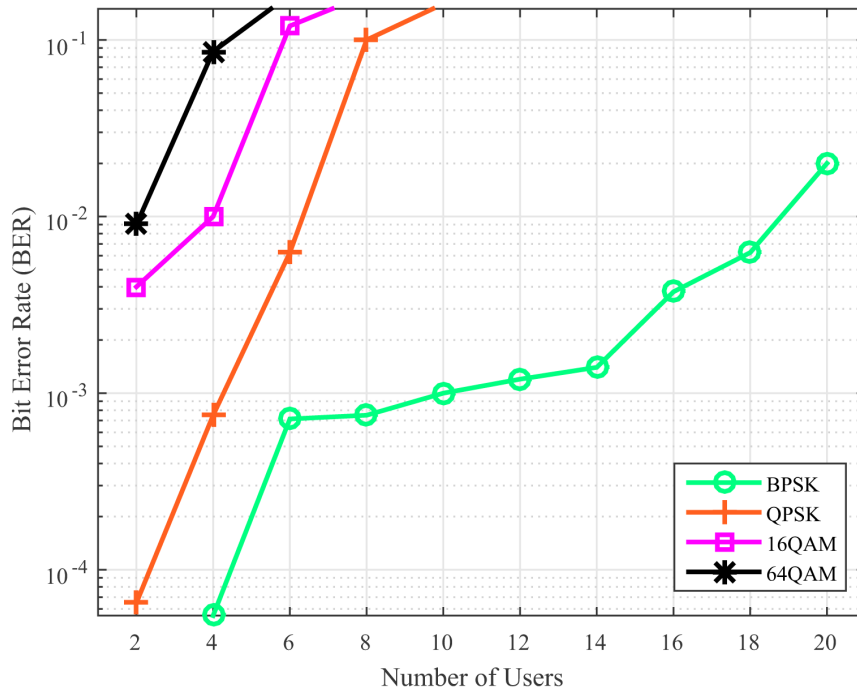


Fig. 4.5 BER vs. the number of users per cell under various modulation schemes

performance degrades with the increase in the number of users per cell. Initially, at lower user densities, The proposed QA ML decoder demonstrates a capability to maintain relatively low BER values, suggesting effective decoding when the system is under lighter loads. However, as the number of users per cell increases, there is a noticeable increase in BER across all modulation schemes, indicating the growing complexity that QA needs to manage for decoding superposed signals from multiple users.

BPSK, being the simplest modulation, shows the least increase in BER as the number of users grows, maintaining an average BER of 10^{-3} until the user count exceeds 16. This points to a more efficient decoding by the proposed QA Maximum-Likelihood under BPSK modulation at higher user densities compared to more complex modulations. QPSK, while performing comparably to BPSK for fewer than 6 users, also begins to show performance decline as the number of users increase. Conversely, complex schemes like 16QAM and 64QAM achieving an average BER exhibit rapid increases in BER with just a small addition of users, highlighting their sensitivity to decoding errors under QA due to their denser signal constellations.

The performance of the QA ML decoder is highlighted by its ability to maintain relatively low BER for a reasonable number of users in simpler modulation schemes. However, for higher-order modulation schemes, the BER increases rapidly, suggesting a limitation in handling more complex modulation with an increasing number of users. By setting a threshold BER for outage probability at say 10^{-2} , we can strategically determine an optimal number of users for each modulation type for further detailed analysis. For instance, BPSK can support up to 16 users, and QPSK can support up to 6 users while maintaining the acceptable BER. Higher-order modulation schemes like 16QAM and 64QAM can not achieve acceptable BER for more than 3 users.

Table 4.3 Simulation Parameter Settings

Parameter	Value
Modulation Schemes	BPSK, QPSK, 16-QAM, 64-QAM
Distance from User to Base Station	100 – 500 m
Channel Fading	Rayleigh
Pathloss Model	Log-distance
Environment	Urban cellular radio
Pathloss exponent	2.7
Transmit Power	-30 – 30 dBm
AWGN Noise Power	-30 dBm
Number of Users per Cell	2 – 20

4.4.3 Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with BPSK modulation

After analyzing the previous plot for BER versus the number of users, A 10 user scenario was selected for detailed analysis based on the performance degradation observed beyond this point, particularly under BPSK modulation. The decoding effectiveness of Maximum Likelihood (ML) brute force solution BF, ML Quantum Annealing (QA), and Successive Interference Cancellation (SIC) in terms of BER performance across varying transmit powers were compared against each-other. with the noise power set to -30dbm QA was simulated for particular transmit power levels, i.e., $P = \{-30, -20, -10, 0, 10, 20, 30\}$, other simulation parameters showcased in Table 4.3

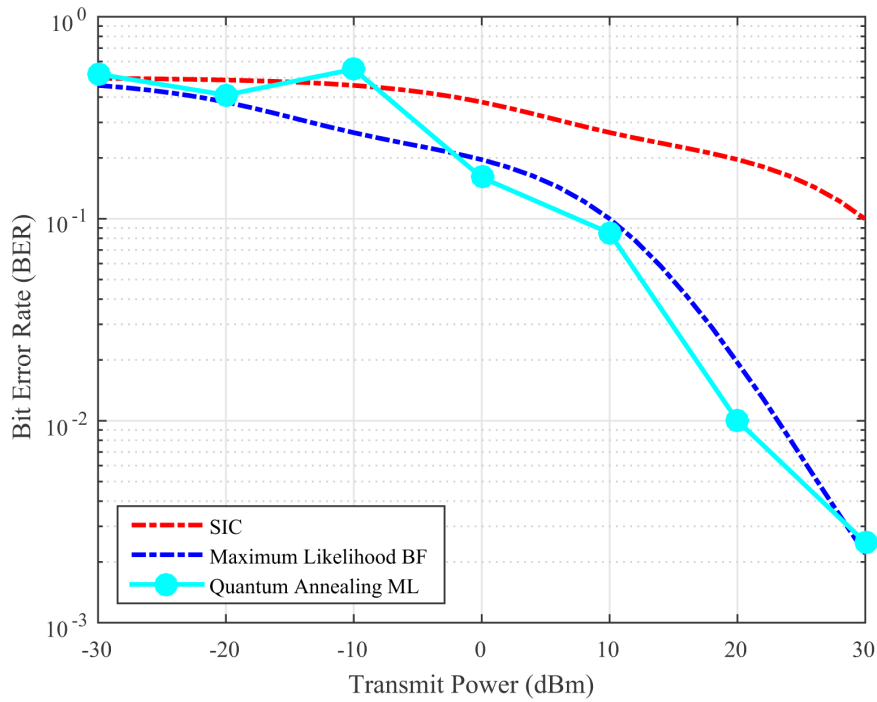


Fig. 4.6 The average BER performance vs. the transmit power for a 10 user NOMA scenario employing BPSK, with different decoding techniques.

Figure 4.6 illustrates the BER performances of each decoding method. The BER curves can be characterized by different behaviour. The findings show that the 3 methods performances coincide up to -20 dBm, and then the brute force ML starts outperforming both SIC and QA

over a short transmit power range (i.e., 15 dbm to -5 dbm). However, afterwards, SIC starts experiencing noticeable performance degradation while ML brute force consistently reduces BER with increased power, demonstrating robust decoding, ML QA improves significantly at higher power levels, providing identical or even lower BER than BF. In general, the QA, BF and SIC curves follow the same trend till 10 dBm, but after this level, the SIC performance degrades substantially, while the BER result of QA is approximately the same as BF. The reason for the SIC's low performance could be insufficient differences between the end-users' power levels.

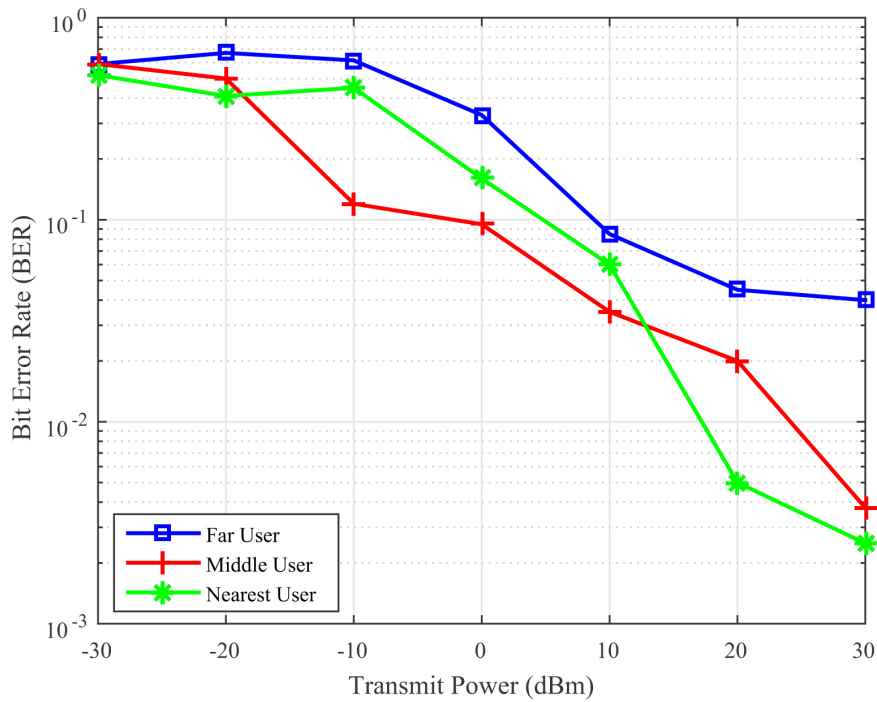


Fig. 4.7 The BER performance vs. the transmit power per user at different cell positions—near, middle, and far— for BPSK with Quantum Annealing based maximum likelihood decoder.

Figure 4.7 shows in-depth analysis of ML QA performance employing BPSK modulation for each user at different cell positions—near, middle, and far— the plot revealed that the 3 users have similar performance upto -25 dbm, the middle user performance enhances considerably, while the far and near user follow the same trend. A sharp decrease in BER

is noticed the near user performance after 10 dbm power, while the Far user demonstrates a slowly improving trend and the middle user exhibits a steady improvement in BER as the transmit power increases. Overall, while both near and far users experience better performance in terms of BER, the difference in improvement between them is small.

4.4.4 Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with QPSK modulation

For QPSK, a simulation identical to that conducted for BPSK was performed for 5-user cell. In Figure 4.8, the comparison reveals that SIC, BF, and QA perform similarly up to approximately -10 dBm. After this threshold, BF begins to outperform the other methods. QA and SIC maintain similar BER levels until 0 dBm, after which QA experiences a substantial improvement, reaching parity with BF at 10 dBm. Meanwhile, SIC starts to experience noticeable performance degradation.

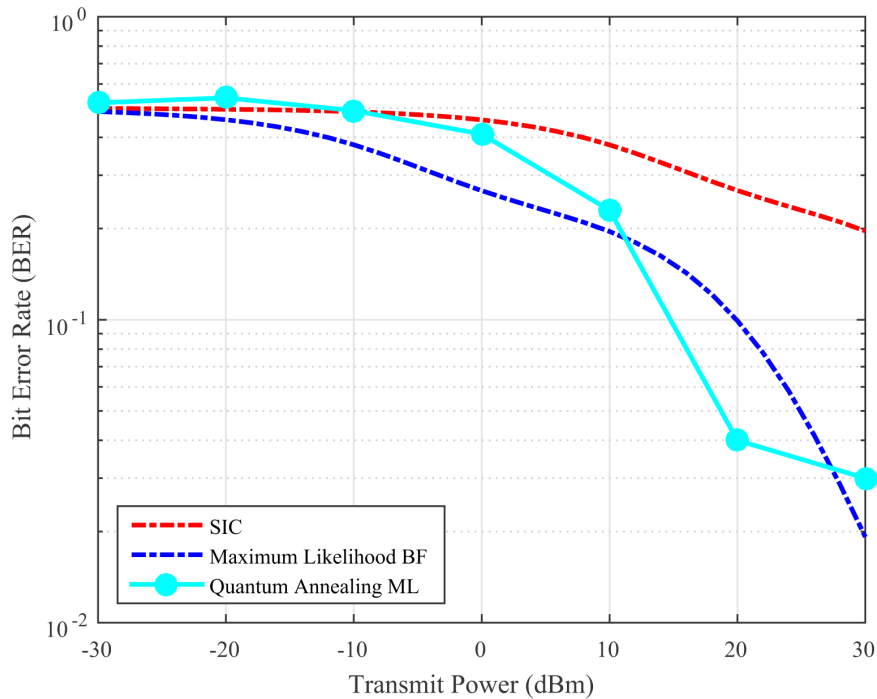


Fig. 4.8 The average BER performance vs. the transmit power for a 5 user NOMA scenario employing QPSK, with different decoding techniques.

the performance of the ML QA decoder employing QPSK modulation for users located at different positions within a cell is depicted in Figure 4.9. Initially, all users types exhibit comparable BER performance up to approximately -20 dBm of transmit power. As the transmit power increases beyond this point, the performance begins to diverge.

For the far user, the BER gradually improves with increasing transmit power, indicating a slow but steady enhancement in performance. The BER remains relatively high until about 10 dBm, where a more pronounced improvement is observed.

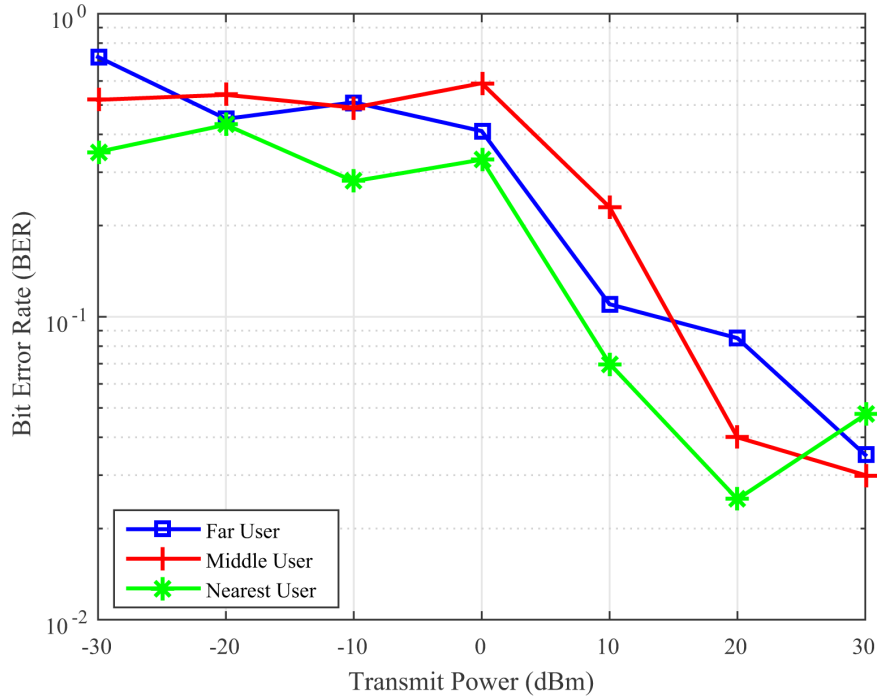


Fig. 4.9 The BER performance vs. the transmit power per user at different cell positions—near, middle, and far- for QPSK with Quantum Annealing based maximum likelihood decoder.

The middle user, shows a consistent improvement in BER as the transmit power increases. This user's performance begins to diverge slightly from the far user's around -20 dBm, with a more noticeable improvement emerging after 10 dBm, eventually aligning closely with the far user's trend. The near user, experiences a significant reduction in BER after 10 dBm transmit power, indicating a substantial enhancement in performance. This user maintains

the lowest BER among the three, improving more rapidly compared to the middle and far users.

Overall, the analysis reveals that the near user demonstrates the most substantial improvement with increasing transmit power, followed by the middle user, and finally the far user. Although the performance of the users is almost identical at high transmit power e.g 30 dbm.

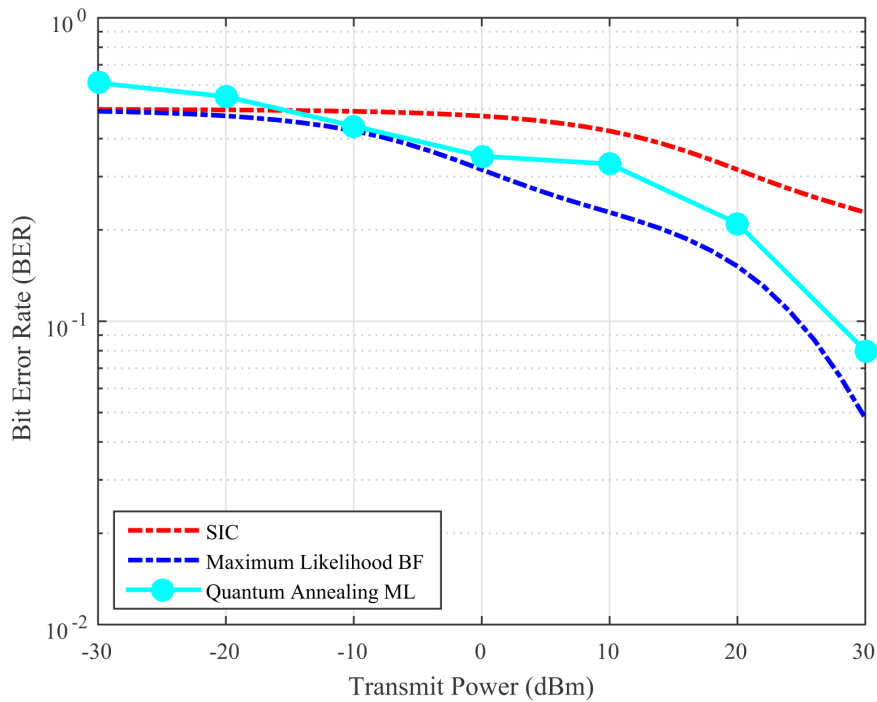


Fig. 4.10 The average BER performance vs. the transmit power for a 3 user NOMA scenario employing 16-QAM, with different decoding techniques.

4.4.5 Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with 16-QAM modulation

Based on Figure 4.5, a 3-user cell was chosen for testing 16-QAM performance. Figure 4.10 demonstrate the performance of the proposed QA method against BF and SIC employing 16-QAM modulation for the system. Initially, the SIC and BF provide identical and lower BER than QA until -15dbm, afterwards Consistent improvement in BER is observed for QA method coinciding with BF. At higher transmit power ML brute force solution clearly

outperform both SIC and QA, although the gap between QA and ML is marginal. similar to BPSK and QPSK, SIC curve enhances minimally until 15 dBm with the following saturation after this point.

Similarly, A thorough breakdown of the BER performance using the ML QA decoder with 16-QAM for users positioned at different cell locations is presented in Figure 4.11. the 3 users at diffrenet locations having different path loss experience almost the same BER until about 10 dbm. After that, the difference between them begins to grow considerably. The performance of the near user improves remarkably, where the middle user curve enhances minimally. The far user exhibit the same overall trend but has the worst performance of the 3.

In general, the BER performance of QA maximum likelihood 16-QAM is comparable with the bruteforce solution.

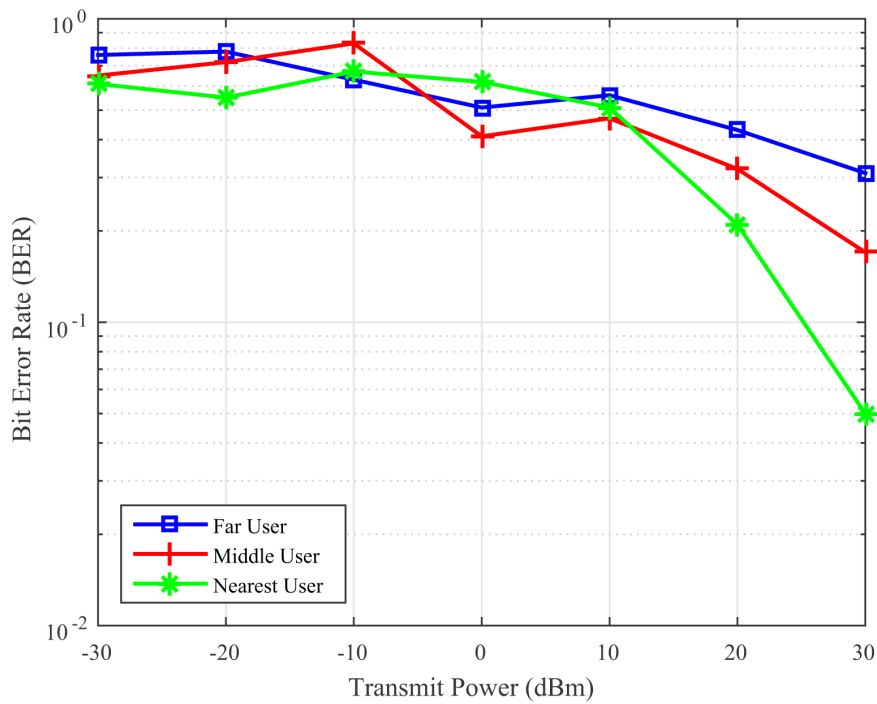


Fig. 4.11 The BER performance vs. the transmit power per user at different cell positions—near, middle, and far- for 16-QAM with Quantum Annealing based maximum likelihood decoder.

4.4.6 Bit-Error Rate Performance for QA Based Maximum-Likelihood detection with 64-QAM modulation

In this study, we chose a scenario with only two users for the evaluation of 64-QAM modulation. This decision was based on the performance curve of the ML QA decoder versus the number of users, which indicated unacceptable BER levels for more than two users. Additionally, the brute force (BF) solution for ML detection in the case of 64-QAM is computationally intensive, as it must evaluate all possible symbols, resulting in considerably long processing times for high user count.

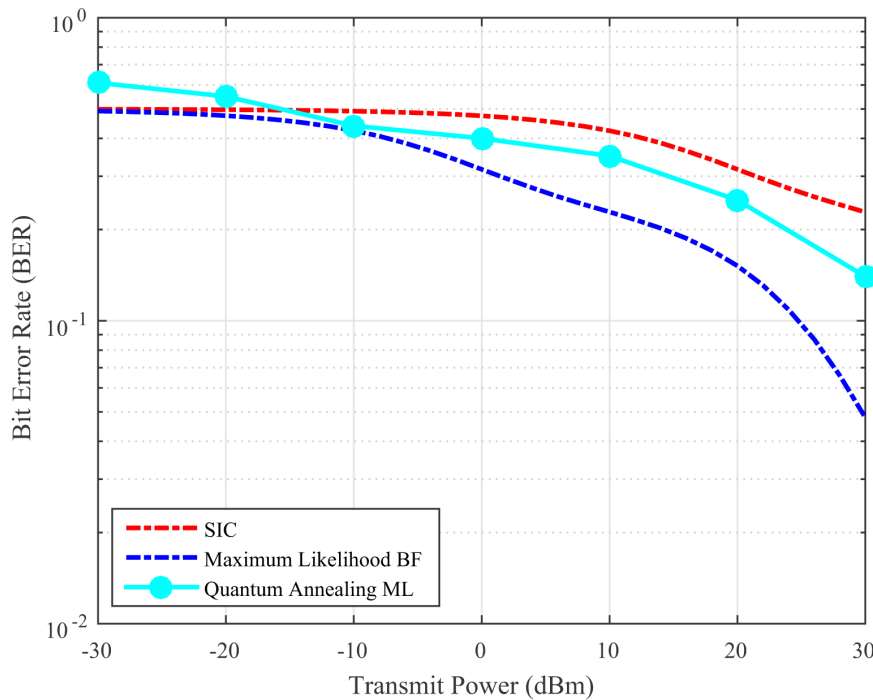


Fig. 4.12 The average BER performance vs. the transmit power for a 2 user NOMA scenario employing 64-QAM, with different decoding techniques.

in Figure 4.12 First, we compared the three methods—BF, SIC, and QA—against each other. The results show a clear advantage of BF across the entire transmit power range. BF consistently provides the lowest BER, demonstrating superior performance. In contrast, SIC suffers from performance degradation, delivering subpar results. Initially, QA exhibits a

higher BER compared to the other methods. However, as the transmit power increases, QA's performance begins to improve. Although QA performs better than SIC, there remains a significant gap between QA and BF.

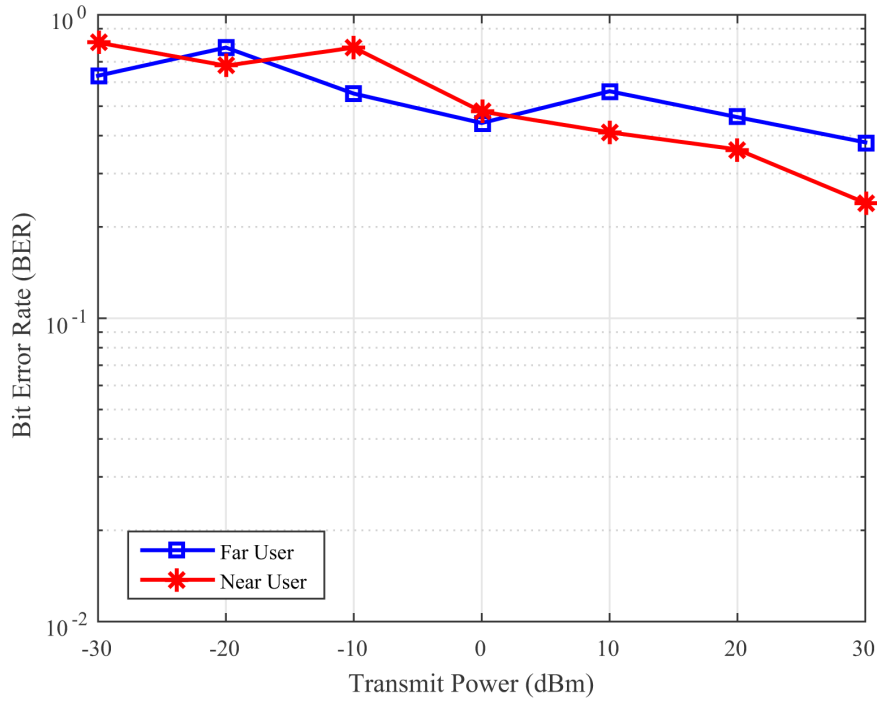


Fig. 4.13 The BER performance vs. the transmit power per user at different cell positions—near, and far- for 64-QAM with Quantum Annealing based maximum likelihood decoder.

Unlike the previous analysis that averaged BER across users, in this evaluation, the BER of each user was assessed independently in Figure 4.13. The far user, positioned at 500 meters, and the near user, positioned at 100 meters, displayed mixed performance initially. Up to 0 dBm, the far user demonstrated slightly better or identical performance compared to the near user. However, beyond this point, the near user began to recover. The BER for the near user improved in a slow yet steady manner, ultimately providing better BER at higher transmit power levels. In summary, while BF maintains the best performance in terms of BER across all transmit power levels, QA shows promise by improving with increased

power, while still lagging behind BF. SIC, on the other hand, fails to deliver competitive performance, highlighting its limitations in the context of 64QAM modulation.

4.4.7 Comparison of Computation Time

Figure 4.14 presents the timing data for each technique. From the bar graph, it is evident that SIC exhibits the fastest execution time among the techniques presented. As expected, the second fastest technique is BF, as it needs to iterate over all possible combinations.

The QA execution time interval consists of several timing sub-intervals, including T_p , T_A , T_D , and T_R which denotes the different timing of QPU access time as explained in subsection 3.4.5. While T_p and T_A remain constant across all cases, T_R and T_D shows slight variations. This variation arises because the reading time depends on the position of the exploited physical qubits on the chip and the number of qubits used (more time is needed for a larger number of qubits). the total post-processing time Δ is too low and can be neglected in most cases.

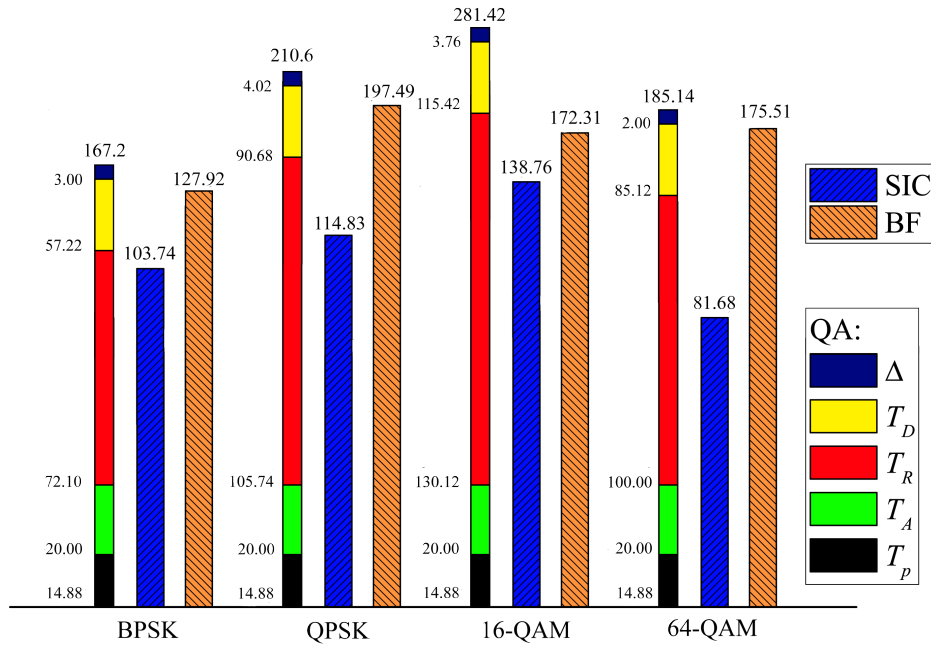


Fig. 4.14 Comparison between the decoding techniques in terms of simulation time for different modulations (in ms)

The brute-force solution of ML decoding method, which involves an exhaustive search over all possible signal combinations, also requires considerable computation time. Although it is less time-consuming than QA, ML decoding is still computationally expensive due to the sheer volume of potential combinations it needs to evaluate, especially as the number of users and modulation complexity increase.

Among the three methods, SIC stands out as the fastest. This method leverages the differences in signal power to sequentially decode and cancel out the contributions of the strongest signals first, thus progressively simplifying the decoding of remaining signals. Due to its iterative nature and the avoidance of an exhaustive search, SIC significantly reduces the computation time required for signal decoding in NOMA systems.

Overall, the total time taken for QA execution is about 2 times more than the SIC simulation. while BF is marginally faster than QA and is comparable. It is important to note that QA needs to simulate the same problem R number of times (set to 1000 in our case). However, this problem can be addressed by employing a parallelization method. Therefore, one could potentially conclude that the overall QA simulation time can be decreased by simulating multiple problem instances simultaneously, while the.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

NOMA has been widely recognized as a potential multiplexing scheme for next-generation networks due to its ability to improve spectrum efficiency. However, the implementation complexity of SIC receiver at the NOMA receiver has limited its full adoption in current 5G standards. As the number of users increases, SIC's latency and reliability performance deteriorates significantly. The error made in each iteration propagates through other successive iterations, and the power levels of each signal must be sufficiently different for successful detection. Various techniques such as user pairing/grouping, user scheduling, and multichannel transmission have been proposed to mitigate these issues. However, these approaches often consume additional network resources and somewhat contest the idea of transmission on overlapping resources on which the entire NOMA concept relies.

Maximum Likelihood (ML) detection is the optimal method for signal detection but is computationally heavy and often unfeasible on classical computers due to its NP-hard nature. In this thesis, we explored the QA-aided ML decoder for multi-user NOMA networks as an alternative to the SIC method. In our envisioned scenario, quantum processors will be co-located with C-RAN computational resources in a data center, partitioning the compute

roles for hundreds of base stations or Remote Radio Heads (RRH) connected via low-latency optical fiber. Here, quantum computing will handle computationally demanding tasks.

Towards this objective, we first reformulated the signal estimation problem as a quadratic unconstrained binary optimization (QUBO) problem and ran it using D-Wave's libraries on the newly released Advantage 5.2 QPU. We formulated the ML detection problem and derived the QUBO coefficients for different modulation schemes, including BPSK, QPSK, 16-QAM, and 64-QAM, enabling their solution using a QPU. To assess performance, we simulated a practical scenario involving a single base station, considering practical system parameters such as channel randomness and possible transmit power levels for all individual signals of all involved users.

In the obtained results, the QA-assisted ML decoder achieved similar Bit Error Rate (BER) performance as the brute force (BF) method and outperformed the SIC technique. Specifically, for BPSK and QPSK, the QA-aided decoder maintained low BER across various transmit power levels. For 16-QAM and 64-QAM, the QA-assisted decoder showed significant improvement in BER compared to SIC, but slightly higher BER compared to BF especially under higher modulation complexities. However, QA's execution time was longer than both BF and SIC.

Even though the annealing time is in the order of microseconds, the total processing time for QA is much higher than classical computing due to the time required for sampling and readout. However, this can be enhanced as quantum engineering advances in superconducting qubit technology. Despite the relatively high programming time, QA presents an advantage over classical approaches for problems that are infeasible for the latter. QA does not require trying all possible cases to find the minimum distance solution.

Our performance results establish a baseline for future fully integrated systems within the context of the centralized RAN architecture. We show that once engineering efforts optimize the integration between quantum and conventional computation, quantum computation should be considered a competitive technology for the future design of high-capacity wireless networks.

5.2 Future Work

- In this project, the proposed method used forward annealing. Future work could employ reverse annealing to enhance solutions. Reverse annealing initiates from a programmable classical state, enabling a quasi-nonlocal search by increasing quantum fluctuations, which refine the search around the initial state. This method could synergize conventional classical detectors with quantum annealing-based detectors, allowing reverse annealing to probabilistically enhance initial solutions, akin to error correction using quantum effects.
- Exploration of more efficient embedding techniques, such as manual embedding, is recommended. Adjusting parameters like the number of reads, chain strength, and annealing time may better suit latency-sensitive tasks. Employing post-processing techniques, such as Single Qubit Correction (SQC), could also enhance results by iteratively optimizing each solution bit to lower Quantum Annealing energies.
- Investigating the use of gate-model quantum hardware for Maximum-Likelihood decoding is advised. Despite current devices' limitations in qubit numbers, future enhancements in quantum hardware might facilitate larger-scale experiments. Gate-model systems provide additional controls that could improve performance. Furthermore, exploring algorithms like the Quantum Approximate Optimization Algorithm (QAOA) and Grover's search algorithm may prove beneficial, despite the current qubit limitations.
- Future work should explore and compare hybrid, quantum-inspired, and classical solutions, such as D-Wave's Leap Hybrid service, which can handle up to 20,000 QUBO variables using problem partitioning. Other technologies like Coherent Ising Machines (CIM)¹, digital annealing, and simulated annealing, offer potential quantum

¹A CIM is a network of optical parametric oscillators (OPOs) programmed to solve Ising model-based problems, representing competitive interactions in magnetic systems.

advantages on classical platforms, especially in environments where quantum hardware may not be available.

- Exploring the integration of NOMA with other wireless technologies—such as MIMO-NOMA, cooperative relaying, mmWave communications, beamforming, network coding, and advanced error correcting codes like LDPC and turbo codes—promises significant performance improvements and is a promising direction for future work.
- Upcoming investigations should consider the performance of QA-based ML detection in NOMA systems under imperfect channel state information (CSI) and challenging channel conditions, possibly modeled using ray tracing techniques.
- Future research should extend the application of QA to other NP-hard optimization problems in wireless communications, such as resource allocation, precoding, and scheduling.

References

- [1] P. Yang, Y. Xiao, M. Xiao, and S. Li, “6g wireless communications: Vision and potential techniques,” *IEEE Network*, vol. 33, no. 4, pp. 70–75, 2019.
- [2] International Technology Roadmap for Semiconductors (ITRS), “International technology roadmap for semiconductors 2.0, executive report.” Online, 2015. April, 2024.
- [3] M. Kim, D. Venturelli, and K. Jamieson, “Leveraging quantum annealing for large mimo processing in centralized radio access networks,” in *Proceedings of the ACM Special Interest Group on Data Communication, SIGCOMM ’19*, ACM, Aug. 2019.
- [4] S. Kasi and K. Jamieson, “Towards quantum belief propagation for ldpc decoding in wireless networks,” in *Proceedings of the 26th Annual International Conference on Mobile Computing and Networking, MobiCom ’20*, ACM, Sept. 2020.
- [5] M. Kim, S. Kasi, P. A. Lott, D. Venturelli, J. Kaewell, and K. Jamieson, “Heuristic quantum optimization for 6g wireless communications,” *IEEE Network*, vol. 35, no. 4, pp. 8–15, 2021.
- [6] S. Kasi, P. A. Warburton, J. Kaewell, and K. Jamieson, “A cost and power feasibility analysis of quantum annealing for nextg cellular wireless networks,” 2022.
- [7] S. Namba, T. Warabino, and S. Kaneko, “Bbu-rrh switching schemes for centralized ran,” in *7th International Conference on Communications and Networking in China*, pp. 762–766, 2012.
- [8] Y. Liu, Z. Qin, and Z. Ding, *Non-Orthogonal Multiple Access for Massive Connectivity*. Springer, 2020.
- [9] E. Gabdulsattarov, K. Rabie, X. Li, and G. Nauryzbayev, “Towards quantum annealing for multi-user noma-based networks,” 2023.
- [10] R. C. Kizilirmak, “Quantum annealing approach to noma signal detection,” in *2020 12th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP)*, pp. 1–5, 2020.
- [11] Cisco, “Cisco annual internet report (2018–2023) white paper,” Mar 2020. [Online]. Available: <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>.
- [12] “6g flagship,” 2021. Accessed: April, 2024. [Online]. Available: <https://www.oulu.fi/6gflagship/>.

References

- [13] Z. Zhang *et al.*, “6g wireless networks: Vision, requirements, architecture, and key technologies,” *IEEE Vehicular Technology Magazine*, vol. 14, pp. 28–41, Sept. 2019.
- [14] Y. Liu, C. Ouyang, Z. Ding, and R. Schober, “The road to next-generation multiple access: A 50-year tutorial review,” 2024.
- [15] Y. Chen, A. Bayesteh, Y. Wu, B. Ren, S. Kang, S. Sun, Q. Xiong, C. Qian, B. Yu, Z. Ding, S. Wang, S. Han, X. Hou, H. Lin, R. Visoz, and R. Razavi, “Towards the standardization of non-orthogonal multiple access for next generation wireless networks,” 2018.
- [16] Y. Liu, Z. Qin, M. El Kashlan, Z. Ding, A. Nallanathan, and L. Hanzo, “Non-orthogonal multiple access for 5g and beyond,” 2018.
- [17] S. M. R. Islam, N. Avazov, O. A. Dobre, and K.-s. Kwak, “Power-domain non-orthogonal multiple access (noma) in 5g systems: Potentials and challenges,” *IEEE Communications Surveys Tutorials*, vol. 19, no. 2, 2017.
- [18] Y. Liu, S. Zhang, X. Mu, Z. Ding, R. Schober, N. Al-Dhahir, E. Hossain, and X. Shen, “Evolution of noma toward next generation multiple access (ngma) for 6g,” *IEEE Journal on Selected Areas in Communications*, vol. 40, p. 1037–1071, Apr. 2022.
- [19] Z. Ding, X. Lei, G. K. Karagiannidis, R. Schober, J. Yuan, and V. K. Bhargava, “A survey on non-orthogonal multiple access for 5g networks: Research challenges and future trends,” *IEEE Journal on Selected Areas in Communications*, vol. 35, no. 10, pp. 2181–2195, 2017.
- [20] R. C. Kizilirmak, *Non-Orthogonal Multiple Access (NOMA) for 5G Networks*. IntechOpen, 2016.
- [21] S. M. R. Islam, M. Zeng, O. A. Dobre, and K.-S. Kwak, “Non-orthogonal multiple access (noma): How it meets 5g and beyond,” 2019.
- [22] H. Tabassum, M. S. Ali, E. Hossain, M. J. Hossain, and D. I. Kim, “Uplink vs. downlink noma in cellular networks: Challenges and research directions,” in *2017 IEEE 85th Vehicular Technology Conference (VTC Spring)*, pp. 1–7, 2017.
- [23] B. Makki *et al.*, “A survey of noma: Current status and open research challenges,” *IEEE Open Journal of the Communications Society*, vol. 1, pp. 179–189, 2020.
- [24] B. Sklar, “Rayleigh fading channels in mobile digital communication systems .i. characterization,” *IEEE Communications Magazine*, vol. 35, no. 7, pp. 90–100, 1997.
- [25] T. S. Rappaport, *Wireless Communications: Principles and Practice*. Prentice Hall, 2 ed., 2001.
- [26] P. Botsinis, D. Alanis, Z. Babar, H. V. Nguyen, D. Chandra, S. X. Ng, and L. Hanzo, “Quantum search algorithms for wireless communications,” *IEEE Communications Surveys Tutorials*, vol. 21, pp. 1209–1242, 2019.
- [27] M. Al-Imari, P. Xiao, M. A. Imran, and R. Tafazolli, “Uplink non-orthogonal multiple access for 5g wireless networks,” in *2014 11th International Symposium on Wireless Communications Systems (ISWCS)*, IEEE, Aug. 2014.

References

- [28] J. S. Yeom, H. S. Jang, K. S. Ko, and B. C. Jung, “Ber performance of uplink noma with joint maximum-likelihood detector,” *IEEE Transactions on Vehicular Technology*, vol. 68, no. 10, pp. 10295–10300, 2019.
- [29] M. O. Damen, H. E. Gamal, and G. Caire, “On maximum-likelihood detection and the search for the closest lattice point,” *IEEE Trans. Inf. Theory*, vol. 49, pp. 2389–2402, 2003.
- [30] K. Sundaresan, M. Y. Arslan, S. Singh, S. Rangarajan, and S. V. Krishnamurthy, “Fluidnet: A flexible cloud-based radio access network for small cells,” *IEEE/ACM Transactions on Networking*, vol. 24, no. 2, pp. 915–928, 2016.
- [31] M. A. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*. AAPT, 2002.
- [32] G. Benenti, G. Casati, and G. Strini, *Principles of Quantum Computation and Information: Volume I: Basic Concepts*. World Scientific, 2004.
- [33] R. Pereira da Silva, “Annealing quantum computing: An overview,” 07 2023.
- [34] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*. Cambridge University Press, 2018.
- [35] Y. Chen, A. Bayesteh, Y. Wu, B. Ren, S. Kang, S. Sun, Q. Xiong, C. Qian, B. Yu, Z. Ding, S. Wang, S. Han, X. Hou, H. Lin, R. Visoz, and R. Razavi, “Towards the standardization of non-orthogonal multiple access for next generation wireless networks,” 2018.
- [36] J. Preskill, “Quantum computing in the nisq era and beyond,” *Quantum*, 2018.
- [37] E. Farhi, J. Goldstone, S. Gutmann, J. Lapan, A. Lundgren, and D. Preda, “A quantum adiabatic evolution algorithm applied to random instances of an np-complete problem,” *Science*, vol. 292, no. 5516, pp. 472–475, 2001.
- [38] A. Rajak, S. Suzuki, A. Dutta, and B. K. Chakrabarti, “Quantum annealing: an overview,” *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 381, Dec. 2022.
- [39] S. Yarkoni, E. Raponi, T. Bäck, and S. Schmitt, “Quantum annealing for industry applications: introduction and review,” *Reports on Progress in Physics*, vol. 85, no. 10, p. 104001, 2022.
- [40] C. McGeoch and P. Farré, “The d-wave advantage system: An overview technical report,” 2020.
- [41] C. McGeoch and P. Farré, “Advantage processor overview,” technical report, 1 2022.
- [42] S. Yarkoni, E. Raponi, T. Bäck, and S. Schmitt, “Quantum annealing for industry applications: introduction and review,” *Reports on Progress in Physics*, vol. 85, p. 104001, Sept. 2022.
- [43] D-Wave Systems, “D-wave qpu architecture: Topologies.” https://docs.dwavesys.com/docs/latest/c_gs_7.html, 2021. Accessed: April 2024.

References

- [44] Á. Marosits, Z. Tabi, Z. Kallus, P. Vaderna, I. Gódor, and Z. Zimborás, “Exploring embeddings for mimo channel decoding on quantum annealers,” *Infocommunications journal*, 2021.
- [45] Z. Tabi, Á. Marosits, Z. Kallus, P. Vaderna, I. Gódor, and Z. Zimborás, “Evaluation of quantum annealer performance via the massive mimo problem,” *IEEE Access*, vol. 9, pp. 131658–131671, 2021.
- [46] D-Wave Systems, “Constraints example: Minor-embedding.” https://docs.dwavesys.com/docs/latest/c_gs_7.html, 2021. Accessed: April 2024.
- [47] D-Wave Systems, “Solver properties and parameters.” https://docs.dwavesys.com/docs/latest/doc_solver_ref.html, 2022. Accessed: May 2024.
- [48] D-Wave Systems, “Operation and timing.” https://docs.dwavesys.com/docs/latest/c_qpu_timing.html, 2022. Accessed: May 2024.

Appendix A

QUBO MODEL COEFFICIENTS

A.1 16-QAM

The Q_{ij}^{16Q} values can be found using

$$Q_{(4i-3),(4i-3)}^{16Q} = \frac{P}{9} \left[-4h_{i,R} \left(h_{i,R} + 3 \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) - \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right. \\ \left. -4h_{i,I} \left(h_{i,I} + 3 \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right] - \frac{\sqrt{2P}}{3} (4y_R h_{i,R} + 4y_I h_{i,I}) \quad (\text{A.1})$$

$$Q_{(4i-2),(4i-2)}^{16Q} = \frac{P}{9} \left[-4h_{i,R} \left(h_{i,R} + \frac{3}{2} \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) - \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right. \\ \left. -4h_{i,I} \left(h_{i,I} + \frac{3}{2} \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right] - \frac{\sqrt{2P}}{3} (2y_R h_{i,R} + 2y_I h_{i,I}) \quad (\text{A.2})$$

$$Q_{(4i-1),(4i-1)}^{16Q} = \frac{P}{9} \left[-4h_{i,R} \left(h_{i,R} + 3 \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right. \\ \left. -4h_{i,I} \left(h_{i,I} + 3 \left(- \left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right] - \frac{\sqrt{2P}}{3} (-4y_R h_{i,I} + 4y_I h_{i,R}) \quad (\text{A.3})$$

$$\begin{aligned}
 Q_{(4i),(4i)}^{16Q} = & \frac{P}{9} \left[-4h_{i,R} \left(h_{i,R} + \frac{3}{2} \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right. \\
 & \left. -4h_{i,I} \left(h_{i,I} + \frac{3}{2} \left(- \left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right] - \frac{\sqrt{2P}}{3} (-2y_R h_{i,I} + 2y_I h_{i,R})
 \end{aligned} \tag{A.4}$$

$$\begin{aligned}
 Q_{(4i-3),(4i-1)}^{16Q} &= Q_{(4i-3),(4i)}^{16Q} \\
 &= Q_{(4i-2),(4i-1)}^{16Q} = Q_{(4i-2),(4i)}^{16Q} = 0,
 \end{aligned} \tag{A.5}$$

$$Q_{(4i-3),(4i-2)}^{16Q} = Q_{(4i-1),(4i)}^{16Q} = \frac{8P}{9} (|h_{i,R}|^2 + |h_{i,I}|^2), \tag{A.6}$$

$$\begin{aligned}
 Q_{(4i-3),(4j-2)}^{16Q} &= Q_{(4i-2),(4j-3)}^{16Q} = Q_{(4i),(4j-1)}^{16Q} \\
 &= Q_{(4i-1),(4j)}^{16Q} = \frac{P}{9} (8h_{i,R}h_{j,R} + 8h_{i,I}h_{j,I}),
 \end{aligned} \tag{A.7}$$

$$Q_{(4i-2),(4j-2)}^{16Q} = Q_{(4i),(4j)}^{16Q} = \frac{P}{9} (4h_{i,R}h_{j,R} + 4h_{i,I}h_{j,I}), \tag{A.8}$$

$$\begin{aligned}
 Q_{(4i-1),(4j-1)}^{16Q} &= Q_{(4i-3),(4j-3)}^{16Q} \\
 &= \frac{P}{9} (16h_{i,R}h_{j,R} + 16h_{i,I}h_{j,I}),
 \end{aligned} \tag{A.9}$$

$$Q_{(4i-2),(4j-1)}^{16Q} = Q_{(4i-3),(4j)}^{16Q} = \frac{8P}{9} (-h_{i,R}h_{j,I} + h_{i,I}h_{j,R}), \tag{A.10}$$

$$\begin{aligned}
 Q_{(4i-1),(4j-2)}^{16Q} &= Q_{(4i),(4j-3)}^{16Q} \\
 &= \frac{P}{9} (-8h_{i,I}h_{j,R} + 8h_{i,R}h_{j,I}),
 \end{aligned} \tag{A.11}$$

$$Q_{(4i-2),(4j)}^{16Q} = \frac{P}{9} (-4h_{i,R}h_{j,I} + 4h_{i,I}h_{j,R}), \tag{A.12}$$

$$Q_{(4i),(4j-2)}^{16Q} = \frac{P}{9} (-4h_{i,I}h_{j,R} + 4h_{i,R}h_{j,I}), \tag{A.13}$$

$$Q_{(4i-1),(4j-3)}^{16Q} = \frac{P}{9} (-16h_{i,I}h_{j,R} + 16h_{i,R}h_{j,I}), \tag{A.14}$$

$$Q_{(4i-3),(4j-1)}^{16Q} = \frac{P}{9} (-16h_{i,R}h_{j,I} + 16h_{i,I}h_{j,R}). \tag{A.15}$$

A.2 64-QAM

The corresponding Q_{ij}^{64Q1} values are presented below, where $\{c_1, c_2, c_3\} = \{6, 14, 8\}$ for $m = 6i - 5$ or $l = 6i - 2$, $\{5, 7, 4\}$ for $m = 6i - 4$ or $l = 6i - 1$, and $\{3, \frac{7}{2}, 2\}$ for $m = 6i - 3$ or $l = 6i$.

$$Q_{(m),(m)}^{64Q} = \frac{P}{49} \left[-4h_{i,R} \left(c_1 h_{i,R} + c_2 \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) - \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right. \\ \left. - 4h_{i,I} \left(c_1 h_{i,I} + c_2 \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right] - \frac{\sqrt{2P}}{7} c_3 (y_R h_{i,R} + y_I h_{i,I}) \quad (\text{A.16})$$

$$Q_{(l),(l)}^{64Q} = \frac{P}{49} \left[-4h_{i,R} \left(c_1 h_{i,R} + c_2 \left(\left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right. \\ \left. - 4h_{i,I} \left(c_1 h_{i,I} + c_2 \left(- \left(\sum_{k=1}^N h_{k,R} - h_{i,R} \right) + \left(\sum_{k=1}^N h_{k,I} - h_{i,I} \right) \right) \right) \right] - \frac{\sqrt{2P}}{7} c_3 (y_I h_{i,R} - y_R h_{i,I}) \quad (\text{A.17})$$

$$Q_{(6i-5),(6i-4)}^{64Q} = Q_{(6i-2),(6i-1)}^{64Q} = \frac{P}{49} (32|h_{i,R}|^2 + 32|h_{i,I}|^2), \quad (\text{A.18})$$

$$Q_{(6i-5),(6i-3)}^{64Q} = Q_{(6i-2),(6i)}^{64Q} = \frac{P}{49} (16|h_{i,R}|^2 + 16|h_{i,I}|^2), \quad (\text{A.19})$$

$$Q_{(6i-4),(6i-3)}^{64Q} = Q_{(6i-1),(6i)}^{64Q} = \frac{8P}{49} (|h_{i,R}|^2 + |h_{i,I}|^2), \quad (\text{A.20})$$

$$Q_{(6i-5),(6i-2)}^{64Q} = Q_{(6i-5),(6i-1)}^{64Q} = Q_{(6i-5),(6i)}^{64Q} \\ = Q_{(6i-4),(6i-2)}^{64Q} = Q_{(6i-4),(6i-1)}^{64Q} = Q_{(6i-4),(6i)}^{64Q} \\ = Q_{(6i-3),(6i-2)}^{64Q} = Q_{(6i-3),(6i-1)}^{64Q} = Q_{(6i-3),(6i)}^{64Q} = 0, \quad (\text{A.21})$$

$$Q_{(6i-5),(6j-5)}^{64Q} = Q_{(6i-2),(6j-2)}^{64Q} \\ = \frac{P}{49} (64h_{i,R}h_{j,R} + 64h_{i,I}h_{j,I}), \quad (\text{A.22})$$

$$Q_{(6i-5),(6j-2)}^{64Q} = \frac{P}{49} (-64h_{i,R}h_{j,I} + 64h_{i,I}h_{j,R}), \quad (\text{A.23})$$

¹Note that the equations involving the positive integer variables i and j must satisfy the condition $i < j$.

$$Q_{(6i-2),(6j-5)}^{64Q} = \frac{P}{49} (64h_{i,R}h_{j,I} - 64h_{i,I}h_{j,R}), \quad (\text{A.24})$$

$$\begin{aligned} Q_{(6i-5),(6j-4)}^{64Q} &= Q_{(6i-4),(6j-5)}^{64Q} = Q_{(6i-2),(6j-1)}^{64Q} \\ &= Q_{(6i-1),(6j-2)}^{64Q} = \frac{P}{49} (32h_{i,R}h_{j,R} + 32h_{i,I}h_{j,I}), \end{aligned} \quad (\text{A.25})$$

$$\begin{aligned} Q_{(6i-5),(6j-1)}^{64Q} &= Q_{(6i-4),(6j-2)}^{64Q} \\ &= \frac{P}{49} (-32h_{i,R}h_{j,I} + 32h_{i,I}h_{j,R}), \end{aligned} \quad (\text{A.26})$$

$$\begin{aligned} Q_{(6i-2),(6j-4)}^{64Q} &= Q_{(6i-1),(6j-5)}^{64Q} \\ &= \frac{P}{49} (32h_{i,R}h_{j,I} - 32h_{i,I}h_{j,R}), \end{aligned} \quad (\text{A.27})$$

$$\begin{aligned} Q_{(6i-5),(6j-3)}^{64Q} &= Q_{(6i-4),(6j-4)}^{64Q} = Q_{(6i-3),(6j-5)}^{64Q} \\ Q_{(6i-2),(6j)}^{64Q} &= Q_{(6i-1),(6j-1)}^{64Q} = Q_{(6i),(6j-2)}^{64Q} \\ &= \frac{P}{49} (16h_{i,R}h_{j,R} + 16h_{i,I}h_{j,I}), \end{aligned} \quad (\text{A.28})$$

$$\begin{aligned} Q_{(6i-5),(6j)}^{64Q} &= Q_{(6i-4),(6j-1)}^{64Q} = Q_{(6i-3),(6j-2)}^{64Q} \\ &= \frac{P}{49} (-16h_{i,R}h_{j,I} + 16h_{i,I}h_{j,R}), \end{aligned} \quad (\text{A.29})$$

$$\begin{aligned} Q_{(6i-2),(6j-3)}^{64Q} &= Q_{(6i-1),(6j-4)}^{64Q} = Q_{(6i),(6j-5)}^{64Q} \\ &= \frac{P}{49} (16h_{i,R}h_{j,I} - 16h_{i,I}h_{j,R}), \end{aligned} \quad (\text{A.30})$$

$$\begin{aligned} Q_{(6i-4),(6j)}^{64Q} &= Q_{(6i-3),(6j-1)}^{64Q} \\ &= \frac{P}{49} (-8h_{i,R}h_{j,I} + 8h_{i,I}h_{j,R}), \end{aligned} \quad (\text{A.31})$$

$$\begin{aligned} Q_{(6i-4),(6j-3)}^{64Q} &= Q_{(6i-3),(6j-4)}^{64Q} = Q_{(6i-1),(6j)}^{64Q} \\ &= Q_{(6i),(6j-1)}^{64Q} = \frac{P}{49} (8h_{i,R}h_{j,R} + 8h_{i,I}h_{j,I}), \end{aligned} \quad (\text{A.32})$$

$$Q_{(6i-1),(6j-3)}^{64Q} = Q_{(6i),(6j-4)}^{64Q} = \frac{8P}{49} (h_{i,R}h_{j,I} - h_{i,I}h_{j,R}), \quad (\text{A.33})$$

$$Q_{(6i-3),(6j-3)}^{64Q} = Q_{(6i),(6j)}^{64Q} = \frac{P}{49} (4h_{i,R}h_{j,R} + 4h_{i,I}h_{j,I}), \quad (\text{A.34})$$

$$Q_{(6i-3),(6j)}^{64Q} = \frac{P}{49} (-4h_{i,R}h_{j,I} + 4h_{i,I}h_{j,R}), \quad (\text{A.35})$$

$$Q_{(6i),(6j-3)}^{64Q} = \frac{P}{49} (4h_{i,R}h_{j,I} - 4h_{i,I}h_{j,R}). \quad (\text{A.36})$$